Generating Clinically Meaningful Patient Networks from Therapy Transcripts

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Darmstadt, 15. Juni 2025

E. Owtscharow

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Abstract

In recent years, there has been a notable increase in the demand for mental health support, driven by greater public awareness, reduced stigma, and large-scale events such as the COVID-19 pandemic. As more individuals seek help, it becomes increasingly important that therapeutic approaches adapt to the diverse and nuanced psychological needs of each person. Traditional therapies like Cognitive Behavioral Therapy (CBT) and Acceptance and Commitment Therapy (ACT) typically follow standardized treatment protocols, but they often fail to account for individual psychological differences. As a result, psychologists are increasingly personalizing treatments to better meet each person's unique needs and circumstances. Process-Based Therapy (PBT) addresses this issue by creating personalized networks of psychological processes that guide tailored interventions. However, manually constructing these networks after each therapy session for every patient is both labor-intensive and mentally demanding for therapists.

This thesis presents an end-to-end pipeline that automates the construction of patient-specific networks from annotated therapy session transcripts, leveraging large language models (LLMs). The clinical usefulness and relevance of these networks are evaluated, particularly in the context of case conceptualization and treatment planning. The proposed method involves prep-rocessing session data, identifying clinically meaningful themes, grouping psychological processes observed in the therapy dialogue under these themes, and determining the strength and direction of inter-theme relationships that describe the patient's psychological dynamics. We evaluate our approach through four experiments, including expert assessments by clinical psychologists specialized in PBT. Results show that the generated networks are consistently rated as clinically relevant, novel, helpful, and trustworthy, outperforming simultaneous prompting and a baseline in pairwise comparisons.

Our contributions include a clinically grounded evaluation framework co-developed with PBT experts, as well as the release of 77 patient networks derived from six therapy cases. These networks abstract psychological functioning without disclosing personal data, offering a valuable resource for both practice and future research. This work provides a scalable and interpretable tool to enhance therapist efficiency and deepen understanding of personalized mental health care.

In den letzten Jahren ist die Nachfrage nach Therapie deutlich gestiegen, was auf ein größeres öffentliches Bewusstsein, den Abbau von Stigmatisierung und großflächige Ereignisse wie die COVID-19-Pandemie zurückzuführen ist. Da immer mehr Menschen Hilfe suchen, wird es zunehmend wichtiger, dass therapeutische Ansätze sich an die vielfältigen psychologischen Bedürfnisse jedes Einzelnen anpassen.

Bei traditionellen Therapien wie der kognitiven Verhaltenstherapie (CBT) und der Akzeptanz- und Commitment-Therapie (ACT) wird mit standardisierten Behandlungsprotokollen gearbeitet. Dabei werden individuelle psychologische Unterschiede jedoch häufig übersehen. Um Interventionen besser an die individuellen Bedürfnisse und Umstände von Patientinnen anzupassen, personalisieren Psychologinnen ihre Behandlungen zunehmend. Process-Based Therapy (PBT) schließt diese Lücke, indem sie personalisierte Netzwerke psychologischer Prozesse modelliert, die gezielte Interventionen unterstützen. Die manuelle Erstellung solcher Netzwerke nach jeder Therapiesitzung ist jedoch arbeitsintensiv und stellt eine hohe kognitive Belastung für Therapeut*innen dar.

In dieser Arbeit wird eine End-to-End-Pipeline zur automatisierten Erstellung patientenspezifischer Netzwerke aus annotierten Transkripten von Therapiesitzungen mithilfe großer Sprachmodelle (LLMs) präsentiert. Die klinische Nützlichkeit und Relevanz dieser Netzwerke wird insbesondere im Hinblick auf Fallkonzeptualisierung und Behandlungsplanung evaluiert. Der vorgeschlagene Ansatz umfasst die Vorverarbeitung der Sitzungsdaten, die Identifikation klinisch bedeutsamer Themen, die Zuordnung psychologischer Prozesse innerhalb des Therapiedialogs zu diesen Themen sowie die Bestimmung von Stärke und Richtung der Beziehungen zwischen den Themen, um die psychologische Dynamik der Patient*innen zu beschreiben.

Wir evaluieren unseren Ansatz anhand von vier Experimenten, darunter Expertenbewertungen durch auf PBT spezialisierte klinische Psycholog*innen. Die Ergebnisse zeigen, dass die generierten Netzwerke durchweg als klinisch relevant, neuartig, hilfreich und vertrauenswürdig eingestuft werden. Die Ergebnisse zeigen, dass die simultane Durchführung von Prompting und der anschließende paarweise Vergleich mit der Baseline zu besseren Ergebnissen führt.

Unsere Beiträge umfassen ein klinisch fundiertes Bewertungsframework, das gemeinsam mit PBT-Expertinnen entwickelt wurde, sowie die Veröffentlichung von 77 Patientennetzwerken aus sechs Therapiefällen. Diese Netzwerke abstrahieren psychichse funktionsweise, ohne persönliche Daten offenzulegen, und stellen eine wertvolle Ressource für Praxis und zukünftige Forschung dar. Diese Arbeit bietet ein skalierbares und interpretierbares Werkzeug zur Steigerung der Effizienz von Therapeutinnen und zur Vertiefung des Verständnisses personalisierter psychischer Therapie.

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1. Introduction

In recent years, mental health issues have emerged as a significant public health challenge. This increase is primarily driven by rising economic pressures [34] global conflicts [39, 41] and also the impact of COVID-19 [43]. At the same time, societal attitudes toward mental health are evolving. People are more open about mental health issues and more willing to seek therapy, which further contributes to the growing visibility of mental health concerns [6]. To address these challenges, traditional treatment methods, such as Cognitive Behavioral Therapy (CBT) and Acceptance and Commitment Therapy (ACT), are commonly employed. Many traditional therapeutic approaches often employ standardized protocols that adhere to one-size-fits-all strategies. Consequently, they may fail to address the specific changes required by an individual to improve their well-being [13].

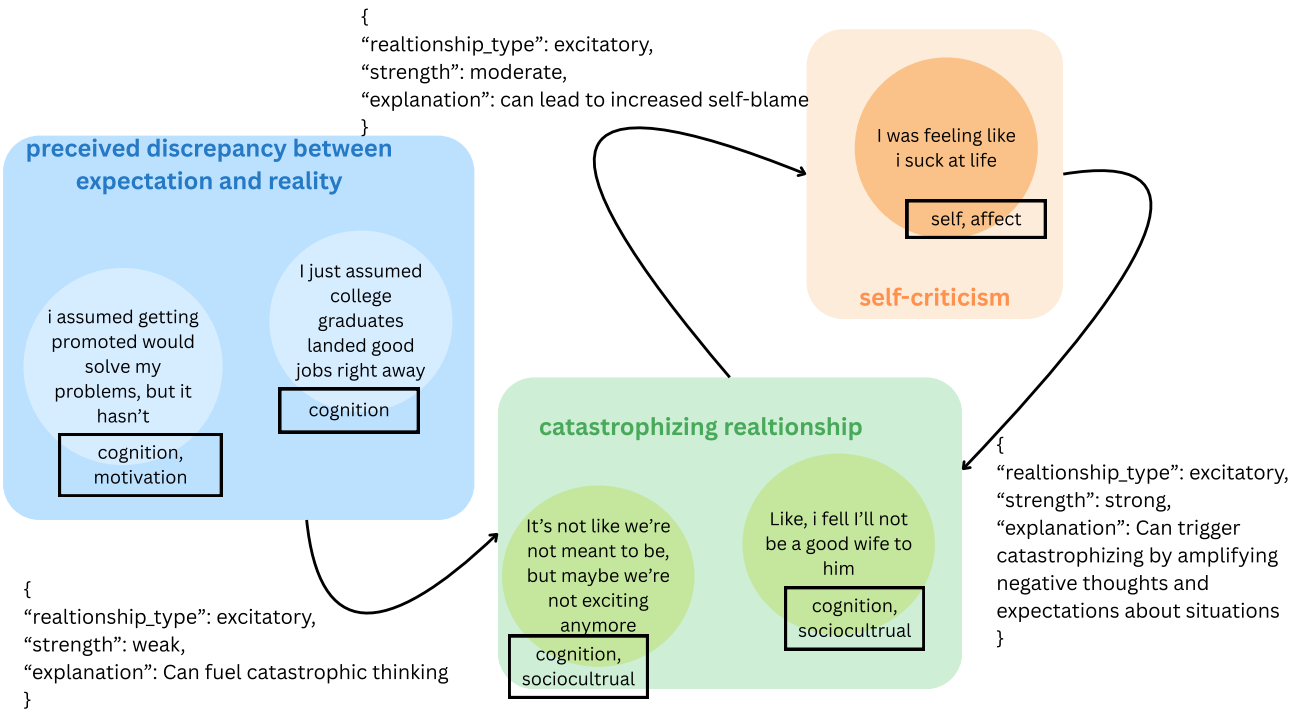


Figure 1.1.: **Patient Network:** A simplified example of a patient network following a therapy session comprises five psychological processes, each with associated dimensions, grouped into three distinct, clinically meaningful categories. Expressive relationships between these groups indicate the type of connection, its strength, and an explanation for their connection.

In contrast, Process Based Therapy (PBT) offers a compelling alternative by adopting a more personalized

approach. PBT aims for more effective treatment by tailoring strategies to meet the specific needs of each individual. It focuses on identifying and targeting core psychological processes [22]. A psychological process refers to a recurring, meaningful pattern that reflects underlying cognitive, emotional, or behavioral functions within a patient's experience. It captures how an individual perceives, interprets, reacts to, or regulates internal or external events. These processes can span a range of mental functions, including emotional responses, thoughts, decisions, memories, social interpretations, and personal reflections, and are often categorized under one or more predefined psychological dimensions [13, 22]. Together, they provide insight into the patient's psychological functioning and therapeutic needs.

To apply this approach effectively, therapists create a patient network that maps out these psychological processes [8, 13], as illustrated in Figure 1.1. To construct this network, the therapist first identifies the relevant psychological processes. These processes are then grouped into clinically meaningful themes, which reflect patterns that reveal underlying psychological functions or dynamics in the patient's experience. Finally, the therapist connects these groups using expressive relationships that reflect psychological dynamics. This patient network supports both the monitoring of therapeutic progress and the development of targeted interventions for future sessions [13, 22].

Constructing patient networks is both time-consuming and cognitively demanding, as therapists must manually generate and revise them after each session for every individual. These networks are dynamic in nature, continuously evolving as therapy progresses. For clinicians managing multiple patients, the ongoing effort required to maintain and update personalized networks can quickly become overwhelming. To reduce this burden and make PBT more practical, we have developed a pipeline that automates the creation of these patient networks. Our approach leverages language models (LM) and large language models (LLM) to analyze session data and generate meaningful, individualized networks. This automation not only boosts efficiency but also uncovers valuable insights that might otherwise go unnoticed. By easing the cognitive load on therapists, it allows them to focus more fully on delivering effective, personalized care.

This thesis aims to address the following three research questions:

1. How can we leverage LLMs to find and organize recurring clinical themes from annotated transcripts of therapy sessions?
2. In what ways can LLMs establish meaningful connections between identified clinical themes to construct a patient network that enhances therapeutic understanding?
3. How insightful are the created patient networks to clinical psychologists?

In this context, *insightful* refers to the ability of LLM-generated patient networks to provide clinical psychologists with a deeper understanding of a patient's psychological functioning. These networks provide enhanced insight into a patient's condition, contributing to a clearer understanding of therapeutic progress.

Figure 1.1, for example, illustrates how clinical psychologists can gain insight into a patient's psychological state. It presents three clinically meaningful themes extracted from a single therapy session: *Perceived Discrepancy Between Expectation and Reality, self-Criticism*, and *Catastrophizing Relationships.* In this session, the patient expressed, "*I was feeling like i suck at life,*" which was categorized under *Self-Criticism* as a representative statement of that theme. Another utterance, "*i feel I'll not be a good wife to him,*" was grouped under *Catastrophizing Relationship*. A connection between these two themes reveals how self-critical thoughts can intensify negative expectations in relational contexts.

The LLM generated both the thematic categorization and the expressive connections between themes. This example demonstrates how LLMs can effectively organize psychological processes into clinically meaningful themes and how the relationships between these themes can uncover valuable therapeutic insights.

The contributions of this work are as follows:

1. **We pre-process the provided annotated therapy session transcripts**

   We implement a two-step pre-processing approach: (1) reducing redundancy by identifying and removing repetitive utterances that naturally occur in spoken dialogue, and (2) generating and using human-annotated ground truth data to distinguish between filler and non-filler sentences, as well as to identify semantically similar clusters. These steps produce a cleaner, more compact dataset that is semantically rich, thereby enabling more effective analysis in later stages.

2. **We cluster psychological processes into clinically meaningful themes**

   To ensure both interpretability and clinical utility, we first established clear annotation guidelines and developed custom evaluation metrics in collaboration with clinical psychologists. This step is crucial, as it represents the first attempt to generate patient networks from annotated and transcribed therapy sessions. The guidelines help evaluators consistently assess the quality and coherence of thematic clusters that emerge from the psychological processes identified in the transcripts. Additionally, we developed a comprehensive evaluation framework in collaboration with clinical psychologists. Their expert input was essential in shaping the metrics, determining what constitutes clinically valuable content, and assessing the quality of the generated clusters. This framework leverages expert-annotated ground truth data, providing examples of both clinically valuable and non-valuable themes.

3. **We generate expressive relationships between the psychological processes**

   In addition to identifying clinically meaningful themes, we use an ensemble approach with LLMs to generate expressive links between previously isolated themes. Unlike simple co-occurrence methods that link themes based on proximity within the text, our approach captures deeper conceptual relationships between psychological processes. This enables the construction of coherent and interpretable patient networks that reflect the psychological dynamics underlying a patient's experience, providing clinicians with a more structured and nuanced understanding of the patient's condition.

4. **We are providing openly available Patient Networks**

   We release 77 openly available patient Networks. Each network includes clinically meaningful themes and the expressive relationships between them, along with a fine-grained expert evaluation based on a rubric co-developed with clinical psychologists. This ensures that the networks are not only interpretable but also clinically grounded and reliable for future research. To safeguard patient privacy, we do not include direct quotes from therapy sessions in the released materials.

Through these contributions, we facilitate the automated generation of patient networks that provide clinically meaningful insights for psychologists. This automation not only reduces the considerable effort required to construct such networks manually but also meets the growing demands on clinicians by supporting more efficient and structured analysis of therapeutic processes.

The structure of this thesis is as follows: Section 2 presents a review of related work covering the application of LLMs in mental health, natural language processing (NLP) clustering techniques, and the challenges of personalized therapy. Section 3 outlines the proposed methodology, detailing the construction of the pipeline used to generate patient networks. Section 4 provides a comprehensive account of the experiments, including results, analysis, and error evaluation. Finally, Section 5 concludes the thesis with a summary of key findings, a discussion of implications, and suggestions for future research.

2. Related Work

In this chapter, we review prior work relevant to our approach. First, we explore how LLMs have been applied in the mental health domain, particularly in contexts such as therapy support, diagnostic assistance, and mental motioning. We then examine existing methods for clustering in NLP, focusing on how clustering techniques have evolved from a traditional approach to more advanced strategies, leveraging LM and LLMs. We then briefly touch on approaches to relationship generation. Finally, we provide an overview of PBT, outlining its theoretical foundations, clinical application, and relevance.

2.1. LLMs in Mental Health Applications

LLms have demonstrated significant potential in clinical and mental health contexts. Their application in this domain has attracted growth, with notable examples including the use of social media data to detect signs of depression [7]. The files are increasingly moving toward the development of explainable models for mental health applications. Achieved through prompting strategies that enable the extractions from the model [45] or developing LLMs, such as a fine-tuned version of LLaMA 2 [37], designed explicitly for explainable mental health task using text generation rather than traditional binary or multi-class classification approaches [44].

Additionally, recent studies have evaluated newer large language models (LLMs) in various mental health scenarios. For instance, open-source models like LLaMA 3.1 [37] have demonstrated accuracy rates of approximately 90% in psychiatric knowledge [12], underscoring the increasing reliability of LLMs in diverse applications.

There has also been progress in analyzing real-world therapy session transcripts. Aghakhani et al. [3] employed LLMs to annotate Problem-Solving Therapy (PST) strategies within such transcripts and evaluated the alignment of model-generated annotations with those made by human experts. Their findings suggest that LLMs, such as GPT-4, can reliably identify therapeutic strategies. The authors also noted that research leveraging real-life counseling conversations remains relatively limited, pointing to an important area for further exploration.

Building upon this progress, the present thesis introduces a novel approach: the development of patient networks grounded in PBT. Unlike prior studies that rely on social media data or use transcribed therapy sessions solely for annotation or classification tasks, this work constructs patient networks directly from transcribed therapy sessions. To the best of our knowledge, no existing work has developed a complete pipeline for generating patient networks from such transcripts. This thesis, therefore, fills a critical gap by proposing a structured, end-to-end methodology tailored specifically for real-world therapeutic data.

At the same time, it retains the core principle of interpretable text generation, leveraging it to produce outputs that offer meaningful insights into underlying psychological processes.

2.1.1. LLMs for Therapy Sessions Summarization

Previous research, such as the study by [2], has examined the use of LLMs for summarizing therapy sessions, with a primary focus on generating textual summaries of session content. In contrast, our work extends beyond mere summarization. While we also utilize transcribed therapy sessions, our approach involves constructing patient networks that offer deeper insights into the patient's psychological functioning and reveal dynamic interactions between psychological processes. This network-based methodology enables clinicians to develop a more nuanced and integrative understanding of the patient.

2.2. Semantic Clustering in NLP

LM-based Clustering One of the initial steps in our pipeline involves the semantic clustering of utterances marked as psychological processes within transcribed therapy sessions. Semantic clustering refers to the task of grouping similar utterances based on their underlying meaning. In the early stages of NLP, traditional techniques such as Bag of Words (BoW) [28] and TF-IDF [33] were commonly used for this purpose. However, these methods rely on word frequency and do not capture the semantic context of text [17]. With the advent of LM, it is now possible to transform texts or sentences into dense vector representations, known as embeddings, that encode semantic information. These embeddings enable more accurate clustering of semantically related utterances, even when the exact words are not used [1].

Abdalgader et al. [1] conducted extensive experiments on short text clustering using modern embeddings techniques in combination with various clustering algorithms including *k-means* [35] and *agglomerative clustering* [24]. Their workflow inspired our approach to semantic clustering in this thesis. Following a similar methodology, we first generate embeddings using the language model (LM) and then apply different clustering algorithms to group semantically similar utterances. We compare the outcomes to determine the most effective combination. Notably, Abdalgader et al. [1] that LM such as SentenceBERT [30] significantly outperformed traditional similarity measures. Specifically, the combination of *paraphrase-distilroberta-base-v1* and hierarchical (agglomerative) clustering yields the best results.

A significant limitation of widely used clustering algorithms, such as k-means and standard agglomerative clustering, is that they necessitate the specification of a predefined number of clusters in advance. In the context of therapy sessions, however, this number is not known beforehand. This uncertainty poses challenges when attempting to accurately group utterances based on their semantic content. Therefore, we explore alternative clustering methods that do not rely on a fixed number of clusters. As described in Section 4, we conducted our experiments using various LM for embedding generation and clustering methods such as BERTopic [11], which do not require a predefined number of clusters. To guide our choice of embedding model, we consulted the Massive Text Embedding Benchmark (MTEB) [23], a large-scale evaluation framework that covers eight embedding tasks, including clustering and classification.

Although several studies have addressed the challenge of short text clustering, the task remains complex [46, 17]. Short texts often span a broad range of categories with diverse distributions, leading to class imbalance and making clustering more difficult [46]. Notably, Zheng et al. and Kamthawee et al. demonstrate that fine-tuning models like SentenceBERT [30] on domain-specific data can significantly improve clustering performance. Their findings suggest that domain-specific embeddings can capture more meaningful semantic relationships, resulting in more robust clustering. However, fine-tuning requires a large volume of data, which is not feasible in our case due to the relatively small size of our dataset.

LLM-based Clustering After reducing redundancy through semantic clustering, we incorporate LLM-based clustering at a later stage in our pipeline to assist in generating higher-level themes. Unlike clustering methods that rely on vector-based similarity and fixed algorithms, this approach prompts an LLM to actively group or label utterances based on thematic conceptual similarity. By leveraging the model's internal knowledge and reasoning capabilities, this method facilitates the identification of overarching psychological themes and organizes utterances accordingly. This step is crucial for moving beyond surface-level semantic similarity, allowing more abstract and interpretable structures to emerge. In doing so, LLM-based clustering supports the thematic synthesis of patient utterances and aligns with the broader objectives of PBT.

Huan and He [16] introduced a two-step approach to LLM-based clustering. In the first stage, the model is provided with a set of sentences and example labels and is prompted to generate new thematic labels for the input sentences. After these labels are aggregated and redundancies removed, the second stage transforms the task into a classification problem, where the model assigns each sentence to one of the generated themes. Inspired by this method, we adapted and integrated it into our pipeline (see Section 3.3), tailoring it to the specific needs of the clinical psychology domain. To ensure domain relevance and clinical precision, we refined the original prompts by incorporating expert-written definitions, psychological theory, and domain-specific instructions. Additionally, we compare this two-step method to a one-step variant, in which the model performs both theme generation and sentence assignment within a single prompt. This comparison allows us to assess the differences in performance and suitability for the clinical context.

2.3. Relationship Generation

At the end of our pipeline, we also generate expressive relationships between clinically meaningful themes. In contrast to prior work such as He et al. [14], which focuses on generating missing links between entities, our approach requires a deeper level of interpretability and domain-specific reasoning. Specifically, their method emphasizes link prediction based on ranking relevance, similar to a PageRank-style recommendation system, without offering insight into why entities are connected. In our case, the LLM must construct relationships without pre-existing links and instead be grounded in psychological knowledge. This requires the model not only to determine whether a connection exists but also to explain why it exists and to characterize it with attributes such as connection type and strength. As a result, our generated relationships go beyond statistical associations to support clinically meaningful interpretation and therapeutic use.

2.4. Personalized Therapy

Historically, mental health treatment has been based on the idea that mental health problems are caused by hidden diseases in the brain. This idea has shaped the way mental health conditions are officially classified in systems such as the Diagnostic and Statistical Manual of Mental Disorders (DSM) and the International Statistical Classification of Diseases (ICD), which classify mental health problems like depression or anxiety as specific disorders. Another approach used early in behavior therapy focused more on how their environment and situation shape a person's behavior. Later, in the 1980s, this idea merged with the notion that our thoughts also play a significant role. This led to what we now call cognitive behavioral therapy (CBT) [22, 13, 9]. CBT became the most widely accepted method in psychology. It focuses on how our thoughts, feelings, and actions are connected. Over time, therapists developed special CBT programs for specific mental health diagnoses, such as Obsessive-Compulsive Disorder (OCD) or depression. These programs were tested in controlled trials

to ensure their effectiveness, and this type of testing has shaped most psychological research over the past several decades [13].

However, there have been problems with using a one-size-fits-all approach to general diagnoses. Many people have more than one mental health condition at a time, making it hard to treat just one without considering other important issues. Additionally, people are unique, so a standard treatment may not be effective for everyone. As a result, the field is shifting toward personalized treatments that focus on how biological, psychological, and social factors interact to bring about change. Another primary reason for this shift is that studies conducted on large groups do not always accurately reflect how individuals change over time. Psychologists are moving away from general treatment models that overlook individual differences. Instead, the field is shifting toward treatments that consider what works for each person individually and then build general ideas from there without losing what makes each person unique. This is part of what is called PBT.

PBT shifts the question from "*what treatment works for this disorder*" to "*what specific process should we focus on for this person in this situation and how can we change it effectively?*" [22, 15]. PBT looks at the personal patterns and reasons behind a person's mental health struggles. This approach is more helpful because it considers how life history, culture, relationships, and current life situations have shaped the person rather than just focusing on symptoms. Simply using labels such as depression and anxiety does not capture how different people experience and cope with these problems. People with the same diagnosis can have very different life stories and reasons for their symptoms. Because PBT focuses on understanding how a person changes rather than just treating a diagnosis, therapists can mix and match different strategies from various types of therapy. They are not limited to one method and can choose whatever works best for each individual. The therapist selects tools that align with the person's unique problems and goals, monitors whether these tools are effective, and adjusts the approach as necessary. It is not random or unstructured but rather a thoughtful, science-based approach to help someone improve. For example, suppose a therapist believes a client feels lonely, and that is a significant part of their struggle. In that case, they might encourage them to engage in more social activities or pursue hobbies they enjoy. They would then carefully track if those changes are helping, using detailed tracking to see patterns in the client's progress.

PBT is not about randomly trying different things. It involves using real data from a person's life over time to make informed, testable predictions about what will be effective. By closely tracking changes, the therapist can determine whether the chosen approach is effective. PBT also builds on ideas from evolutionary science. It does not view mental health problems as diseases, as traditional treatments have done, but as ways of thinking, feeling, or behaving that do not work well in a person's current life. These problems come from difficulties in how people change, choose, or maintain specific patterns over time. To track these patterns, therapists utilize patient networks. This network helps organize all the factors that influence a person's mental health.

3. Methodology

In this section, we outline the methodology used in this thesis to generate patient networks from annotated therapy transcripts automatically. Specifically, we describe the overall pipeline and explain each step involved in transforming transcribed and annotated therapy sessions into structured patient networks.

3.1. Overview of the Pipeline

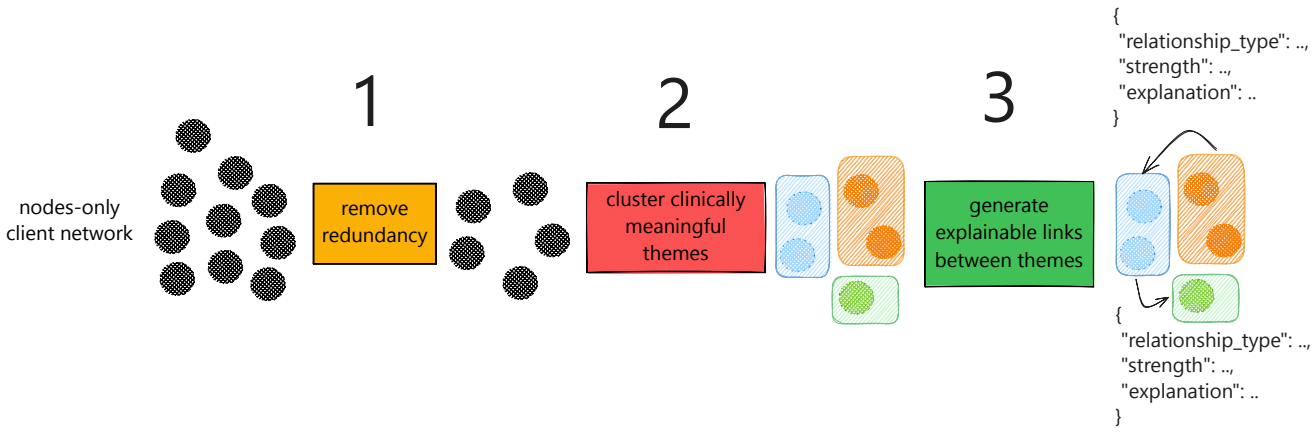


Figure 3.1.: **The Pipeline for Automatically Constructing Patient Networks:** The pipeline consists of three steps for constructing a patient network from an annotated therapy transcript. The first step removes redundancy and noise. The second step generates themes and classifies the psychological processes under each theme. The third step establishes expressive relationships between themes, including the type, strength, and explanatory context of the relationship.

An overview of the pipeline is presented in Figure 3.1. which consists of three major work packages: 3.2 **Redundancy Removal & Preprocessing**, 3.3 **Generating Clinically Meaningful Process clusters**, 3.4 **Generatinc Clinically Meaningful Connections**.

As input, we received anonymized therapy session transcripts, in which identifiable information such as names and locations had been replaced with placeholder tokens. Clinical psychologists had previously annotated these transcripts by identifying and labeling psychological processes. The dataset consists of 77 transcribed and annotated sessions from six patients and was provided by the The Department of Psychology at the University of Toledo [38].

Two clinical psychologists performed the annotations. For each utterance by either the patient or therapist, they indicated whether it contained a psychological process *(yes/no)* and if so, assigned one or more predefined

labels corresponding to nine psychological dimensions (e.g., Affect, Cognition, Attention, Sense of Self, Motivation, Overt Behavior, Sociocultural, Biophysical, Moderators) [13]. These annotations were applied at the utterance level, with each marked span representing a complete spoken segment.

From an NLP perspective, the dataset is structured as a sequence of timestamped utterances, each including the speaker identity (patient or therapist), the exact text spoken, and the time range during which it occurred. Further details about the dataset are provided in Section 4.1.

The goal of the first work package is to reduce redundant information. Specifically, psychological processes that have already been annotated but are either repeated by the patient or lack inherent meaning within the context of the therapy session. In the second work package, the filtered psychological processes serve as input to an LLM, which generates clinically meaningful themes and classifies each process accordingly. The third and final work package builds upon these themes by applying ensemble learning to generate expressive links between them. These connections establish meaningful relationships among the themes, collectively forming the resulting patient network.

3.2. Work Package 1: Redundancy Removal & Preprocessing

The Redundancy Removal & Preprocessing work package consists of three sequential steps, as illustrated in Figure 3.2. These steps are: (1) "Sting Matching with Context Expansion," (2) "Majority Vote-Based Filler Detection", and (3) "Embedding-Based Sentence Clustering".

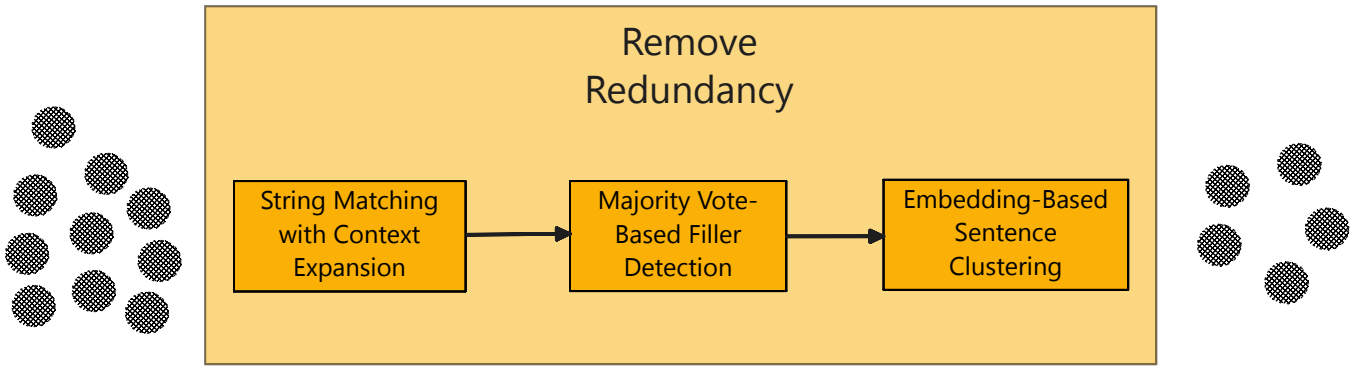


Figure 3.2.: **Overview of Work Package 1: Redundancy Removal & Preprocessing.** This pipeline step includes three sub-steps. The first involves adding surrounding client context to duplicate processes in order to identify and remove redundancies. The second applies an ensemble learning approach, where a majority vote determines whether a process is a filler utterance or a meaningful one. The third step uses only the non-filler utterances and groups them semantically using language models and clustering algorithms. The resulting preprocessed data is then passed to the next stage of the pipeline.

String Matching with Context Expansion This step identifies duplicate psychological processes within the annotated therapy session transcripts. When it detects duplicates, it adds the surrounding context to enrich the captured information. This contextual expansion ensures that subsequent processing steps operate on more meaningful and relevant content.

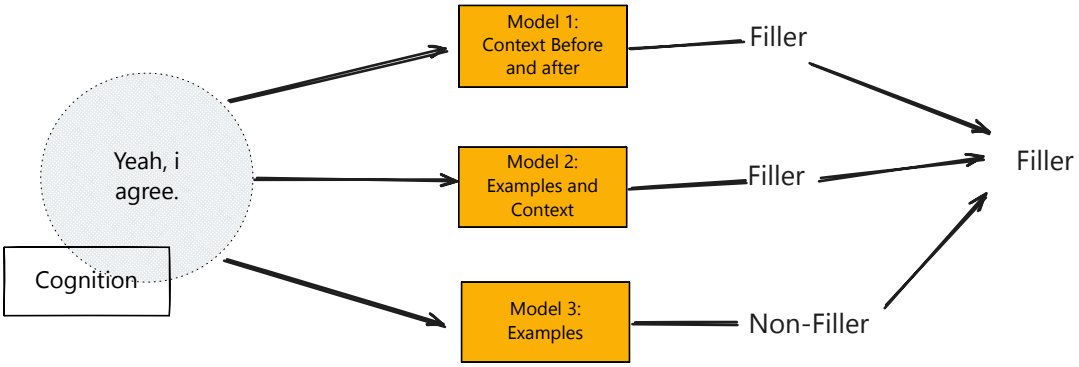


Figure 3.3.: **Ensemble Learning Majority Vote Approach:** We apply the same model three times, each time using a different prompt to perform binary classification of utterances previously identified by clinical psychologists as psychological processes. The majority vote classifies each utterance as either a filler or a non-filler based on a majority vote across the three model outputs. The prompts vary in the type of information they provide: the first includes the immediate context before and after the utterance from the patient; the second includes both context and examples of filler and non-filler utterances; and the third provides only examples without context.

Majority Vote-Based Filler Detection We apply an ensemble learning strategy, specifically a Max Voting ensemble approach [21], to perform binary classification on each psychological process using a large language model, LLaMA 3.1 70B Instruct [37]. The model labels each process as either a filler or a non-filler. We define a filler sentence as an utterance that lacks inherent meaning and does not contribute substantive value to the therapeutic conversation *(e.g., "Yeah, I would say," "It's like, oh, well," "Okay, nice")*.

As illustrated in Figure 3.3, the ensemble comprises three models prompted differently. Each model receives the process to be classified, along with varying supplementary information:

- We provide the **first model** with the immediate conversational context, approximately six utterances from the client both before and after the target sentence (see prompt in Appendix A.2)
- The **second model** uses a few-shot prompting approach, incorporating illustrative examples of filler and non-filler content (The examples are found in Appendix A.1 and the Prompt A.3)
- The **third model** classifies the utterance using only illustrative examples, without any additional conversational context (Prompt found in Appendix A.4).

At least two models must agree to determine the final classification through a majority vote. When the models produce ambiguous outputs and fail to reach a clear majority, the system classifies the process as non-filler. This fallback strategy preserves potentially meaningful content for later analysis.

Embedding-Based Sentence Clustering The final step of this work package involves using an embedding model to generate a vector representation for each psychological process. We then use clustering algorithms on the embeddings to cluster them semantically. In our case, we employ the extraction model *paraphrase-mpnet-base-v2* from SentenceBERT [30] and apply *Agglomerative Clustering* [24] with a distance threshold of 2.6. We determined the threshold empirically based on optimal performance observed using ground truth data. We provide a more detailed explanation of the model selection and clustering algorithm in the Experiments Section 4.

This process yields a reduced set of distinct psychological processes while preserving the core informational content. Each semantic cluster is assigned a frequency count, indicating the number of individual psychological processes it represents. This enables the use of a single representative per cluster while still reflecting the underlying distribution of the data. The pre-processed data then proceeds to the next stage of the pipeline: the generation of clinically meaningful themes and the classification of psychological processes under these themes.

3.3. Work Package 2: Generating Clinically Meaningful Process Clusters

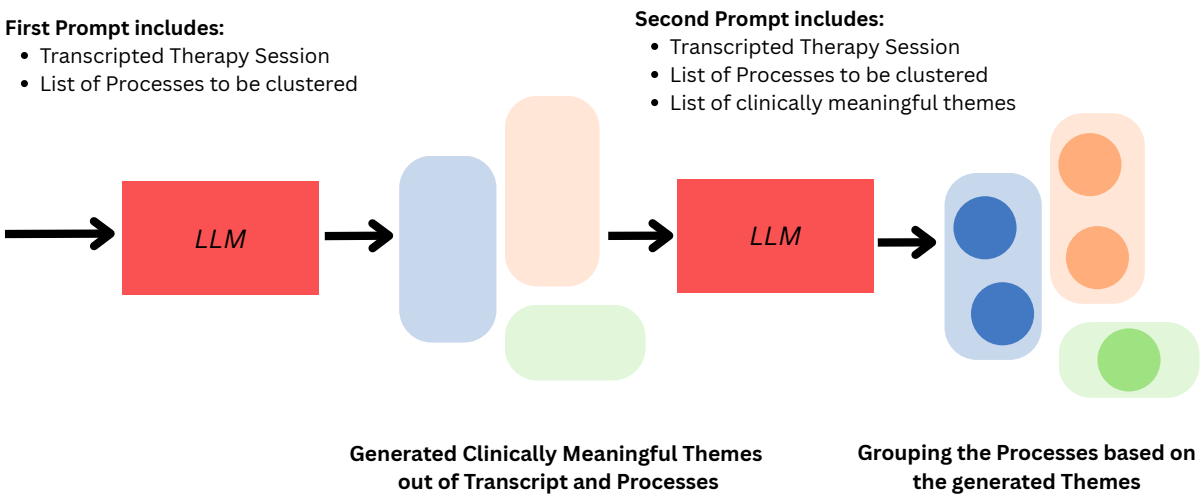


Figure 3.4.: **Stepwise Approach for Generating Clinically Meaningful Themes and Classifying Psychological Processes:** This prompting strategy separates the text generation and classification procedures [16]. In the first step, clinically meaningful themes are generated based on a provided therapy transcript and a list of psychological processes. In the second step, we use the generated themes to classify the psychological processes under the most appropriate theme.

In this work package, we aim to leverage an LLM to group psychological processes into **clinically meaningful themes**. Unlike purely semantic clustering, this approach requires the LLM to demonstrate a deeper understanding of behavioral patterns that are relevant to clinical psychology. The generated themes must reflect psychological insights and patterns. At the same time, it serves as the basis for assigning each psychological process to a corresponding theme based on its conceptual and contextual alignment.

The LLMs enable a more nuanced and clinically informed categorization of processes, transcending surface-level semantic similarity. As input, we use the pre-processed psychological processes obtained from the previous work package 3.2. The output consists of clinically meaningful themes derived from the current therapy session, with each psychological process grouped under its respective theme.

We conduct the procedure in two stages, following a method similar to that proposed by Huang and He [16], as illustrated in Figure 3.4. We adapted the original prompts in each step to better align with our objective of generating clinically meaningful themes and classifying the provided psychological processes accordingly.

In the first stage, we prompt the model to generate themes by providing a set of psychological processes and the transcribed therapy session. The shortened version of this prompt appears in Figure 3.5, while the complete version is in Appendix A.9.

In the second stage, we categorize the psychological processes under the corresponding themes generated. The Figure 3.6 illustrates the prompt, with the full version available in Appendix A.10.

Prompt for Generating Clinically Meaningful Themes

You are a clinical psychologist analyzing a therapy session transcript.

Task: Generate clinically meaningful themes in a clear, one-sentence description.

Definitions: `{placeholder}`

Input:

- A transcript excerpt from a therapy session (for context only).
- List of psychological processes.

Your Goal:

1. Read the transcript excerpt to understand the emotional and interpersonal context. Do not quote or extract specific utterances.
2. Generate clinically meaningful themes based on the listed psychological processes.

Guidelines for Writing Themes: `{placeholder}`

Examples `{placeholder}`

Transcript: `{prompt_str}`

Listed Processes: `{formatted_data_process_data}`

Now, generate clinically meaningful themes without additional explanations and no modifications to the provided data. Stick to the Guidelines.

Figure 3.5.: **Prompt for Generating Clinically Meaningful Themes:** This prompt generates clinically meaningful themes based on a provided therapy transcript and a list of psychological processes. We supply the model with expert-written definitions, the transcribed therapy session, and the corresponding list of psychological processes. Additionally, we include qualitative examples of both valuable and non-valuable themes to demonstrate how themes should be formulated and how they should not. The output is a list of clinically meaningful themes. The full version of this prompt is in the Appendix A.9

Prompt for Clinical Process Classification

Task: You are a clinical psychologist classifying psychological processes based on provided themes.
Definitions: `{placeholder}`
Input: `{placeholder}`
Your Goal:

1. Review the therapy session transcript for context.
2. Classify each listed psychological process under one or more of the provided themes based on their thematic relevance.
3. Do not assign a process to multiple themes unless it is highly relevant for that theme.
4. Ensure classification accurately reflects clinical relevance.

Guidelines for Clustering: `{placeholder}`
Here is the Output Structure: `{placeholder}`
Therapy Session Transcript (Context Only!): `{prompt_str}`
List of Processes with Frequency (To Be Classified): `{formatted_data_process_data}`
Provided Themes: `{generated_labels}`
Now, classify all of the given processes into the provided themes. Do not generate any explanation or modification to the provided data. Stick to the Guidelines.

Figure 3.6.: **Prompt for Classification of Psychological Processes Under Themes:** This Figure illustrates the prompt used to classify psychological processes under the generated clinically meaningful themes. It includes definitions of what the themes represent and what psychological processes are, along with clear instructions outlining the criteria for classification. The input to the prompt includes the transcribed therapy session, a list of psychological processes to be clustered, and the generated themes from the text generation component of the pipeline. The full version of this prompt can be found in the Appendix A.10

As shown in both prompts Figure 3.5 and 3.6, we provide the LLM with the complete transcribed therapy session, along with a list of psychological processes and their corresponding frequencies. The rationale for including the full transcript is that, like psychologists, the model requires access to a broader conversational context in order to make informed and clinically relevant judgments. Just as a clinician gains insight by being present during a session, supplying the full transcript enables the LLM to approximate this contextual awareness.

With input from clinical psychologists, we also supplied clear definitions of what constitutes a psychological process and what qualifies as a clinically meaningful theme. Additionally, we employed a few-shot prompting strategy, incorporating examples of both valuable and less valuable themes to guide the model toward generating precise, relevant, and concise thematic outputs [45]. To preserve the integrity of the input data, we include explicit instructions prohibiting the model from introducing new psychological processes or altering existing content. Based on further consultation with the psychologist, we established additional constraints. For example, each theme must contain at least two psychological processes, and the model should avoid generating a single, overly broad theme that encompasses all processes. We also include specific guidelines on how themes should be written, emphasizing that they should go beyond simple rephrasing of psychological

processes and must not include placeholder titles such as "*Theme 1*" or similar generic labels in the prompt.

We deliberately excluded information about the psychological dimension assigned to each psychological process from the prompts used throughout our pipeline. Although this information was available in the provided dataset and retained internally during processing, we observed that when we provided the model with the dimension labels, it tended to group the psychological processes primarily based on these predefined categories. This behavior resulted in overly rigid or semantically narrow clusters, which did not align with the clinically meaningful themes identified by the clinical psychologists. We aimed to generate those grounded in reflecting psychological functioning. As a result, to encourage the model to form more nuanced and conceptually coherent themes, we decided to withhold the dimension labels from all generation and classification prompts. This decision enabled the model to reason more freely and rely on the semantic and contextual information within the therapy transcripts, resulting in themes that better align with human clinical intuition.

Evaluation Framework The thesis introduces a novel approach for generating patient networks from annotated therapy session transcripts, supporting psychologists in conducting PBT. Since no prior research exists in this specific area, it was necessary to develop a custom evaluation framework from scratch. In close collaboration with clinical psychologists, we developed a set of evaluation metrics specifically designed to assess the quality and clinical relevance of the generated themes and classification results. These self-developed metrics enable a structured assessment of whether the outputs meet key criteria such as clinical usefulness, relevance, specificity, and coverage. The resulting evaluation guideline, along with the scoring spreadsheet, is provided in the Appendix D.1.

In collaboration with clinical psychologists, we organized the evaluation framework into three levels: **Network level, Cluster Level**, and **General Level**, each containing distinct metrics. Evaluators assign scores from 1 to 3, where 1 represents poor quality, and 3 indicates an ideal outcome. To ensure clarity and consistency in scoring, the guidelines include illustrative examples corresponding to each score level.

At the **Network Level**, the focus is on assessing the overall structure and coherence of the whole network, encompassing all clusters. Definitions and Scores are illustrated in Table 3.1. The network level includes Metrics like "Specificity," where we ask the psychologist whether the generated clusters are overall specific enough. We also ask the psychologist, after reading through the therapy transcript, whether the generated clusters provide any new or unexpected insights via "Novelty." Coverage helps determine if any important themes are missing from the network, while Redundancy assesses whether the generated themes significantly overlap with one another. We automatically compute "Completeness" by checking whether each psychological process has been assigned to a cluster and identifying any that have been left out. Another central metric is Clinical Relevance, where psychologists judge whether each theme is meaningful and accurately reflects the patient's psychological functioning. Finally, we provide psychologists with the option to add new themes if they believe any topic was overlooked, enabling them to expand or refine the generated network as needed.

The **Cluster Level** focuses on the detailed evaluation of each cluster. Definitions and scoring criteria are provided in Table 3.2. At this level, clinical psychologists assess how well the psychological processes assigned to a theme align with the theme itself. A key metric here is Intruder, which helps identify whether any assigned process seems out of place or inconsistent with the theme.

The **General Level** captures an overall assessment of the output, considering both the quality of individual clusters and the coherence of the network as a whole. The general assessment is measured using the Usefulness metric, which reflects how clinically helpful the entire structure is from the psychologist's perspective. Definitions and scoring for this metric are outlined in Table 3.3.

Metric	Scores	Definition
Specificity	1: Too Broad 2: Moderately Specific 3: Highly Specific	Is the network too broad or sufficiently specific to be clinically useful?
Novelty	1: None 2: Moderately Insightful 3: Highly Novel	Does the network reveal any new, unexpected, or useful insights?
Coverage	1: Incomplete 2: Partial Coverage 3: Full Coverage	Is the network missing any clinically relevant themes?
Completeness	Calculated Automatically	Are all processes listed in the network? (Automatically computed.)
Redundancy	1: Highly Redundant 2: Moderately Redundant 3: Minimal Redundancy	Are there redundant clinically relevant themes in the network?
Clinical Relevance	1: Clinically Insignificant 2: Partially Meaningful 3: Clinically Significant	Is the identified clinical theme meaningful?

Table 3.1.: **Evaluation Metrics for Network-Level Assessment:** Custom-designed metrics used to evaluate the generated clinically meaningful themes written in collaboration with domain experts. Each metric includes a definition in the form of a guiding question, along with a description of the associated score values.

Metric	Scores	Definition
Intruder	1: Multiple Intruders 2: Few Intruders 3: All Relevant	Is there an intruder node in the cluster?

Table 3.2.: **Evaluation Metrics for Cluster-Level Assessment:** This Table presents the custom-designed metrics used to evaluate theme clusters written in collaboration with domain experts. Each metric includes a definition framed as a guiding question, along with an explanation of what the scores represent.

Following the formal evaluation, we conducted a debriefing discussion with the clinical psychologists to determine which metrics they considered most important. Based on their input, we developed a **ranking** and used it to calculate a weighted **Total Score**, enabling a more nuanced overall assessment. The metrics were ranked in the following order of importance (from most to least): *Clinical Relevance, Novelty, Usefulness, Specificity and Coverage (tied in importance), Completeness, Intruder,* and *Redundancy.*

In collaboration with the domain experts, we also organized the evaluation metrics into two overarching categories:

- **Insightfulness:** Comprising *Clinical relevance, Novelty* and *Usefulness*. This category reflects how well the generated themes support psychologists in gaining deeper insights into the patient's psychological

Metric	Scores	Definition
Usefulness	1: Limited Insight 2: Some Insight 3: Valuable Insight	Can this network provide insight or support clinical decision-making?

Table 3.3.: **Evaluation Metrics for General-Level Assessment:** This Table presents the custom-designed metrics used to evaluate the overall quality of the generated themes and clustering procedure written in collaboration with domain experts. Each metric includes a guiding question and an explanation of what the scores represent.

dynamics during therapy.

- **Trustworthiness:** Including *Specificity, Coverage, Completeness, Intruder,* and *Redundancy*. This category captures how faithfully the network represents the provided data and how structurally reliable it is for clinical interpretation.

This categorization enables a more straightforward interpretation of both the *interpretative value* and the *structural validity* of generated patient networks. The Formula used for calculating these scores can be found in Appendix C.1.

We also tracked inter-rater agreements between the clinical psychologists who evaluated the generated clinically meaningful themes and their classifications. For each evaluation metric, we calculated agreement values except for Completeness, which is automatically computed and, therefore, not subject to subjective assessment. As shown in Table 3.4, we introduced four custom agreement metrics. *Label-Based Agreement*, *Hard Agreement*, *Selection Agreement*, and *F1 Agreement*. *Label-based Agreement* compares the rounded mean scores assigned by two evaluators for a given metric. *Hard Agreement* compares the exact (non-rounded) mean scores between evaluators. *Selection Agreement* applies specifically to metrics that allow for multiple score selections, such as Clinical Relevance and Intruder. *F1 Agreement* is calculated by treating one evaluator's labels as the ground truth and the other as predicted labels.

It determines whether both evaluators assigned the same score to the same value. For single-score metrics, the Selection Agreement behaves similarly to the Label-Based and Hard Agreement. We provide an illustrative example of how we calculate each of these self-made agreement measures in Appendix C.2.

In addition to our custom agreement metrics, we also employed widely used statistical measures to assess inter-rater reliability: Cohen's Kappa [10], and Spearman's Rank Correlation [19]. Cohen's Kappa evaluates the extent to which two raters agree in their categorical ratings beyond what would be expected by chance. A Kappa value of 1 indicates perfect agreement, zero indicates agreement equivalent to chance, and values below zero suggest agreement worse than chance. Spearman's Rank Correlation, on the other hand, assesses the degree to which two raters rank items in a similar order. This metric is handy when raters may use different absolute scores but still agree on the relative quality of items. A correlation of 1 indicates identical ranking, 0 indicates no correlation, and –1 indicates opposite rankings.

Agreement Type	What It Measures	How It Works	Value Range
Label-based Agreement	Whether two evaluators are *approximately aligned* in judgment by comparing the **rounded mean scores**.	For each session and metric, round both evaluators' means; if equal, count as agreement.	0 or 1 (then averaged)
Hard Agreement	Whether two evaluators gave **the same score** (mean or scalar).	For each session and metric, compare means or scalar values directly for equality.	0 or 1 (then averaged)
Selection Agreement	The **degree of item-level agreement** in scoring or labeling across a list of values.	Compares individual item scores between evaluators and calculates the proportion of exact matches.	0.0 – 1.0
F1 Agreement	Measures the **harmonic mean of precision and recall** between two evaluators' selections.	One evaluator is treated as *ground truth*, the other as *predictions*.	0.0 – 1.0
Cohen's Kappa [10]	Measures **inter-rater agreement on categorical data**, adjusting for chance agreement.	Computes observed vs. expected agreement based on label distributions.	-1 to 1
Spearman Rank Correlation [36]	Measures **correlation of rank ordering** between two sets of scores, regardless of absolute values.	Converts scores to ranks and computes the correlation between rank sequences.	-1 to 1

Table 3.4.: **Overview of Agreement Metrics:** This Table summarizes both standard and custom agreement metrics used in the evaluation process. It details what each metric measures, how it is calculated, and the corresponding value ranges. Examples illustrating the calculation of the self-made agreement metrics can be found in Appendix C.2.

3.4. Work Package 3: Generating Clinically Meaningful Connections

In the last work package, we utilize the clinically meaningful themes generated from the designated therapy session to create expressive relationships that connect the themes discussed during the session.

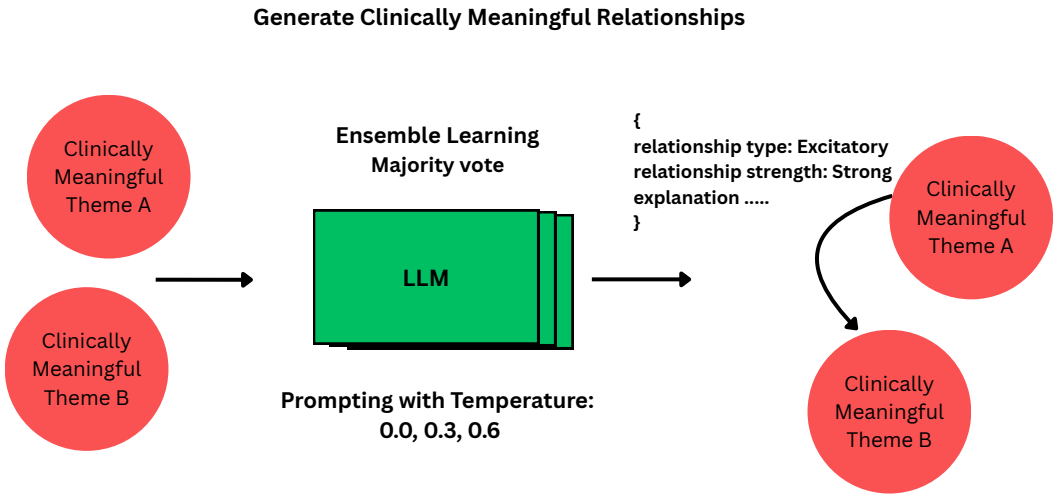


Figure 3.7.: **Pipeline for Generating Clinically Meaningful Connections:** The LLM is provided with a pair of themes and generates a directional connection from Theme A to Theme B (i.e., not bidirectional). We repeated the process three times using different temperature settings (0.0, 0.3, and 0.6). If any of the three outputs indicate a strong connection, we classify it as such and select it. A majority vote determines the relationship type, while the explanation is selected from among the three generated outputs.

These relationships include the following attributes:

- **Relationship Type:**
  - **Excitatory:** One Theme amplifies or reinforces another Theme (e.g., Theme A leads to Theme B, Theme A causes Theme B).
  - **Inhibitory:** One Theme suppresses or counteracts another Theme (e.g., Theme A stops Theme B, Theme A diminishes Theme B).
- **Strength of the Relationship:**
  - **Strong:** The two themes are closely and consistently related across contexts. One strongly influences or is inherently linked to the other in psychological functioning.
  - **Moderate:** The two themes are meaningfully related, but the connection is less consistent or conditional. They are associated in many situations but not tightly bound.
  - **Weak:** The relationship between the two themes is minimal, indirect, or highly context-dependent. They may co-occur at times, but the link is loose or peripheral.

- **Explanation:** A concise, natural language explanation of why this relationship exists.

An example of an **excitatory connection** can be seen in Figure 1.1. There is a strong connection from the theme *Self-Criticism* to the theme *Catastrophizing Relationships*. The relationship is considered a strong excitatory link because self-criticism tends to amplify negative thoughts and expectations in relational contexts, thereby contributing to catastrophizing. Additionally, a moderate exhortatory connection exists in the opposite direction, reflecting how catastrophizing about relationships can lead to increased self-blame and reinforce self-critical thinking.

To determine such connections, we evaluate every pair of clusters within a therapy session and prompt the model to assess whether a clinically meaningful link exists. If so, the model also provides the corresponding connection attributes.

The generation of clinically meaningful relationships follows an ensemble learning approach, similar to the method applied in Work Package 1 (see Section 3.2) [21]. Specifically, we employ the *LLaMA 3.1 70B Instruct* model [37], querying it under three different temperature settings (0.0, 0.3, and 0.6). For each pair of clusters, we prompt the model to determine whether a clinically meaningful connection exists and, if so, to specify the attributes of that connection. In consultation with clinical psychologists, we determined that if any of the three model outputs indicate a strong connection, the final relationship should be marked as strong. The connection type (e.g., excitatory or inhibitory) is decided via a majority vote across the three outputs. For the *Explanation* attribute, one of the model-generated explanations is randomly selected from the available outputs.

This procedure for generating clinically meaningful relationships is illustrated in Figure 3.7. As shown, the model receives a pair of clinically meaningful themes and, through the ensemble voting process, generates expressive and clinically grounded connections between them. These connections represents underlying psychological dynamics and enhance the interpretability of the resulting patient network.

The prompt used to generate expressive, clinically meaningful relationships between theme pairs is shown in Figure 3.8. This prompt incorporates definitions provided by clinical psychologists, including precise descriptions of what constitutes an excitatory or inhibitory connection and how the strength of such connection should be classified. Additionally, we provide the model with clear guidelines: it must always indicate whether a connection exists and, if so, provide a brief explanation of the connection. To avoid redundancy, the Explanation should not include the names of the themes. The full version of this prompt is available in Appendix A.13.

Prompt for Generating Clinically Meaningful Connections

You are a psychologist and your task is to identify relationships between psychological processes.
Task: Your task is to determine whether a relationship exists between **Process A** and **Process B**, and to generate information on the nature of that relationship.
Definitions: `{placeholder}`
Input: Two psychological processes
Guidelines:

- Given process A and process B, determine whether or not a relationship exists between two processes.
- Provide a **brief explanation (a few words)** of the connection between Process A and Process B. Avoid restating the processes themselves

Output Structure: If relationships exist from process A to process B, return a structured output containing details for each identified relationship:

```
json
{{"relationship":
[{{ "input_processes": ["Process A", "Process B"],
"connection": [1],
"relationship_type": ["excitatory", "inhibitory"],
"strength_of_relationship": ["strong", "moderate", "weak"],
"explanation":
"A concise natural language explanation of why this relationship exists."}}]
}}
```

If no relationships exist from Process A to Process B, return:

```
 json
{{"relationship":
[{{ "input_processes": ["Process A", "Process B"],
"connection": [0]}}]
}}
```

Process A: `{process_1_str}`
Process B: `{process_2_str}`
Now, generate the clinically meaningful relationship between the provided Processes in the specified format. Only output the relationship, and nothing else.

Figure 3.8.: **Prompt for Generating Clinically Meaningful Connections Between Themes (Short Version):** This Figure presents the prompt used to generate clinically meaningful relationships from Theme A to Theme B. The prompt includes definitions of what constitutes a psychological process, explanations of the relationship types, and the meaning of relationship strength. The full version of this prompt is in the Appendix A.13

4. Experiments

This chapter presents all the experiments conducted throughout the thesis. These include the Filler Detection task, Semantic Clustering, the generation of clinically meaningful themes, and the classification of psychological processes, all of which are evaluated by clinical experts. Additionally, we report the results of a pairwise comparison between the patient networks generated by our pipeline and those generated by a baseline approach, also assessed by clinical psychologists.

4.1. Dataset

The Department of Psychology at the University of Toledo [38] owns the dataset used in this thesis. It includes transcripts from 6 patients. Four of these patients completed 15-16 therapy sessions, while the remaining two completed 10 or fewer sessions, as shown in Table 4.1.

Patient ID	Session Count
P1	15
P2	15
P3	16
P4	15
P5	6
P6	10

Table 4.1.: **Session Counts per Patient:** This Table displays the patient IDs along with the number of therapy sessions available for each patient in the provided dataset.

Natural Language Analysis We calculated the total number of words spoken by both the therapist and the patient, as presented in Figure 4.1. For patients who completed 15 sessions, the word count remained relatively consistent, ranging from approximately 15.000 to nearly 17.500 words. The patient who attended only six sessions spoke approximately 7.500 words, while Patient P6, who completed 10 sessions, exceeded 9.000 words. In some sessions, such as those with Patients P1 and P3, the therapist spoke the same number of words as the patient. For Patient P4, the case is fascinating, as the therapist spoke more words than the patient.

We also computed the total number of utterances made by each patient and therapist, as shown in Figure 4.2. Overall, patients produced more utterances than therapists. For example, patients who completed around

15 sessions, such as P1 and P3, made approximately 1.500-1.600 utterances. Patients with fewer sessions, such as P5 and P6, had approximately 600 and 800 utterances, respectively. Interestingly, although Patient P2 spoke more words overall, the therapist had a higher number of utterances. This suggests that this patient tend to express themselves in longer, more complete statements, while therapists contribute shorter responses probably aimed at guiding the conversation or reflecting on the patient's input.

Figure 4.3 shows the total speaking time for each patient. Those who completed 15 sessions typically spoke for a similar duration, around 120 minutes, and often spoke longer than the therapist. When comparing the total word count is particularly interesting. For instance, Patient P1 used fewer words than Patient P2 but had a similar speaking duration. This pattern suggests that some patients may require more time to form their thoughts during emotionally charged discussions. The contrast becomes even clearer when comparing the number of words spoken by therapists to the time they spent speaking.

For a more detailed breakdown of the number of words spoken B.1, utterances B.3, and corresponding speaking time per patient and therapist per session B.2, refer to the Appendix B.

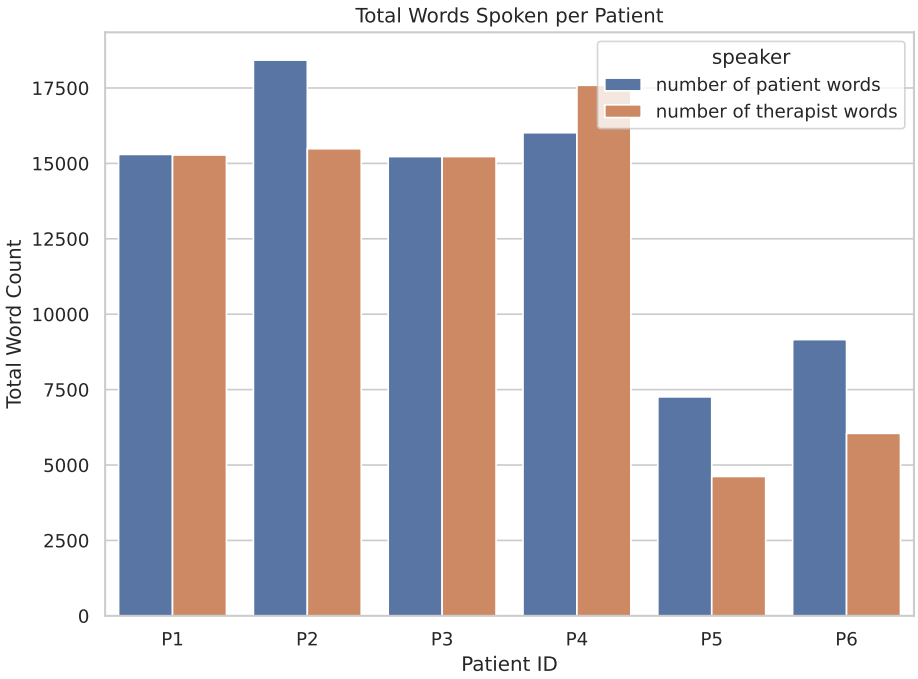


Figure 4.1.: **Total Word Count for Patients and Therapists:** This Figure displays the total number of words spoken by each patient and therapist across all therapy sessions.

Psychological Process & Dimension Three psychologists annotated the dataset by identifying psychological processes within each therapy session transcript and assigning a corresponding psychological dimension to each identified process. Inter-annotator agreement on whether an utterance represents a psychological process is approximately 79%, while agreement on the assigned dimensions is around 71%

Figure 4.5 displays the number of psychological processes annotated in each session for every patient. The number of processes varies across sessions, with some sessions containing significantly more than others. Interestingly, for certain patients, such as P1 and P3, the first session includes the highest number

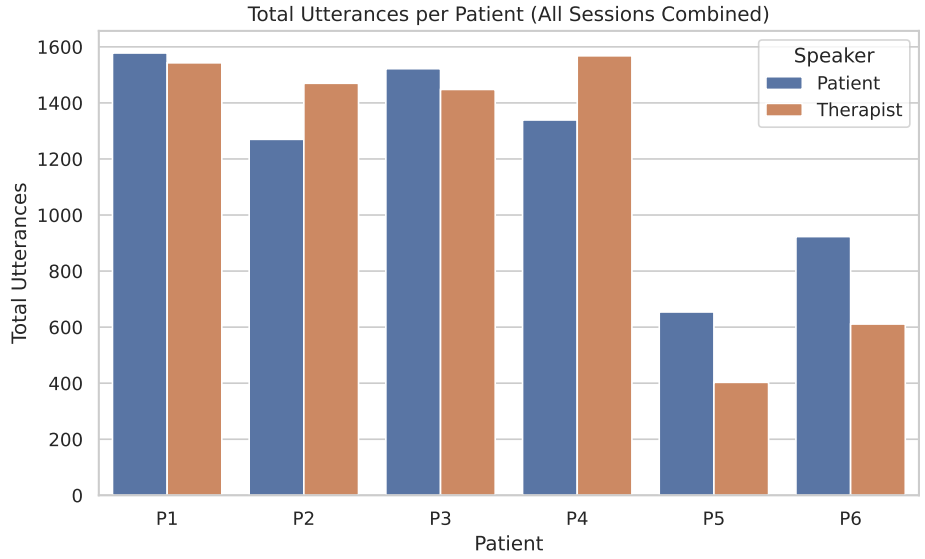


Figure 4.2.: **Total Utterance Count for Patients and Therapist:** This Figure displays the total number of utterances spoken by each patient and therapist across all therapy sessions.

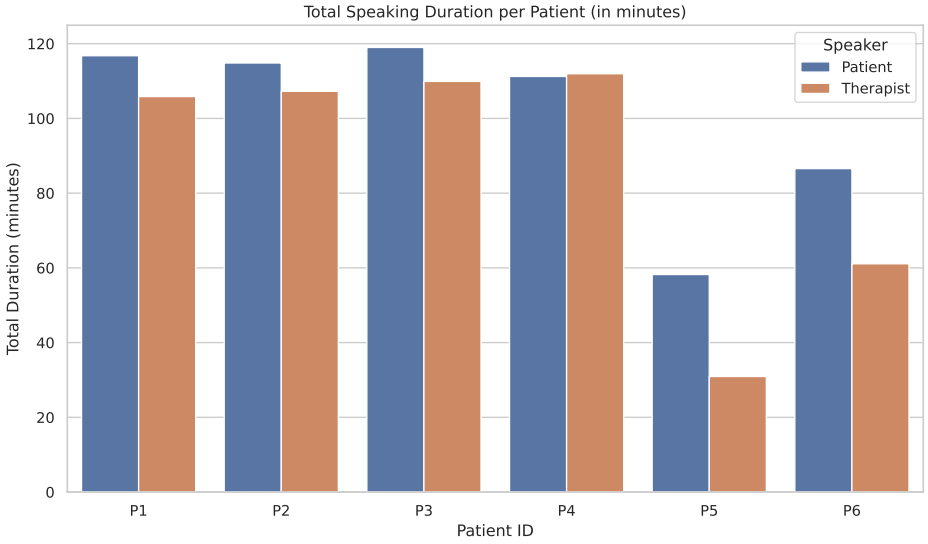


Figure 4.3.: **Speaking Duration per Patient and Therapist in Minutes:** This Figure displays the total speaking duration (in minutes) of utterances spoken by each patient and therapist across all therapy sessions.

of psychological processes across the entire therapy timeline. Additionally, sessions toward the end also exhibit high process counts. This pattern may indicate that the beginning and closing phases of therapy are particularly significant for understanding psychological functioning, potentially reflecting early engagement and later breakthroughs in the therapeutic process. In contrast, some sessions contain fewer than 20 annotated psychological processes, suggesting variability in therapeutic depth or focus across sessions.

When we examine the overall number of psychological processes per patient in Figure 4.4, we observe an

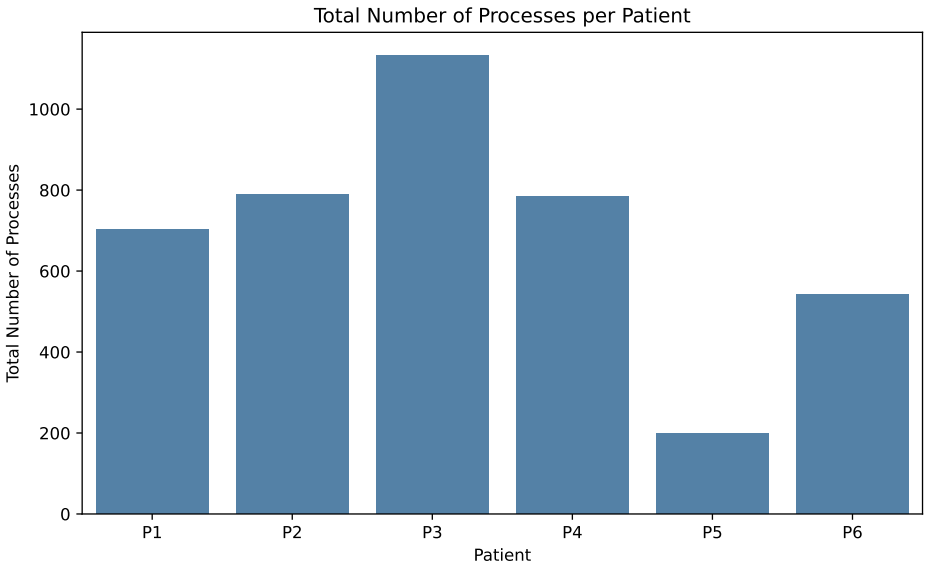


Figure 4.4.: **Total Count of Psychological Processes per Patient:** This Table presents the total number of psychological processes identified by clinical psychologists for each patient across all therapy sessions.

additional variation. Even among patients who completed the same number of sessions, the total number of psychological processes differs markedly. for instance, P1 and P2 both completed 15 sessions, yet P1 has approximately 700 annotated processes, while P2 has nearly 800, similar to P4. Meanwhile, P3, who completed 16 sessions, shows the highest total, with over 1.000 annotated psychological processes.

Comparing P1 and P3 highlights an interesting contrast: although P1 produced more utterances than P3, P3 had a substantially higher number of psychological processes. This suggests that P3's utterances were denser in clinically relevant content, indicating qualitative differences, such as therapeutic discourse, across patients.

When examining the dimensions assigned to each psychological process, it is important to note that a single process can be associated with multiple dimensions. For example, the dimension *Affect* captures how a patient felt in a given situation, while *Cognition* reflects how the patient interprets or assigns meaning to specific events. *Attention* describes how a patient focuses on or perceives experiences, and *Motivation* refers to the underlying goals or desires driving behavior. *Sociocultural* refers to the patient's social context, including relationships and cultural background. *Sense of Self* captures how the patient perceives their own identity. *(Overt) Behavior* refers to observable, repetitive actions taken by the patient. *Context/Moderator* reflects external circumstances that are difficult to change, and *Biophysiological* pertains to the patient's physical or bodily experiences. A complete overview and description of all nine predefined psychological dimensions can be found in Table 4.2.

We also tracked the number of psychological processes assigned to specific dimensions. Figure 4.6 illustrates the distribution of these dimensions across patients. The dimension Cognition appears most frequently, accounting for approximately 40% of all identified processes. This is followed by (Overt) Behavior, which comprises around 20% of the total. Other dimensions, such as Attention, Sociocultural, Contextual/Moderator, Affect, Sense of Self, Biophysiological, and Motivation, appear with varying frequency across patients. This variation reflects differences in therapeutic focus and individual psychological functioning. Additional details regarding the assignment of psychological dimensions to processes in each session per patient and also therapist are provided in Appendix B.5.

Dimension	Description	Example
Affect	How the patient feels about their situation.	*I always feel like it all depends on the day and how I'm feeling.*
Cognition	How the patient thinks and assigns meaning to events.	*Like, prioritize someone else's needs over and above my own.*
Attention	How the patient directs or shifts focus during experiences.	*I go from time to stuff all the time, the holding branches about something that happened years ago that I've apologized for a million times over.*
Motivation	The goals or aims the patient pursues.	*I want to make change for my life.*
Sociocultural	The social and cultural relationships and contexts the patient engages with.	*Everybody makes a comment about my weight.*
Sense of Self	How the patient perceives and conceptualizes themselves.	*I'm not perfect and not everything that I'm going to produce is always going to be perfect*
Overt Behavior	Observable or repetitive actions the patient takes.	*Well, so in those situations, sometimes I'll gravitate towards watching TV, and I'm not really sure that's true self-care.*
Context / Moderator	Situational factors that are static or difficult to change.	*Um, he actually had been going to one therapist for many, many years who helped him with his school stress.*
Biophysiological	Aspects related to sleep, diet, exercise, and chronic health conditions.	*It's just very hard for me to go to sleep.*

Table 4.2.: **The Nine Predefined Psychological Dimensions and Their Descriptions:** This Table lists the nine predefined psychological dimensions used to label psychological processes, along with a description of what each dimension represents and examples.

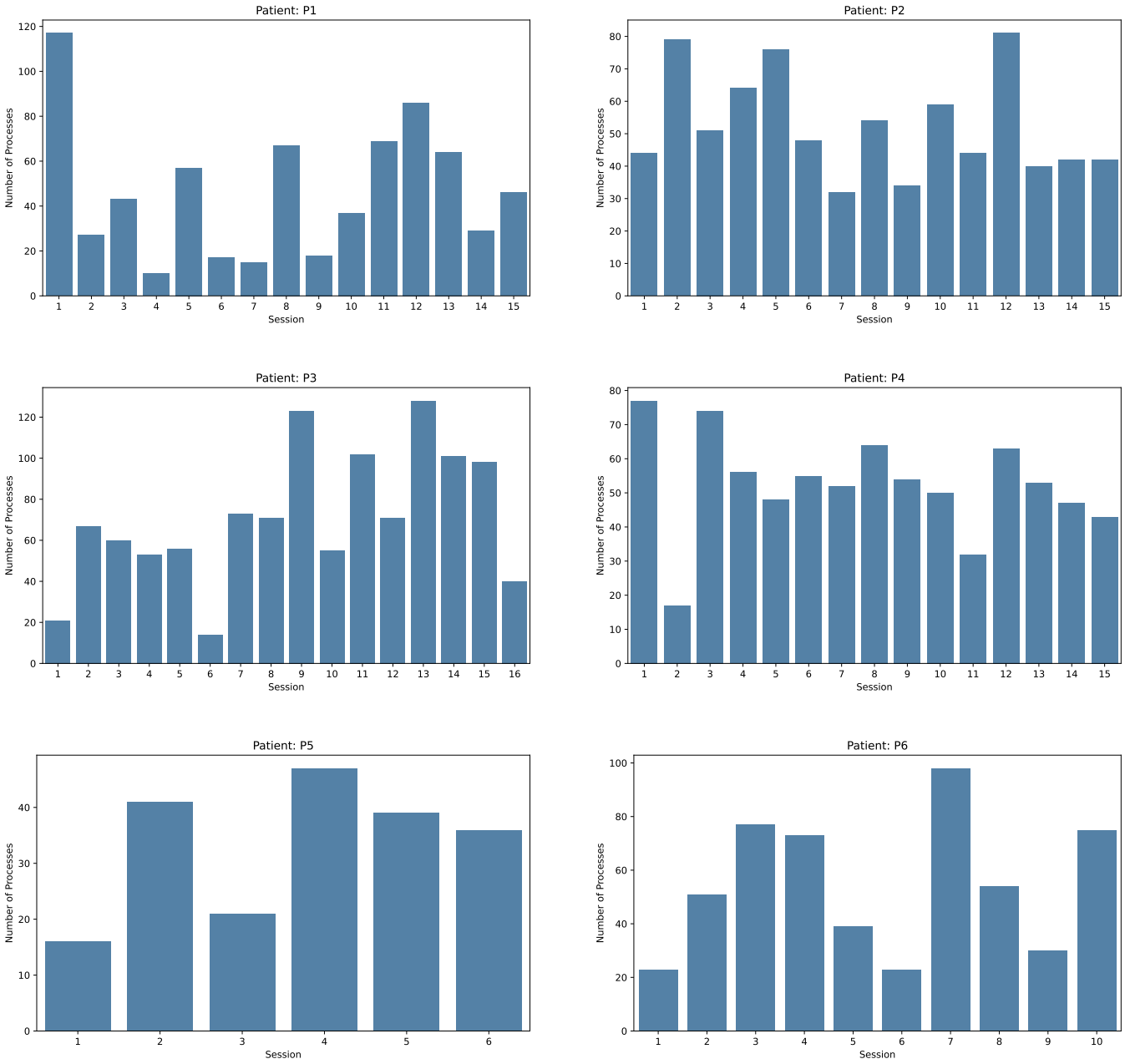


Figure 4.5.: **Number of Processes for Each Patient Across Sessions:** This Figure provides a detailed breakdown of how many psychological processes were identified for each patient in each therapy session.

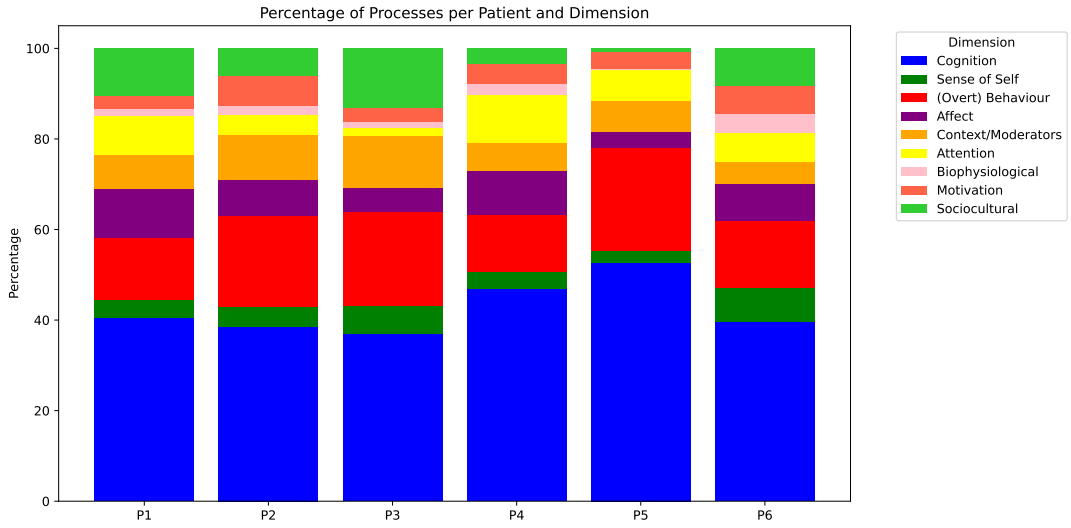


Figure 4.6.: **Total Patient Dimension Percentages:** This Figure shows the percentage distribution of psychological processes across all therapy sessions for each patient based on the predefined psychological dimensions. Each bar sums to 100%, representing how a patient's processes are distributed across the different dimensions. Note that a single psychological process may be associated with multiple dimensions.

4.2. Work Package 1: Redundancy Removal & Preprocessing

As mentioned in Section 3.2, the work package is divided into three steps. The first step is to see if a psychological process has a duplicate. Most often, a duplicate psychological process appears as follows: "*yes*" or "*yeah*". We add context around it to remove the duplicate, thereby providing more information about the specific psychological process.

4.2.1. Majority Vote-Based Filler Detection

In the next step, **Majority Vote-Base Filler Detection**, we apply ensemble learning to determine whether a psychological process constitutes a filler sentence. We define a filler sentence as a sentence that has no inherent meaning. It, therefore, adds no value to the conversation. We introduced this step based on feedback from clinical psychologists, who noted that some of the annotated psychological processes when viewed in isolation, do not convey meaningful information. Since the annotations were made at the utterance level, the process is often tied to the broader conversational context. In such cases, the utterance itself may not inherently reflect the psychological process. It is the surrounding context that provides its clinical significance.

It means that specific processes, on their own, do not contribute valuable information to the construction of the patient network. The purpose of this procedure is to identify and filter out such processes, particularly those that are context-dependent and not self-contained. By doing so, we aim to retain only processes that either convey meaning independently or become meaningful when contextual information is taken into account.

Setup We conducted a human annotation task to create a ground truth for evaluating the performance of our majority vote approach in identifying filler and non-filler psychological processes. We randomly selected 200 processes from various patients and sessions. Each annotator labeled 150 processes, with an overlap of 50 processes used to assess inter-annotator agreement. This overlap was essential for determining the consistency of human judgments regarding what constitutes a filler versus a non-filler process. The inter-annotator agreement was approximately 98%, indicating a high level of consistency.

Results & Discussion After applying majority voting to the human-annotated ground truth, we present sample qualitative results in Table 4.3. To quantitatively evaluate the model's performance, we computed a confusion matrix, shown in Figure 4.7. According to the matrix, the ensemble voting approach classified 16 processes as *Filler* while misclassifying four as *Not filler*. Additionally, 36 *Filler* processes were incorrectly labeled as *Not filler*, and 144 *Not filler* processes were correctly identified. These results yield an overall classification accuracy of approximately 80%, with an F1 score of 66%, precision of 64%, and recall of 80%.

However, the results reveal a performance gap between the two classes. For the *Filler* class, the model achieved a precision of 30.8%, a recall of 80%, and an F1 Score of 44.44%. In contrast, the *Not filler* class achieved a precision of 97.3%, a recall of 80.0%, and an F1 Score of 87.8%.

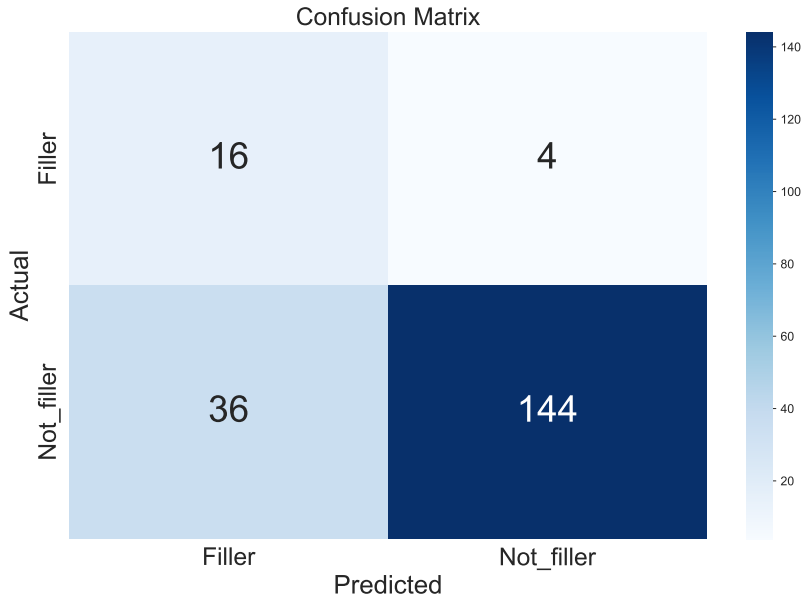


Figure 4.7.: **Confusion Matrix for Filler Detection:** This Figure shows the results of the binary classification task for detecting filler processes, presented as a confusion matrix based on the human-annotated ground truth.

These scores are lower than expected for detecting filler utterances. A likely reason is the class imbalance in the ground truth used for evaluation. Among the 200 annotated processes, *Not filler* instances significantly outnumber *Filler* ones. Therefore, the imbalance impacts the evaluation metrics, such as precision and recall. That said, the imbalance may accurately reflect the real distribution of utterance types in therapy sessions, where fewer statements act as fillers. Another factor affecting performance is the fallback rule used when the ensemble did not reach a clear decision. In such cases, we defaulted to classifying the process as *Not filler*,

which likely contributed to the higher precision and F1 Score for that class. In our case, the fallback rule occurred 5 times.

Since our primary goal is to preserve as much meaningful information as possible, misclassifying *Filler* processes as *Not filler* is less problematic than misclassifying *Not filler* processes as *Filler*. Discarding a *Not filler* process would risk losing valuable clinical content. Therefore, despite the model's moderate performance on the *Filler* class, its current performance is acceptable for our application.

Process	Model 1	Model 2	Model 3	Majority Vote
"*It was really just an instance of when my girlfriend said she had a really good time.*"	Not Filler	Not Filler	Not Filler	**Not Filler**
"*It's good enough.*"	Filler	Filler	Filler	**Filler**
"*Or inappropriate even.*"	Not Filler	Not Filler	Filler	**Not Filler**
"*And in fact, if I think, well, why, why, why?*"	Filler	Not Filler	Filler	**Filler**

Table 4.3.: **Examples of Model Classifications and Majority Vote Outcomes:** This Table presents examples of psychological processes classified by three models, along with the final classification determined through majority voting.

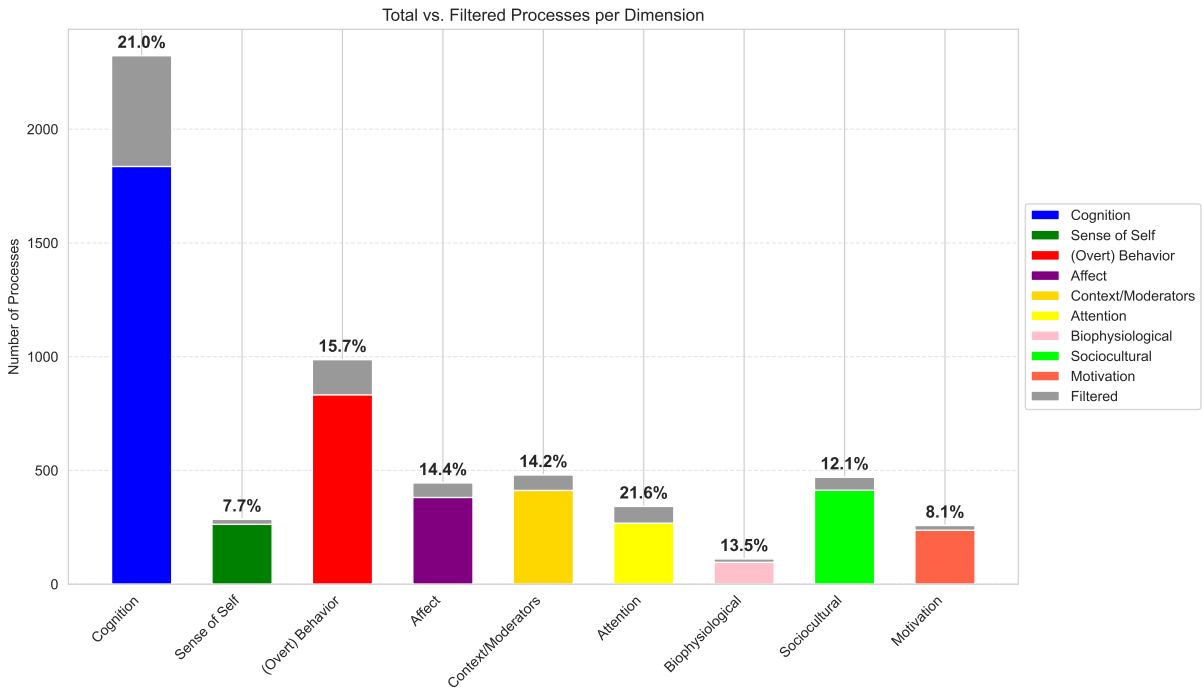


Figure 4.8.: **Filtered Processes per Dimension:** This Figure presents all psychological processes included in the dataset and illustrates the proportion of filtered processes (with percentage) within each predefined psychological dimension.

After evaluating the performance of our filler detection approach, we applied it to the complete set of annotated therapy sessions to identify and remove non-standalone psychological processes. We then analyzed the number and characteristics of the utterances that were filtered out.

Out of a total of 4.153 utterances labeled as psychological processes, the model identified 739 as filler, accounting for approximately 19% of the total. On average, each therapy session contained 53.95 psychological processes, of which 10.72 were removed. The removed utterances had an average token length of 10.13, suggesting they were typically brief and low in content.

Figure 4.3 presents examples of utterances classified as either filler or meaningful. For instance, the phrase "*it's good enough*" was filtered out due to its vague and acknowledgment-like nature, offering little substantive value to the therapeutic exchange. In contrast, the utterance "*It was really just and instance of when my girlfriend said she had a really good time.*" conveys specific information and contributes meaningfully to the session, justifying its inclusion.

A breakdown of the filtered utterances by psychological dimension is shown in Figure 4.8. The *Cognition* dimension had the highest total count with 2.323 processes, of which 20.96 % were filtered out. The *Attention* dimension followed, with a 21.64% removal rate from 342 total processes. Other notable filtering rates include *Overt Behavior* at 15.71%, *Affect* at 14.38%, *Context/Moderators* at 14.17%, and *Biophysiological* at 13.51%. Lower removal rates were observed for *Sociocultural* (12.13%), *Motivation* (8.14%), and Sense of Self (7.71%).

4.2.2. Embedding-Based Sentence Clustering

The final step of this work package involves the semantic clustering of *Not Filler* psychological processes using sentence embeddings and the clustering algorithm that performed best during evaluation.

Setup Before performing semantic clustering, we conducted a human annotation task to create a ground truth dataset with the help of LM. This dataset comprises 30 randomly selected therapy sessions from the entire corpus. Its primary purpose was to evaluate different combinations of sentence embedding models and clustering algorithms in order to identify the configuration that most effectively groups semantically similar psychological processes. This thereby reduces redundancy while preserving informational content.

In this experiment, we evaluated four sentence embedding models. The first, all-mpnet-base-v2, was identified as the best-performing model in Muennighoff et al. [23]. The other models distilbert-base-nli-stsb, paraphrase-mpnet-base-v2, and paraphrase-multilingual-mpnet-base-v2 were also evaluated, having been used in RSTC [46] and MIST [17] [30, 29]. To cluster the resulting embeddings, we tested several algorithms. The first was BERTopic [40], which combines sentence embeddings with class-based TF-IDF to form dense and interpretable topic clusters. We also applied Community Detection [31], which uses cosine similarity to group sentences into communities of highly similar members. Additionally, we evaluated DBSCAN [42], a density-based algorithm that forms clusters based on local density patterns. Lastly, we employed Agglomerative Clustering [24], a bottom-up hierarchical method where each sentence initially forms its cluster and is iteratively merged based on a distance threshold. For this method, we selected a threshold of 2.6, as it showed the highest agreement with human-annotated ground truth in terms of clustering quality. Definitions of each clustering method are provided in Table 4.5.

We also included two baseline methods for comparison: a *baseline with k*, which assumes a predefined number of clusters, and a *baseline without k*, which operates without any prior knowledge of the number of clusters. These baselines serve as reference points to assess whether semantic clustering significantly outperforms random or naive grouping strategies.

To evaluate each model-algorithm combination, we employed a variety of metrics. External evaluation metrics, which rely on human-annotated ground truth, include the Adjusted Rand Index (ARI) [4], which compares two clusterings by counting how many pairs of samples are assigned to the same or different clusters in the predicted and true clustering; Normalized Mutual Information (NMI) [46, 17, 20], which measures the similarity between two sets of clusters; as well as Precision, Recall, and the F1 Score, and Accuracy, which is computed using the Hungarian method to optimally align predicted clusters with the ground truth [46, 17]. Additionally, we used the Silhouette Score [32], an internal evaluation metric that does not require ground truth. It assesses how well each sample fits within its assigned cluster relative to other clusters, thereby providing a measure of both cohesion and separation. Table 4.4 summarizes the definitions, value ranges, and interpretive meanings of each metric used in the semantic clustering evaluation.

Metric	Definition	Range	Values
Adjusted Rand Index (ARI)	It evaluates the similarity between two clusterings by examining how well the predicted clustering corresponds to the actual labels.	-1 to 1	> 0: worse than random 0: random 1: perfect match with ground truth
Normalized Mutual Information (NMI)	quantifies the amount of information that is shared between the actual clustering (ground truth) and the predicted clusters.	0 to 1	0: no mutual information 1: perfect mutual information
Accuracy	Represents how well the clustering results align with the true class labels (Hungarian Method).	0 to 1	1: perfect label alignment (after optimal mapping) 0: no predicted cluster labels match the ground truth labels
F1-Score	Harmonic mean of precision and recall.	0 to 1	1: perfect precision and recall
Silhouette Score	Measures how well a clustering algorithm has separated data points into clusters. Compares how similar data points are within a cluster to how different they are between clusters.	-1 to 1	1: well-clustered 0: overlapping -1: misclassified

Table 4.4.: **Clustering Metrics: Definitions, Ranges, and Interpretation of Values.** This Table presents the clustering metrics used to evaluate the performance of each LM and clustering algorithm combination.

Results & Discussion Table 4.6 shows us the results for each model-algorithm combination. For each metric (i.e., column), the bold value indicates the best performance, while the underlined value represents the second-best performance within that same column.

Focusing on the Silhouette Score [32], we observe that none of the combinations achieved powerful results. This score suggests a moderate degree of overlap between clusters. Based solely on the Silhouette Score, the combination of *all-mpnet-base-v2* [30] with *Agglomerative Clustering DT 2.6* [24] appears to be the most suitable. The low scores observed across all configurations are likely influenced by the characteristics of the data, which consists of transcribed sentences from therapy sessions. This data tends to be noisy and often consists of relatively short sentences. These factors may hinder the reliability of the Silhouette Score [32]

Clustering Method	Definition
Agglomerative Clustering [24]	A hierarchical clustering technique that uses a bottom-up approach starts with each sentence as an individual entity. It iteratively merges sentences into nested clusters by combining the closest pairs of clusters based on a specified distance metric (Distance Threshold).
BERTopic [11]	A topic modeling technique that uses BERT embeddings and class-based TF-IDF to create dense clusters of semantically similar documents.
Community Detection [31]	A fast clustering technique based on cosine similarity of sentence embeddings. Sentences are grouped into communities where each member is highly similar to at least one other member in the community.
DBSCAN [42]	Density-Based Spatial Clustering is a clustering algorithm that groups together points that are close to each other within a specified radius in Euclidean space. It also identifies points that are far away as noise.

Table 4.5.: **Comparison of Clustering Methods:** This table presents the clustering algorithms evaluated in our study. None of these methods require a predefined number of clusters to operate.

in determining the most effective clustering setup. It is also important to note that internal metrics such as the Silhouette Score [32] rely on geometric properties of the embedding space, unlike external metrics that are evaluated against ground truth labels [1, 27]. In datasets with high dimensional or complex structures, internal metrics can often provide misleadingly low scores, even when meaningful clusters exist. In such cases, external metrics are generally more appropriate for assessing clustering quality [26].

Examining the external metric ARI, which assesses the alignment between the predicted clustering and the ground truth, we observe that both baselines, *Baseline w/ k* and *Baseline w/o k*, achieve scores close to zero. This is consistent with the interpretation of ARI, where a score near zero indicates clustering results that are effectively random. In this case, the best-performing combinations are *paraphrase-multilingual-mpnet-base-v2* and *paraphrase-mpnet-base-v2*, both paired with Agglomerative Clustering (DT 2.6), achieving ARI scores of 0.325 and 0.317, respectively. These values indicate a moderate level of alignment with the ground truth. In contrast, clustering algorithms like *BERTopic* appear to struggle with generating semantic clusters that align with the ground truth across all extraction model combinations, as indicated by ARI values that remain close to zero. A similar trend can be observed with the combination of *all-mpnet-base-v2*, *distilbert-base-nli-stsb* and *paraphrase-distilroberta-base-v1* with Agglomerative Clustering (DT 2.6), which yields an ARI score of around 0.07. This contrasts with the previously mentioned best-performing combinations for ARI, highlighting that the choice of extraction model in combination with the clustering algorithm plays a crucial role in performance. Moderate performance in ARI scores is also observed with DBSCAN, ranging from approximately 0.102 to 0.168. In comparison, Community Detection consistently performs better, with ARI scores ranging from 0.131 up to 0.269.

The following external metric is NMI, which we also used to evaluate clustering performance. According to this metric, the best-performing combination is *paraphrase-mpnet-base-v2* with Agglomerative Clustering (DT 2.6), closely followed by the multilingual version *paraphrase-multilingual-mpnet-base-v2* with Community Detection. These combinations achieved NMI scores of 0.951 and 0.950, respectively. However, we observe a similar trend of high NMI scores around 0.9 across various other model and algorithm combinations. For example, combinations such as *mpnet-base-v2* with Community Detection or DBSCAN and *distilbert-base-nli-stsb* with Community Detection also yield comparable results. Similarly, *paraphrase-distilroberta-base-v1* combined with

either Agglomerative Clustering (DT 2.6), Community Detection, or DBSCAN shows consistent performance. These findings are also in line with the results from *paraphrase-mpnet-base-v2* and its multilingual variant across these clustering algorithms. Interestingly, both baselines *baseline w/k* and *baseline w /o k* achieve relatively high NMI scores of around 0.834. Despite this, the top-performing model-algorithm combinations still outperform the baselines, demonstrating the effectiveness of semantic clustering. Notably, one clustering algorithm, *BERTopic*, consistently performs worse than the Baseline across all extraction model combinations. Additionally, the combination of *Agglomerative Clustering (DT 2.6)* with the extraction model *all-mpnet-base-v2* is among the worst-performing in this case. This contrasts sharply with the best-performing setups that use the same clustering algorithm but different extraction models, further emphasizing the crucial interaction between the extraction model and clustering algorithm for achieving strong semantic clustering performance.

Turning to one of the well-known evaluation metrics, the F1 score, which is the harmonic mean of Precision and Recall. We again observe that the combination of Agglomerative Clustering DT 2.6 with *paraphrase-mpnet-base-v2* and its multilingual variant achieves the highest scores, with values of 0.321 and 0.331, respectively. While the mentioned above combinations perform well, *BERTopic*, across its various configurations, consistently ranks among the worst-performing, with F1 scores nearing the Baseline around 0.05. Notably, the extraction model *all-mpnet-base-v2*, regardless of the clustering algorithm used, performs similarly to the Baseline. For example, when paired with Agglomerative Clustering DT 2.6, it achieves a relatively high recall but extremely low precision (e.g., precision as low as 0.025). This suggests that the model is successfully grouping most semantically related pairs (resulting in high recall) but at the cost of forming clusters that also include many unrelated or dissimilar pairs, thus severely lowering precision. The use of Community Detection also appears promising, consistently ranking as the second-best clustering algorithm after Agglomerative Clustering DT 2.6. In contrast, DBSCAN exhibits weak performance and appears less suitable for this task. In our case, higher precision is more important than capturing all semantic groupings, as incorrect semantic groupings can result in the loss of meaningful information within a therapy session. For this reason, we prioritize the combination of *paraphrase-mpnet-base-v2* with Agglomerative Clustering DT 2.6, which achieves the highest precision score of 0.753, along with a recall of 0.237.

The final metric we evaluated in this experiment is accuracy, calculated using the Hungarian method for optimal label matching. While the baseline accuracy is around 0.5, the best-performing combination is once again Agglomerative Clustering DT 2.6, paired with the extraction models *paraphrase-mpnet-base-v2* and its multilingual variant, both of which achieve an accuracy of 0.81. Community Detection and DBSCAN also demonstrate reasonable performance across various combinations, with accuracy scores ranging from 0.7 to 0.8, indicating consistent and reliable clustering behavior. However, this metric should be interpreted with caution. Accuracy relies on finding an optimal permutation between predicted and true labels. In our case, the presence of many single-item clusters or clusters containing only two sentences may limit the reliability of this permutation-based evaluation, potentially leading to misleading conclusions.

Based on the analysis of all evaluation metrics, we selected *paraphrase-mpnet-base-v2* [30] combined with *Agglomerative Clustering* using a distance threshold of 2.6 [24] as the final configuration. This combination consistently demonstrated the best performance on the annotated ground truth data across multiple metrics, making it the most suitable choice for our semantic clustering task.

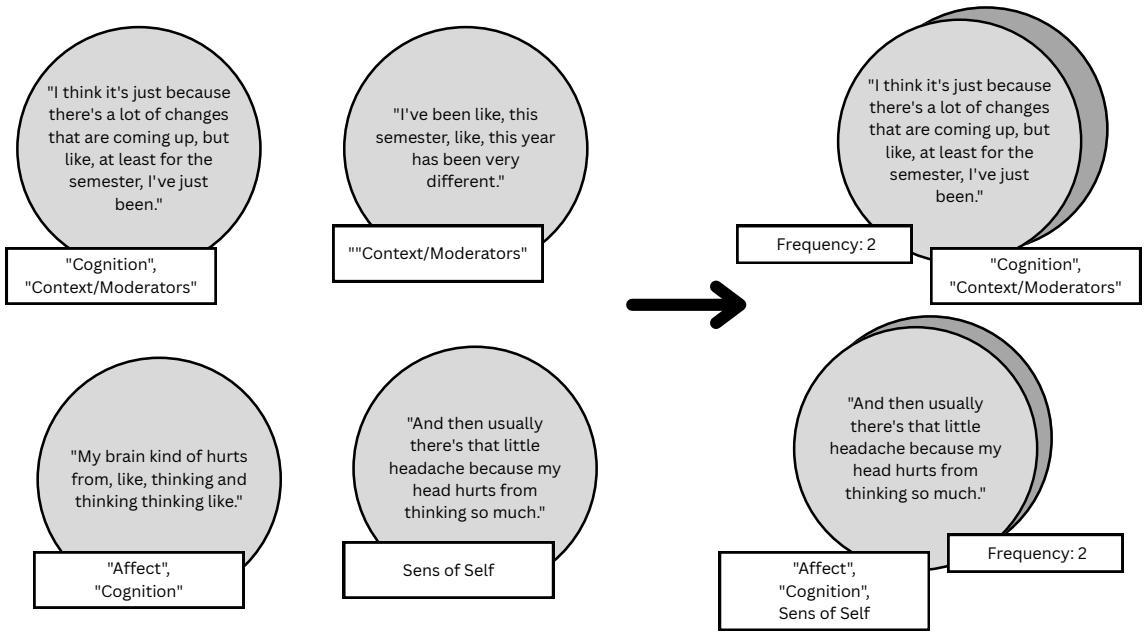


Figure 4.9.: **Example of Semantic Clustering on Psychological Processes:** This Figure illustrates two examples of semantic clustering applied to psychological processes. Within each cluster, the predefined psychological dimensions are represented as a union of all associated dimensions, ensuring that each dimension appears only once. A new attribute, *Frequency*, indicates the number of processes contained within each semantic cluster.

When comparing the best-performing combination, which utilizes a clustering method that does not require prior knowledge of the number of clusters, with algorithms that do rely on this information, such as KMeans [35] and Agglomerative Clustering [24] without using a distance threshold, we find that the best-performing combination still achieves strong results. The performance of our optimal combination demonstrates its effectiveness, even in the absence of prior knowledge about the number of clusters, as shown in Table 4.2.2.

In total, we identified 457 semantic clusters across the provided therapy session transcripts, considering only clusters containing more than one sentence. On average, this results in 5.94 multi-sentence clusters per session. Each of these semantic clusters has an average size of 2.13 sentences.

Figure 4.9 illustrates an example of how semantic clustering is applied. Initially, there are four individual psychological processes. Through semantic clustering, these are grouped into two semantic clusters. The original dimensions associated with the individual processes are aggregated within the semantic cluster, and one representative psychological process is randomly selected. Additionally, we track the number of original processes within each cluster using an attribute called frequency. In this example, each semantic cluster has a frequency of two.

Model	Method	Silhouette	ARI	NMI	Precision	Recall	F1	Accuracy
all-mpnet-base-v2	Agglomerative DT 2.6	**0.112**	0.053	0.32	0.025	**0.963**	0.047	0.236
	BERTopic	0.028	0.05	0.395	0.027	0.933	0.052	0.165
	Community Detection	0.080	0.131	0.946	0.367	0.067	0.099	0.783
	DBSCAN	0.087	0.102	0.944	0.3	0.045	0.070	0.776
	Baseline w/ k	-0.106	-0.005	0.834	0.019	0.018	0.017	0.551
	Baseline w/o k	-0.095	-0.003	0.856	0.021	0.015	0.016	0.587
distilbert-base-nli-stsb	Agglomerative DT 2.6	0.093	0.097	0.943	0.267	0.042	0.065	0.774
	BERTopic	0.046	0.019	0.357	0.026	0.825	0.050	0.142
	Community Detection	0.047	0.269	0.946	0.623	0.180	0.240	0.796
	DBSCAN	0.058	0.168	0.947	0.497	0.095	0.136	0.787
	Baseline w/ k	-0.146	-0.005	0.834	0.019	0.018	0.017	0.551
	Baseline w/o k	-0.131	-0.003	0.856	0.021	0.015	0.016	0.587
paraphrase-distilroberta-base-v1	Agglomerative DT 2.6	0.080	0.111	0.944	0.333	0.050	0.079	0.776
	BERTopic	0.016	0.017	0.353	0.025	0.837	0.047	0.152
	Community Detection	0.056	0.147	0.946	0.517	0.076	0.115	0.786
	DBSCAN	0.076	0.125	0.945	0.367	0.063	0.093	0.780
	Baseline w/ k	-0.118	-0.005	0.834	0.019	0.018	0.017	0.551
	Baseline w/o k	-0.106	-0.003	0.856	0.021	0.015	0.016	0.587
paraphrase-mpnet-base-v2	Agglomerative DT 2.6	0.055	0.317	**0.951**	**0.753**	0.237	0.321	**0.819**
	BERTopic	0.030	0.039	0.419	0.026	0.913	0.050	0.175
	Community Detection	0.052	0.229	0.949	0.642	0.146	0.198	0.800
	DBSCAN	0.067	0.124	0.944	0.433	0.058	0.093	0.780
	Baseline w/ k	-0.127	-0.005	0.834	0.019	0.018	0.017	0.551
	Baseline w/o k	-0.115	-0.003	0.856	0.021	0.015	0.016	0.587
paraphrase-multilingual-mpnet-base-v2	Agglomerative DT 2.6	0.074	**0.325**	0.946	0.582	0.273	**0.331**	0.810
	BERTopic	0.032	0.034	0.389	0.025	0.924	0.048	0.161
	Community Detection	0.058	0.258	0.950	0.611	0.172	0.227	0.805
	DBSCAN	0.064	0.149	0.946	0.500	0.081	0.117	0.786
	Baseline w/ k	-0.136	-0.005	0.834	0.019	0.018	0.017	0.551
	Baseline w/o k	-0.122	-0.003	0.856	0.021	0.015	0.016	0.587

Table 4.6.: **Column-wise Best Clustering Metrics by Model and Method:** This Table presents the results for each evaluation metric across all combinations of LM and clustering methods, tested against the human-annotated ground truth. Bold values indicate the best performance in each column, while underlined values represent the second-best.

4.3. Work Package 2: Generating Clinically Meaningful Process Clusters

We conducted a procedure to generate clinically meaningful clusters across three iterative rounds. After each round, we held consultations with clinical psychologists to identify shortcomings in the model's output and determine areas for refinement. In the first round, we evaluated two prompting strategies, referred to as *Simultaneous* and *Stepwise* Approach. We provide a visual comparison of both strategies in the Appendix A.1. We assessed each strategy using a set of custom-designed evaluation metrics, as explained in Tables 3.1, 3.3,

Model	Algorithm	Silhouette	ARI	NMI	Precision	Recall	F1 Score	Accuracy
paraphrase-mpnet-base-v2	AgglomerativeClustering	**0.08**	**0.44**	0.94	0.51	0.37	**0.42**	0.81
	KMeans	0.07	**0.44**	0.94	0.47	**0.38**	**0.42**	0.81
	Baseline_with_k	-0.13	0.00	0.83	0.02	0.02	0.02	0.55
	AgglomerativeClustering_DT_2.6	0.06	0.32	**0.95**	**0.75**	0.24	0.32	**0.82**

Table 4.7.: **Comparison with Clustering Algorithms Using a Predefined Number of Clusters:** This Table compares the best-performing combination from our experiment without using a predefined number of clusters against clustering algorithms that rely on a predefined number.

and 3.2. In close collaboration with clinical psychologists, we developed metrics to assess the model's ability to generate clinically meaningful process clusters.

As outlined in Section 3.3, the *Stepwise* approach divided the generation process into two distinct stages: first, generating clinically relevant themes and then classifying psychological processes under these themes. In contrast, the *Simultaneous* approach performs both theme generation and process assignment in a single step.

Due to the sensitive nature of our therapy session transcripts, we cannot use closed or cloud-hosted models, such as GPT -4o mini [25]. Instead, we selected the LLaMA 3.1 70B Instruct [37] model, which has demonstrated strong performance in psychiatric and psychological domains [12]

Metric	Simul. R1	Stepwise R1	Stepwise R2	Stepwise R3
Clinically Relevant	1.94	2.04	1.90	2.15
Novelty	1.73	1.87	1.82	2.26
Usefulness	1.67	1.92	1.62	2.24
Specificity	1.80	2.20	1.97	2.61
Coverage	1.93	2.33	1.97	2.21
Completeness	1.87	1.73	2.08	2.28
Intruder	2.23	2.01	2.12	1.87
Redundancy	2.40	1.67	1.90	1.78
Insightfulness	1.81	1.97	1.86	2.22
Trustworthiness	1.97	2.06	2.01	2.24
Total Score	1.88	1.99	1.89	2.21

Table 4.8.: **Domain Experts Ratings for Theme Generation Methods and Classification Across Rounds:** This Table presents expert ratings for the Simultaneous and Stepwise theme generation and assignment methods across multiple evaluation rounds. Scores range from 1 (best) to 3 (worst). The **Total Score** reflects the overall quality ranking provided by domain experts. **Insightfulness** is calculated based on the first three metrics, while **Trustworthiness** is derived from the remaining metrics.

4.3.1. First Round

Setup For the first round, we used a total of 12 randomly selected sessions from the provided therapy transcripts. Each session included both a Stepwise and a Simultaneous output this means we had 24 outputs, and no evaluator assessed the same session twice to avoid bias. Each evaluator reviewed approximately 10 sessions, with overlapping cases incorporated to calculate an agreement

The primary objective of this round was to determine which prompting strategy *Simultaneous* or *Stepwise* should be used in the next two rounds. This decision was supported by a combination of our custom-designed evolution metrics and qualitative feedback obtained from clinical psychologists. The prompt used for the *Simultaneous* is detailed in Appendix A.2.4 and the prompt for the Stepwise procedure is detailed in Appendix A.2.1. A secondary objective of this round was to assess the effectiveness of our self-developed evaluation guidelines and metrics, and to identify and necessary refinements to enhance their validity and utility in evaluating clinical relevance and output quality.

Results & Discussion When comparing the two prompting strategies in the first evaluation round (Table 4.3), the Stepwise approach generally outperforms the Simultaneous strategy across most evaluation metrics. For the Clinically Relevant metric, both approaches achieve similar average scores (close to 2), indicating that each method generates partially meaningful psychological clusters. However, the Stepwise strategy exhibits a higher degree of evaluator agreement. This is reflected in the self-defined F1 Agreement score of 0.411 versus 0.267 for Simultaneous, as well as in Label-based Agreement (1.00 vs. 0.667) and Cohen's Kappa (0.28 vs. 0.174). These values suggest that not only does Stepwise perform better, but evaluators more consistently recognize its clinical relevance.

For Novelty, Stepwise achieves a noticeably higher score (1.87 vs. 1.73), indicating greater insight. While agreement is equal across both strategies for self-defined metrics (as Novelty includes only one score per evaluation), Stepwise again shows marginally better consistency, as indicated by Spearman and Kappa values, which both suggest slight agreement. In contrast, simultaneous shows more frequent disagreement.

In terms of Usefulness, Stepwise outperforms Simultaneous (1.92 vs. 1.67), suggesting better interpretability of psychological functioning. Evaluator agreement supports this: Stepwise shows higher self-defined agreement scores. However, examining the other agreement metrics, Simultaneous performs moderately better, revealing some inconsistencies in how evaluators interpreted the Usefulness metric.

Specificity presents more balanced results. Scores and self-defined agreements are similar across both methods, although Simultaneous achieves a slightly higher Cohen's Kappa, indicating more consistent evaluation across raters.

The structural metrics of Intruder, Redundancy, and Completeness reveal weaknesses in the Stepwise approach. While Stepwise performs better in terms of Coverage, the Simultaneous method shows slightly better results for Completeness. Additionally, agreement scores tend to favor the Simultaneous approach for these structural measures, indicating higher consistency among evaluators. However, Cohen's Kappa remains in the fair-to-moderate range for both strategies.

When comparing the scores for Insightfulness and Trustworthiness, the Stepwise approach shows a distinct advantage. It achieves higher scores of 1.97 for Insightfulness and 2.06 for Trustworthiness. In contrast, the Simultaneous approach scores only 1.81 for Insightfulness and 1.97 for Trustworthiness, which are notably lower. Furthermore, the Total Score also favors the Stepwise approach, indicating its overall superiority in producing more insightful and trustworthy outputs.

Overall, the Stepwise prompting strategy delivers stronger performance across core evaluative metrics, particularly Clinically Relevant, Novelty, Usefulness, and Insightfulness, while also enjoying higher evaluator agreement in these areas. Simultaneous performs more consistently in structural aspects, such as Intruder and Redundancy, but these are considered less important to the psychological insight metrics. This supports the argument for using Stepwise as the more effective prompting strategy.

Although it generates clinically meaningful themes more effectively, there is still room for improvement. Many metrics do not meet clinical standards for reliability and utility. In particular, the first three metrics, Clinically Relevant, Novelty, and Usefulness, need further refinement to be acceptable in applied clinical settings. Additionally, a representative output from the Stepwise approach is shown in Figure 4.10. Clinical psychologists noted that the generated themes, such as "*Struggling with self-improvement*", were often relatively brief, which they identified as a limitation during the evaluation. They suggested that longer and more detailed descriptions of themes could enhance clarity and clinical utility.

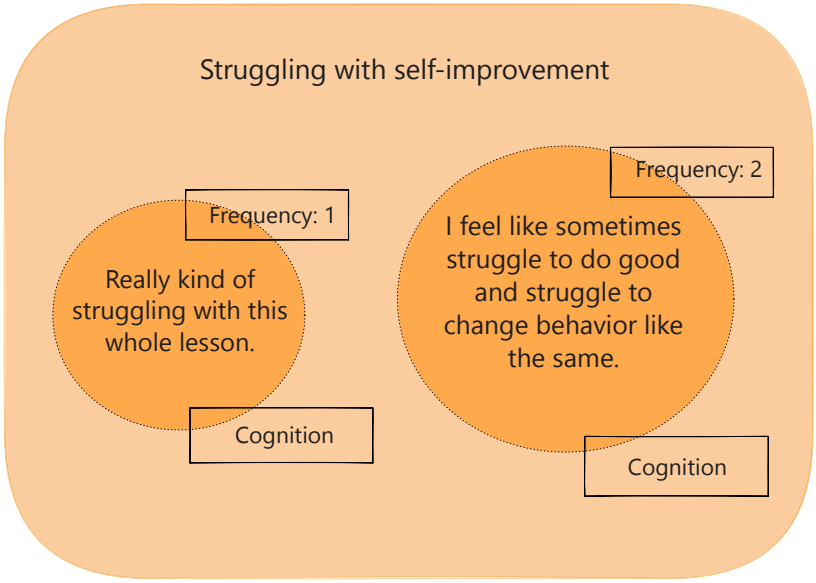


Figure 4.10.: **Example of Generated Theme and Classification from First Round:** An illustrative example showing a generated theme along with the psychological processes classified under it using the stepwise approach. The theme, which is notably brief, was also noted as such by the clinical psychologists.

4.3.2. Second Round

Setup After the first round of evaluation was done, we gathered valuable insights from the clinical psychologists regarding their expectations of the output. They also provided examples of clinically valuable and non-valuable themes, which we incorporated into our chosen prompt, Stepwise, for the following evaluation round. This input helped to improve both the quality and consistency of the generated output. Additionally, some refinements were made to the definition of the evaluation metric. We conducted a feedback-driven prompt engineering. The second round of evaluation included 25 randomly selected sessions that were not previously observed. We included overlap to calculate the agreement between evaluators, and so each clinical psychologist evaluated roughly 13 sessions. The prompt used for this round is provided in Appendix A.2.2.

Results & Discussion In the second evaluation round, we observed an overall decline in performance of the Stepwise prompting strategy compared to the first round. Key interpretive metrics showed notable decreases: the Clinically Relevant score dropped from 2.04 to 1.90, Novelty declined from 1.87 to 1.82, and Usefulness fell more sharply from 1.92 to 1.62. Similarly, Specificity decreased from 2.20 to 1.97, and Coverage from 2.33 to 1.97. Despite these drops, Stepwise still outperforms the Simultaneous strategy from Round 1.

Some structural metrics, however, improved. Completeness increased from 1.73 to 2.08, Intruder from 2.01 to 2.12, and Redundancy from 1.67 to 1.90. These gains suggest that the updated prompt, designed to reduce redundancy and ensure the inclusion and proper categorization of all processes, was partially successful. However, these changes may have unintentionally limited the model's flexibility, resulting in lower scores in more interpretive and subjective areas, such as Usefulness and Insightfulness. Scores for Insightfulness, Trustworthiness, and Total Score also showed slight declines: Insightfulness dropped from 1.97 to 1.86, Trustworthiness from 2.06 to 2.01, and Total Score from 1.99 to 1.89. Although these declines are minor, they indicate a larger trade-off in the model's behavior: improvements in structural completeness came at the expense of interpretive depth in this instance.

Evaluator agreement generally improved in the second round. Most agreement metrics, including the F1 Agreement, Selection Agreement, Spearman's Correlation, and Hard Agreement, showed either stability or modest increases. For instance, Hard Agreement, which had been 0 in the first round, rose to 0.143.

For Novelty, self-defined agreement metrics slightly improved (from 0.33 to 0.357), though Cohen's Kappa dropped into the negative range. This contradiction highlights the inherently subjective nature of novelty, as perceptions of "newness" vary across individuals. Usefulness, another subjective metric, showed similar scores and agreement levels across both rounds.

A more notable improvement is observed in Specificity, where the agreement increased substantially: self-made agreement scores rose from 0.33 to 0.64, and Cohen's Kappa shifted from a negative value to 0.357, indicating fair agreement. Spearman's correlation improved from negative to 0.53, reflecting moderate agreement and better inter-rater consistency.

Coverage also saw improvements in self-made agreement metrics, jumping from 0.0 to 0.5, and F1 Agreement reaching 0.21. However, its Kappa score remained negative, suggesting continued evaluator disagreement. This divergence between agreement types highlights the limits of Kappa in small-sample or high-subjectivity settings despite actual increases in alignment.

Intruder showed minor improvements in agreement, particularly in Hard Agreement, which increased from 0.00 to 0.143. While other agreement scores remained relatively stable, these slight gains point to better alignment in evaluating structural issues.

The findings from the second round indicate a complicated situation. While Stepwise still outperforms Simultaneous overall, its performance has weakened compared to its results from the first round. In particular, declines in Clinically Relevant, Novelty, and Usefulness alongside evaluator agreement pointing in the same direction signal a drop in interpretive quality. At the same time, improvements in Completeness, Intruder, and Redundancy indicate better handling of structural dimensions. These changes likely stem from the modified prompt, which emphasized completeness and proper categorization but may have inadvertently reduced flexibility in generating nuanced and insightful outputs. Additionally, the example included in the prompt may have confused the model, contributing to misinterpretations. Examining a representative theme from the second round in Figure 4.11, we observe that the generated content is now longer than in the first round. However, it still falls short of expert expectations. Clinical psychologists indicated that themes should ideally be complete, descriptive sentences that clearly articulate the psychologically meaningful group. Despite progress in length, the outputs still lack the depth and clarity needed for clinical application.

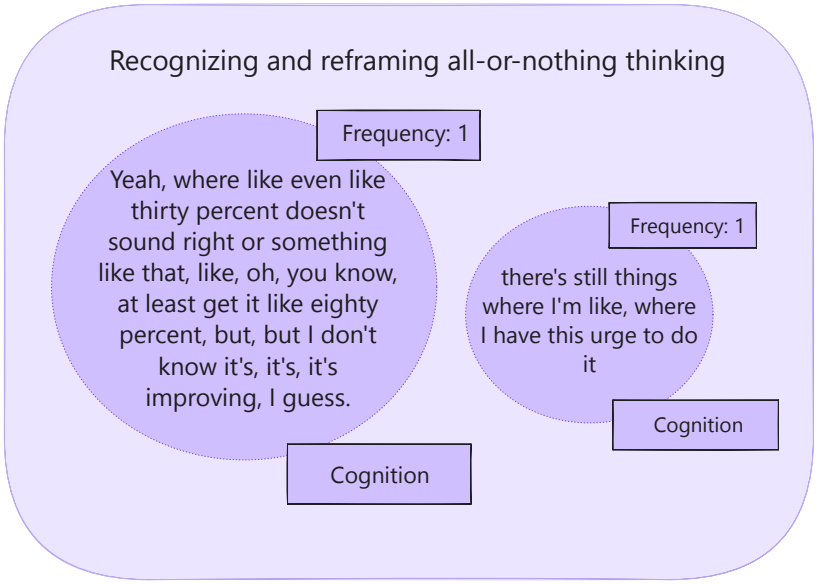


Figure 4.11.: **Example of Generated Theme and Classification from Second Round:** An illustrative example showing a generated theme along with the psychological processes classified under it using the stepwise approach. The theme is longer and provides more information than in the first round, but still does not fully meet the expectations of the clinical psychologist.

4.3.3. Third Round

Setup In the third and final round, we discussed with the clinical psychologists how the generated themes and the classification procedure for the psychological processes performed in the second round. It became clear that there is a challenging balance between creating themes that are too specific and those that become too redundant. The clinical psychologists noted that some redundancy in the themes is not necessarily a problem and, in some cases, might even be expected. For them, it was more important to have specific, clinically meaningful themes with high coverage of the session content.

Based on this feedback, we refined the Stepwise prompt in several ways. First, we included a more thorough definition of what constitutes a psychological process and emphasized the importance of capturing the patient's psychological functioning. Following earlier discussions about the nature of the generated clusters, we also added more explicit instructions to help the model better understand the desired structure and quality of the thematic clusters. Additionally, some guideline sections were condensed in the prompt to prevent it from becoming too lengthy and overwhelming. We restructured the examples to demonstrate and guide the model toward the desired outputs for us and the clinical psychologists. We made these changes to the provided examples because the example themes alone had not been sufficient to improve the quality of the generated themes consistently. For this round, the clinical psychologists evaluated 43 sessions with an overlap of 12 sessions. The prompts used in this round are provided in Appendix A.2.3.

Results & Discussion Table 4.3 presents the final evaluation results for the Stepwise prompting strategy in the third round. Clear improvements are visible across several key metrics. The Clinically Relevant score increased from 1.90 to 2.15, indicating a stronger alignment with meaningful psychological themes. Novelty rose from 1.82 to 2.26, and Usefulness improved significantly from 1.62 to 2.24. These three interpretive

metrics are central to evaluating the clinical utility of the generated themes. Their improvement is also evident in the Insightfulness score, which has now reached 2.22, compared to the previous round's score of 1.86.

Furthermore, Specificity showed a substantial gain, increasing from 1.97 to 2.61, indicating that the refinements made to the prompt helped the model generate more clearly defined and focused themes.

Coverage also improved from 1.97 to 2.21. Interestingly, this score remains lower than in the first round (2.33), which may be attributed to the larger and more diverse evaluation set in this round (43 sessions vs. 12 in Round 1), introducing increased complexity into the task. Only Intruder and Redundancy failed to surpass the 2.0 threshold. These lower scores suggest ongoing difficulties in generating fully distinct, non-overlapping themes. However, a weaker performance on Redundancy was anticipated, as the prompt was optimized to improve Specificity, which is rated as more important by clinical psychologists. The result is that while themes became more narrowly and accurately defined, they also became more prone to overlap, making it harder for the model to differentiate cleanly.

Comparing across all three rounds, the third round yields the highest performance in Insightfulness (2.22), Trustworthiness (2.24), and Total Score (2.21). These results not only exceed those from the second round but also outperform the first round, where Stepwise had already demonstrated superiority over Simultaneous. The continued refinement of the prompting strategy has contributed to outputs that are more meaningful, trustworthy, and clinically useful.

Turning to inter-rater agreement, the third round showed mixed but encouraging developments. For Clinically Relevant, all agreement metrics suggest a moderate level of consensus between evaluators, though the Selection Agreement remained similar to the second round.

For more subjective metrics, such as Novelty and Usefulness, agreement remained moderate, with some indications of increased disagreement likely due to the inherently subjective nature of these evaluations. These fluctuations highlight the challenge of obtaining consistent ratings on metrics that rely heavily on personal perception and domain expertise.

In contrast, Specificity and Redundancy showed the strongest agreement. Self-defined agreement scores reached approximately 0.7, while other agreement types, such as Cohen's Kappa, remained around 0.2, indicating a slight but stable level of agreement. Coverage also demonstrated a slight improvement, maintaining a similar level of inter-rater consistency.

A particularly noteworthy development occurred in the Intruder metric. Selection Agreement improved significantly, suggesting that evaluators frequently selected the same score for specific clusters within a session. This can occur in cases where clusters are particularly strong, clearly capturing a meaningful psychological theme, as well as when clusters are notably weak or irrelevant. In both extremes, evaluators are more likely to agree on the appropriate score. In contrast, Label-based Agreement, which evaluates all clusters across a session, includes more ambiguous or borderline cases where interpretations may differ. The contrast between these two agreement types highlights that while a Selection Agreement can capture consensus around clear cases, good or bad, it does not reflect consistency across the full range of cluster quality. Still, the presence of high Selection Agreement suggests that the model is increasingly capable of producing outputs that evaluators find either clearly valuable or flawed, both of which support higher scoring consistency.

Combining performance results with agreement data, the third round demonstrates the strongest overall performance of the Stepwise strategy. The improvements in Clinically Relevant, Novelty, Usefulness, and Specificity are seen in the elevated scores for Insightfulness, Trustworthiness, and Total Score. While agreement on subjective metrics remained modest, structural metrics such as Specificity, Redundancy, and Intruder, as well as Clinically Relevant, benefited from greater inter-rater reliability. This alignment between higher

scores and improved agreement for more objective metrics suggests that the latest prompt refinements helped produce not only more precise outputs but also more consistently interpretable ones. Nevertheless, continued variability in subjective judgments highlights the challenge of modeling nuanced clinical interpretation and underlines the importance of ongoing expert involvement in evaluation.

To complement these quantitative findings, Appendix C.3.1 and Appendix C.3.2 provide detailed evaluator results and patient-level breakdowns. These data offer transparency into individual variations and potential sources of disagreement.

Metric	Measure	Simul. R1	Stepwise R1	Stepwise R2	Stepwise R3
Clinically Relevant	F1 Agreement	0.267	0.411	0.441	0.409
	Cohen's Kappa	0.174	0.280	0.248	0.113
	Spearman	0.425	0.408	0.369	0.119
	Selection Agreement	0.261	0.463	0.480	0.416
	Label-based Agreement	0.667	1.000	0.643	0.429
	Hard Agreement	0.333	0.000	0.143	0.000
Novelty	F1 Agreement	0.167	0.167	0.261	0.222
	Cohen's Kappa	-0.500	0.000	-0.013	-0.167
	Spearman	-0.500	null	-0.059	-0.167
	Selection Agreement	0.333	0.333	0.357	0.286
	Label-based Agreement	0.333	0.333	0.357	0.286
	Hard Agreement	0.333	0.333	0.357	0.286
Usefulness	F1 Agreement	0.222	0.333	0.357	0.111
	Cohen's Kappa	0.250	0.000	0.097	-0.195
	Spearman	0.500	-1.000	0.078	-0.041
	Selection Agreement	0.333	0.667	0.500	0.143
	Label-based Agreement	0.333	0.667	0.500	0.143
	Hard Agreement	0.333	0.667	0.500	0.143
Specificity	F1 Agreement	0.222	0.167	0.546	0.463
	Cohen's Kappa	0.250	-0.500	0.375	0.533
	Spearman	0.500	-0.500	0.535	0.720
	Selection Agreement	0.333	0.333	0.643	0.714
	Label-based Agreement	0.333	0.333	0.643	0.714
	Hard Agreement	0.333	0.333	0.643	0.714
Coverage	F1 Agreement	0.333	0.000	0.222	0.300
	Cohen's Kappa	0.000	-0.500	-0.140	0.067
	Spearman	-1.000	-1.000	0.000	0.242
	Selection Agreement	0.667	0.000	0.500	0.429
	Label-based Agreement	0.667	0.000	0.500	0.429
	Hard Agreement	0.667	0.000	0.500	0.429
Intruder	F1 Agreement	0.464	0.222	0.412	0.434
	Cohen's Kappa	0.171	0.063	0.116	0.219
	Spearman	0.122	0.285	0.112	0.290
	Selection Agreement	0.522	0.348	0.374	0.485
	Label-based Agreement	1.000	0.667	0.500	0.143
	Hard Agreement	0.000	0.000	0.143	0.143
Redundancy	F1 Agreement	0.444	0.222	0.350	0.750
	Cohen's Kappa	0.400	0.250	0.054	0.562
	Spearman	0.500	0.500	0.177	0.600
	Selection Agreement	0.667	0.333	0.357	0.714
	Label-based Agreement	0.667	0.333	0.357	0.714
	Hard Agreement	0.667	0.333	0.357	0.714

Table 4.9.: **Inter-Evaluator Agreement Metrics by Method and Round:** This Table compares agreement statistics between the Simultaneous and Stepwise theme generation methods across Rounds 1–3. Note that only the Stepwise method was evaluated in Rounds 2 and 3. The *Completeness* metric is not included here, as it is automatically calculated by us and not rated by evaluators.

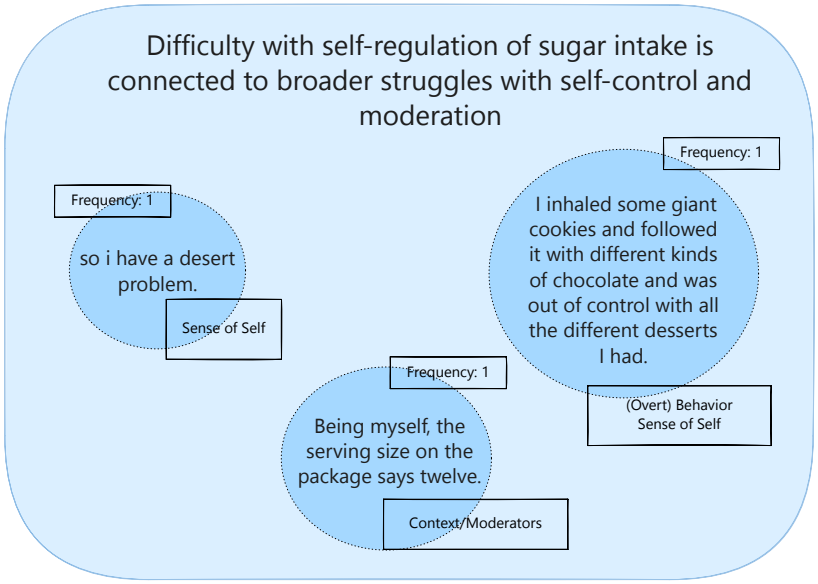


Figure 4.12.: **Example of Generated Theme and Classification from Third Round:** An illustrative example showing a generated theme along with the psychological processes classified under it using the stepwise approach. The theme is now a

A closer look at a representative output from this round (Figure 4.12) provides further insight. The theme "Difficulty with self-regulation of sugar intake is connected to broader struggles with self-control and moderation" exemplifies the model's ability to generate not just longer outputs but also more informative and clinically relevant ones. Unlike earlier rounds, this theme conveys a more abstract psychological connection rather than just a surface-level semantic grouping. The associated processes further support this interpretation, suggesting that the model is beginning to capture patterns that reflect deeper psychological structures rather than mere keyword similarity.

Thematic Overlap in Psychological Processes As outlined in the methodology, we allowed the model to assign a single psychological process to multiple clinically meaningful themes when the process was considered highly relevant to the theme. Figure 4.13 shows the percentage of psychological processes assigned to multiple themes across all therapy sessions with our final version of the stepwise prompt, broken down by patient. Notably, patients P1, P3, and P4 have percentages above the overall average of approximately 63%, suggesting that a significant portion of their psychological processes span multiple thematic contexts. This stands in contrast to Patient P2, who completed a similar number of therapy sessions but exhibited a below-average rate of multiple assignments. This may indicate that P2's psychological content was more narrowly focused or less contextually diverse compared to P1, P3, and P4.

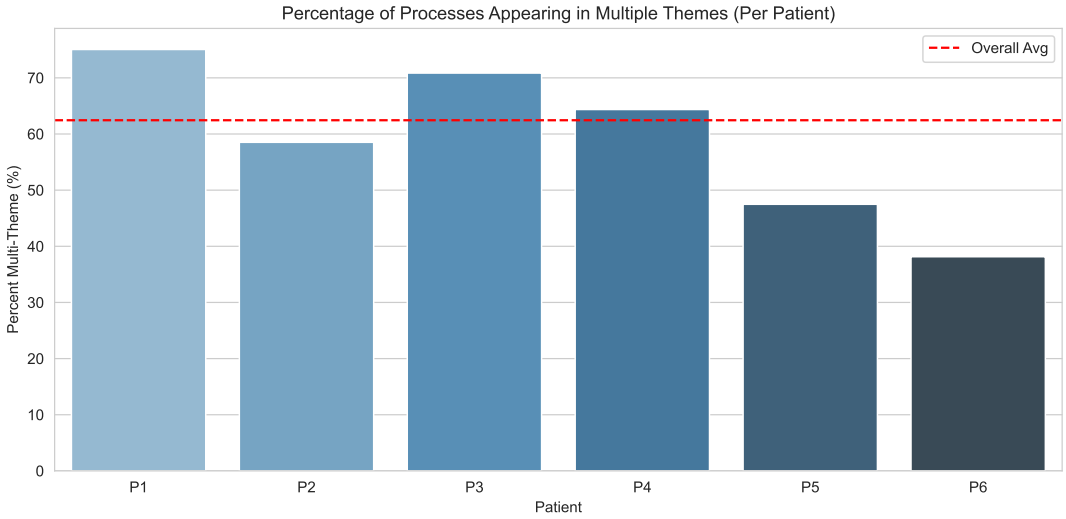


Figure 4.13.: **Percentage of Processes Appearing in Multiple Themes (Per Patient):** This Figure shows the percentage of psychological processes that are classified under more than one clinically meaningful theme for each patient across all therapy sessions. The red line indicates the overall average across patients.

4.4. Work Package 3: Generating Clinically Meaningful Connections

As part of Work Package 3, which focuses on generating clinically meaningful connections within patient networks, we conducted a comparative evaluation of two different approaches to network generation. We generate one network using the full methodology, incorporating all the improvements introduced throughout the project, such as the *Stepwise Approach* 3.3 and the *pairwise generation of clinically meaningful connections 3.4*. This baseline prompt integrates the expert knowledge gathered throughout the thesis without dividing the task into modular steps. The structure of the baseline approach is illustrated in Figure 4.14. The specific prompt used for generating the baseline network is shown in Figure 4.15, and the complete version is provided in Appendix A.12.

The baseline prompt consolidates all steps into a single instruction, asking the model to generate clinically meaningful themes, classify the provided psychological processes under those themes, and create all inter-theme connections in one pass without leveraging our structured, pairwise method. The short version of the prompt can be seen in Figure 4.15 and the full version in the Appendix A.12. Although it includes examples of how themes should look, it lacks the iterative refinement introduced in our pipeline. To ensure unbiased feedback, we presented both networks to expert evaluators without disclosing which was which. We asked them to respond to three questions:

1. Which network contains better clinically meaningful themes?
2. Which network presents clearer or more informative connections between themes?
3. Which network helps you better understand the patient and could more effectively support the therapeutic process?

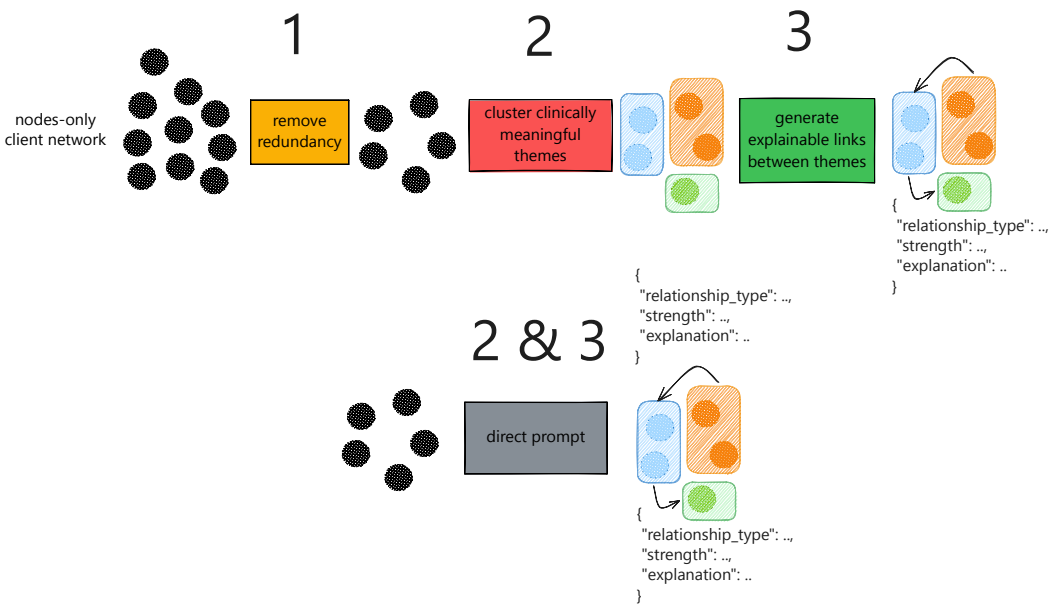


Figure 4.14.: **Comparison of the Pipeline Against the Baseline:** This Figure illustrates how our pipeline generates the patient network in two distinct steps, compared to the baseline method (shown in gray), which simultaneously generates themes, classifies psychological processes and creates relationships between themes. To ensure a fair comparison, the baseline also uses the preprocessed data produced in the first step of our pipeline. The baseline prompt is provided in Appendix A.12..

We designed these questions to evaluate multiple dimensions of the generated networks. The first question targets the quality and clinical relevance of the themes, directly assessing the second step of our pipeline see Section 3.3. The second question focuses on the clarity, coherence, and interpretability of the connections between themes, corresponding to the final step of the pipeline see Section 3.4. The third question integrates both components, providing an overall evaluation of whether the network meaningfully supports clinical psychologists in gaining a deeper insight into the patient and informing future therapeutic sessions.

Setup For the final evaluation round, we reintroduced all 77 therapy sessions to the evaluators, including an overlap of 18 sessions to once again calculate inter-rater agreement. As a result, each evaluator assessed a total of 42 patient networks. This setup enabled a comprehensive comparison across the entire set of provided therapy sessions. We provided each Evaluator an Evaluation Guideline with an Excel Sheet D.2

Results & Discussion When examining the pairwise comparison results in Figure 4.16, which includes two complete evaluations from clinical psychologists and a third partial evaluation covering five patient networks, a consistent and compelling trend emerges across all three evaluation questions: our pipeline clearly outperforms the baseline in terms of evaluator preference, with moderate to strong inter-rater agreement seen in Table 4.10.

For the first question, identifying the network with more clinically meaningful themes, the evaluator preferred it in 89.4% of the cases, with an agreement rate of 44.4% in our pipeline. When we also look at Cohen's Kappa score, which has a value of 0.769, indicating a strong Agreement between evaluators. The result indicates

Baseline Prompt

You are a clinical psychologist and your task is to perform a full, end-to-end clinical reasoning analysis based on a therapy dialogue and a list of psychological processes.
Definitions: Input:

- Transcript: A therapy session dialogue (context only, do not extract utterances)
- Psychological Processes: A list of psychological processes

Guidelines for writing Themes: `{placeholder}`
Guidelines for Process Classification: `{placeholder}`

Inter-Theme Relationships: `{placeholder}`
Your Goals

1. Theme Generation
2. Process Classification
3. Inter-Theme Relationships

Examples of Themes: `{placeholder}`
Output Format: `{placeholder}`
Transcript: `{prompt_str}`

Listed Process to be Clustered: `{formatted_data_process_data}`
Do not reuse the provided examples, generate new Clinically Meaningful Themes. Only return your response as single complete JSON file using the structure above. Do not include commentary or multiple JSON blocks. Now begin your task:

Figure 4.15.: **Baseline Prompt:** This Figure presents the prompt used to generate the baseline patient network. It combines all instructions from our pipeline into a single prompt, including expert-written definitions and collected examples of how clinically meaningful themes should be formulated. The full version of this prompt is found in Appendix A.12

not only a strong overall preference but also a high level of consistency among evaluators, reinforcing the conclusion that our method produces thematically rich and clinically valuable groupings.

In the second question, which addressed the quality of the connections between themes, our approach was favored 77.3% of the time, with approximately 38.9% agreement among evaluators and an Cohen's Kappa Score of 0.55 indicating an moderate agreement. Although slightly lower, these numbers still reflect a clear majority preference and a substantial level of consensus, supporting the view that our method generates more coherent and interpretable relationships between themes.

The third question concerning which network better supports understanding the patient and could more effectively aid therapeutic practice yielded the strongest results: our pipeline was chosen in 92.4% of evaluations, with 50% agreement and a Cohen's kappa score of 1.00 indicating a strong agreement. The unanimous agreement among the evaluators signals a decisive affirmation of the clinical relevance and practical usefulness

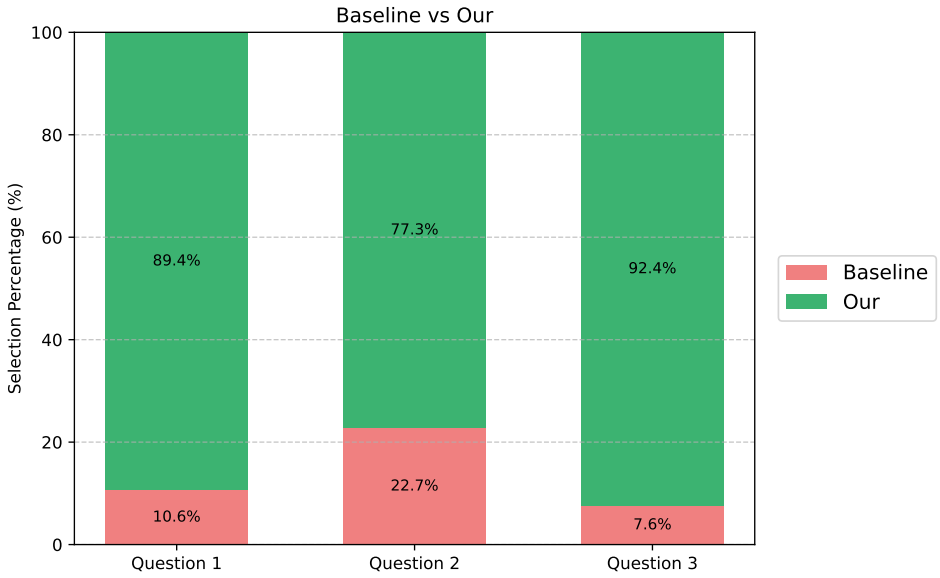


Figure 4.16.: **Evaluation Results of Pairwise Comparison:** This figure presents the results of the pairwise comparison between patient networks generated by our pipeline and the baseline. Evaluators chose between the two networks based on three predefined questions. The results show the distribution of how often each type of patient network was chosen. The evaluation includes two full assessments from clinical psychologists and one partial assessment covering five patient networks.

of the networks produced by our method.

Taken together, the high preference scores and strong agreement levels across all three questions provide robust evidence for the effectiveness of our pipeline. The combination of stepwise prompting for theme generation and classification, along with the pairwise method for relationship modeling, produces outputs that not only resonate with individual evaluators but also align with shared clinical reasoning. This structured approach supports the development of patient networks that are both insightful and actionable in real-world therapeutic contexts.

As part of the evaluation procedure, we instructed the clinical evaluators to flag any patient networks that could potentially disclose personal or identifiable information about the patient. Notably, none of the patient networks generated by our pipeline were flagged during this process, indicating a high level of privacy preservation in the generated outputs.

Question	Agreement Count	Total Comparisons	Agreement %	Cohen's Kappa
Question 1	8	18	44.4	0.769
Question 2	7	18	38.9	0.550
Question 3	9	18	50.0	1.000

Table 4.10.: **Inter-rater agreement statistics per question:** This Table shows how often evaluators selected the same patient network in pairwise comparisons across the three evaluation questions. Higher values indicate stronger inter-rater agreement

Given this strong overall performance, it is essential to examine the instances where our pipeline was not

selected. Understanding these exceptions may offer valuable insights into potential limitations or edge cases where further refinement is needed. A recurring theme in the evaluators' feedback, particularly in response to the second question regarding the quality of the connections between themes, was that our pipeline sometimes generated too many relationships. This high density of connections, while clinically relevant, was occasionally perceived as overwhelming. As a result, the baseline method was chosen sometimes over our patient networks. A particularly insightful comment noted:"*This one is sparser, but I think it has enough to base treatment planning on. There is some balance where the complex network is too complex to be helpful, and the simple network is complex enough to be more helpful than the complex network.*" This feedback highlights a critical trade-off: while our approach excels at capturing nuanced relationships, an excess of detail can reduce interpretability and clinical usability. In some cases, simpler or sparser networks, such as those produced by the baseline, were perceived as sometimes practical for therapeutic planning. This observation highlights a limitation of our current pipeline. While it successfully generates a broad set of clinically relevant connections, the resulting complexity may place a cognitive burden on clinicians reviewing the networks. To enhance the practical utility of these outputs, future work could explore strategies for managing this complexity. One potential refinement of the relationship generation process involves reducing the total number of connections, introducing bidirectional links where appropriate, and encouraging more specific descriptions of relationships. These adjustments could significantly enhance the interpretability and clinical relevance of the resulting patient networks.

Also noted by evaluators throughout the assessment process, the networks generated using the baseline approach generally contained fewer clinically meaningful themes, which were often sparse, vague, and lacked relationships between themes when compared to those produced by our pipeline. This raises the question: *Why is the baseline network smaller and less informative?* One likely explanation is that the baseline attempts to generate the entire patient network, including theme generation, classification, and connection building, in a single step. In contrast, our pipeline divides the process into distinct stages, allowing the model to focus on one task at a time: theme generation, classification of psychological processes, and, finally, pairwise relationship generation. By separating these steps, our approach reduces cognitive and computational overload for the model, leading to more coherent and clinically relevant outputs. The baseline, by contrast, presents the model with too many tasks at once, likely exceeding its capacity to handle adequate context. This problem is further compounded by token limitations. Given the computational constraints and token limits of the model, breaking the task into smaller, more focused prompts enables us to generate more detailed and complete outputs at each stage, a feat that is difficult to achieve with a single, large prompt.

In summary, the pairwise comparison results demonstrate the superior performance of our pipeline over the baseline across all evaluation dimensions: the generated themes are more clinically relevant, the relationships between themes are more coherent, and the overall networks are more useful for therapeutic application. High agreement among clinical psychologists further reinforces the robustness and reliability of our approach. While the richness of relationships occasionally led to cognitive overload, this trade-off highlights valuable areas for future refinement. Significantly, none of the generated patient networks raised privacy concerns, validating the ethical soundness of the method. Overall, our structured, multi-step pipeline effectively produces clinically meaningful, interpretable, and privacy-preserving patient networks suitable for use in psychological therapy. We also provide a detailed evaluation result from each Evaluator in the Appendix C.3.3

5. Conclusion

5.1. Summary

In this thesis, we explored how LLMs can support the construction of patient networks for PBT, a process that traditionally requires significant manual effort from therapists. Starting from annotated therapy session transcripts, we developed a multi-step pipeline that reduces redundancy, extracts clinically meaningful themes, classifies psychological processes under these themes, and constructs expressive relationships between them. The early chapters outlined the motivation for this work, grounded in the increasing demands on therapists using PBT and the lack of automated tools to support them.

Our experimental results show that the proposed approach performs well both technically and clinically. While filler detection and semantic clustering showed solid results, the most meaningful outcomes emerged from the clinical evaluations. Clinical psychologists found that the generated themes and relationships provided valuable insight into patient functioning and were significantly more useful than those produced by a one-shot baseline. The patient networks were rated as clinically relevant, novel, and helpful, though sometimes too complex in structure.

Overall, this work demonstrates that LLMs can meaningfully support the construction of abstract patient representations, potentially easing therapist workload and deepening clinical understanding. The release of 77 patient networks further contributes to the field by offering practical examples of this capability. Looking ahead, this work lays the foundation for integrating LLMs more directly into therapeutic workflows.

5.2. Limitation

When examining the dataset used in this thesis, which could include sensitive information derived from individuals residing in a small city in Ohio, two important questions arise:

First, is there any risk of personal information leakage if we make the 77 generated patient networks publicly available?

We took extensive steps to protect patient privacy. The patient networks used in this thesis are abstract representations of psychological functioning discussed in therapy sessions. They do not include any direct quotations from the transcripts or identifiable personal information. Each network consists only of clinically meaningful themes and the relationships between them, constructed solely at the thematic level. Furthermore, the provided dataset was anonymized for us. While no system can guarantee absolute confidentiality, we have taken all reasonable measures to minimize the risk of identifiable information being retained.

To further mitigate any potential privacy risks, we asked clinical psychologists participating in our final evaluation experiment to identify any network they believed might pose a risk of disclosing sensitive or identifiable patient data. The evaluator flagged none of the networks, confirming that the patient representations remain abstract and de-identified. Based on these measures, the risk of re-identification or leakage of personal information is negligible.

Second, are the data and its findings generalizable?

Although the dataset originates from a specific demographic context, the clinically meaningful themes extracted describe psychological dynamics that are broadly relevant across diverse populations. For example, a theme such as "tension between family obligations and authentic self-expression" reflects a psychological conflict that is not unique to a single region but instead appears in many cultures around the world, especially those with similar social and economic structures to the U.S. Therefore, while the data stems from a localized source, the abstracted nature of the psychological themes supports a degree of generalizability, especially within culturally and economically comparable settings.

Another important consideration is the composition of our evaluator group. Throughout this thesis, three clinical psychologists participated in the evaluation process. They assessed whether the generated themes were clinically relevant and whether they accurately reflected the content of the therapy sessions. Additionally, they were responsible for conducting a pairwise comparison between our proposed pipeline and the baseline approach. As such, they played a central role in validating the clinical utility of the generated patient networks. However, the question arises whether a group of three psychologists is representative enough to generalize the findings. This constitutes a limitation of the study. Despite this, it is essential to note that we have experienced clinical psychologists as evaluators with specialized expertise in PBT. Their domain knowledge, grounded in clinical practice, added significant value to the evaluation process, allowing for in-depth, expert-level feedback. Nevertheless, future work could benefit from involving a larger and more diverse panel of clinicians to further validate the generalizability and robustness of the results.

5.3. Future Opportunities

5.3.1. Generating Clinically Meaningful Theme Relationships via Ensemble LLM Strategies

In our current pipeline setup for generating clinically meaningful relationships between themes, which reflect a patient's psychological dynamic, we rely on a single LLM (LLaMA 3.1 70B Instruct [37]) prompted with the same template under varying temperature settings (0.0, 0.3, and 0.6). We then apply majority voting across these outputs to determine whether a connection exists and to identify its attributes 3.4.

To enhance the process, we could evaluate several alternative ensemble strategies:

1. **Expanded Temperature Range:** Maintain the same model and prompt but use a broader range of temperatures (e.g., 0.0, 0.5, and 1.0). This could increase diversity in the model's responses and potentially surface more nuanced relationships.
2. **Prompt Diversity:** Keep the same model and temperature but apply different prompting templates, each designed to frame the relationship generation task in a slightly different way. Majority voting would again be used to consolidate outputs

3. **Model Diversity:** Use different LLMs (e.g., LLama [37], GPT-4o mini [25], and Qwen [5]) with same prompt and temperature. This would introduce architectural differences as a source of ensemble diversity, possibly improving robustness and reducing model-specific biases.

To assess the effectiveness of these strategies, we would repeat the experiment described in Section 4.4, which involved a pairwise comparison between a baseline model and our pipeline (see Figure 4.14). This time, we would include three different networks, each generated using one of the ensemble strategies above. Clinical psychologists would evaluate all three networks without knowing which strategy produced which network. Since the thematic content would remain consistent across all versions, we would focus on the remaining two questions from the original evaluation. This extended evaluation framework would enable us to assess how various ensemble strategies impact the interpretability, clinical relevance, and practical utility of the resulting patient networks in the context of PBT.

5.3.2. From Static to Dynamic Patient Networks: Capturing Temporal Patterns in Therapy Sessions

The current pipeline developed in this thesis, illustrated in Figure 3.1, processes one therapy session at a time and, therefore, generates static patient networks. In this context, "*static*" refers to networks that are based solely on the content of a single therapy session without incorporating information from previous sessions. We deliberately chose this static approach for the scope of this thesis. Our goal was to investigate whether LLMs are capable of generating clinically meaningful themes and relationships. Therefore, generating insightful patient networks from individual therapy sessions could potentially support psychological treatment and aid in conducting PBT. Based on the results of our experiments, the answer appears to be affirmative. The generated networks have demonstrated clinical relevance and usefulness, as confirmed by expert evaluations.

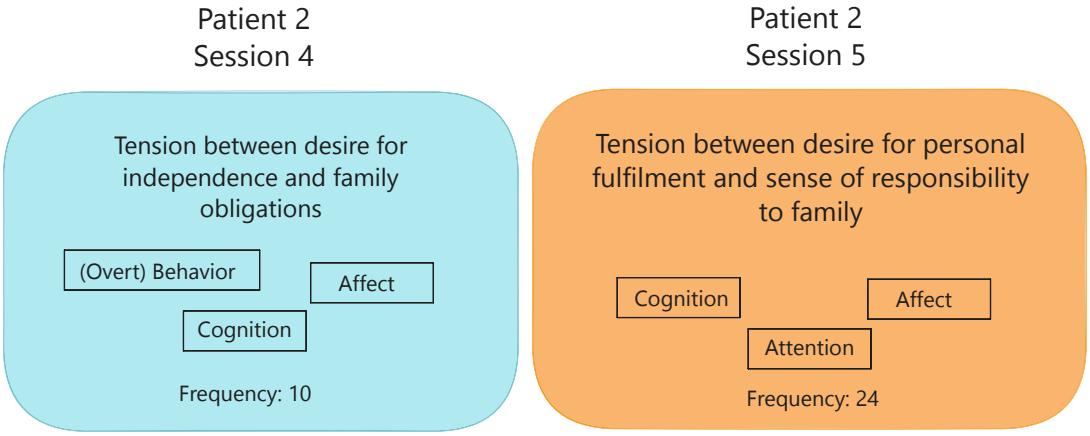


Figure 5.1.: **Recurring Clinically Meaningful Theme Patterns Across Sessions:** This Figure shows an example of a recurring theme appearing across two sessions, indicating that the patient discusses specific clinically meaningful themes multiple times throughout therapy. In this example, the theme recurs in Patient 2's Session 4 and Session 5. We present each clinically meaningful theme along with the three most frequent psychological dimensions derived from the classified processes within that theme. Additionally, we include the *Frequency*, which indicates the number of psychological processes assigned to each theme.

The next step would be to explore the automatic construction of dynamic patient networks that integrate information from previous therapy sessions, thereby capturing temporal patterns in a patient's psychological development. While analyzing static networks, we observed a notable trend: clinically meaningful themes tend to recur across multiple sessions. As shown in Figure 5.1, for instance, the theme "*Tension between desire for independence and family obligation*" appears in Session 4 for Patient 12, with a frequency count of 10 indicating that 10 psychological processes were grouped under this theme. The most prominent dimensions represented in this theme are Overt Behavior, Affect, and Cognition. In the following session (Session 5), a closely related theme emerged: "*Tension between desire for personal fulfillment and sense of responsibility to family*", with a frequency count of 24. The dominant dimensions in this theme are Cognition, Affect, and Attention. This recurrence suggests that important clinically meaningful themes often persist or evolve over time. Therefore, aggregating patient networks across sessions by integrating them into a dynamic framework offers a more comprehensive view of psychological development and provides even deeper insight into the patient's therapeutic journey.

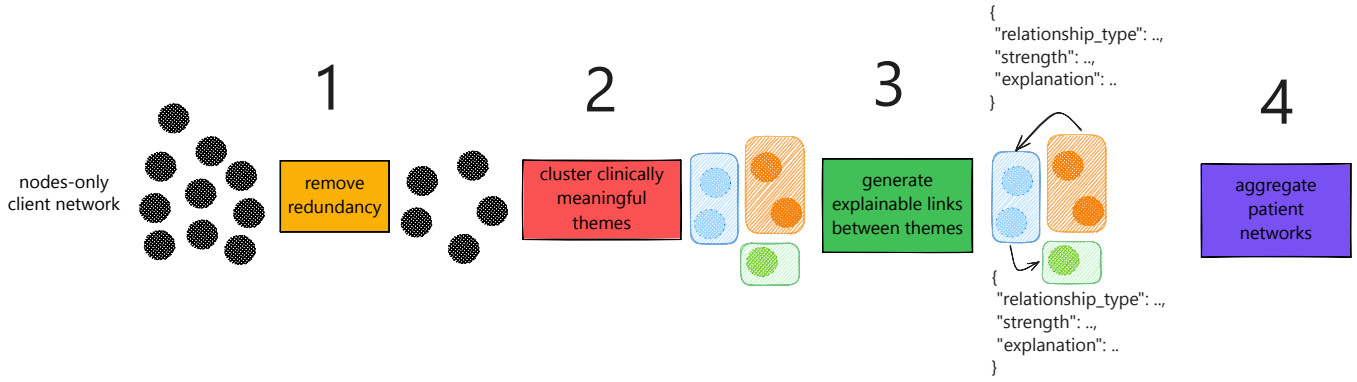


Figure 5.2.: **Future Pipeline Possibility:** This Figure illustrates a potential extension of the current pipeline, which currently generates only static patient networks. A proposed fourth step is added to aggregate two static patient networks into a single network, enabling temporal or session-based analysis over time.

Adding an Additional Step One initial concept for generating dynamic patient networks is illustrated in Figure 5.2. In this approach, we retain the current structure of constructing static networks for each therapy session. However, we would introduce an additional step at the end: we prompt the model to fuse the network from the current session with the network from the previous session. The objective of this fusion step is to preserve clinically relevant insights from earlier sessions while simultaneously capturing psychological development and therapeutic changes over time. This iterative fusion would serve as a foundational mechanism for constructing temporally-aware patient networks, offering a richer, longitudinal perspective on the patient's mental health trajectory. To support this fusion step, we would guide the model by a set of structured instructions, such as:

- **Merge similar themes:** If the two networks contain clinically meaningful themes that are highly similar or overlapping, the model should merge these into a single unified theme and group all associated psychological processes under the fused theme.
- **Incorporate new themes:** For any new themes emerging in the latest session that were not present previously, the model should evaluate whether they form strong conceptual or psychological links with

themes from earlier sessions. If so, it should establish expressive connections to reflect these evolving dynamics.

These guidelines help maintain thematic continuity while allowing the network to adapt and grow in response to new developments in therapy. This fusion-based approach represents a first step toward constructing dynamic patient networks that better capture the longitudinal nature of therapeutic progress and psychological change.

Modifying the Prompt for Dynamic Patient Network Another way to aggregate patient networks across therapy sessions, rather than adding an aggregation step at the end, would be to modify the pipeline itself. Specifically, we could adjust how we generate clinically meaningful themes during the theme clustering phase.

In the current pipeline, as described in Section 3.3, the prompt for generating clinically meaningful themes 3.5 operates independently for each session. To enable longitudinal consistency, we could adapt the prompt by including the themes generated in the previous therapy session. We instruct the model to reuse existing themes unless new, clinically meaningful content emerges in the current transcript. In other words, the prompt would include a statement such as

- "*Here are the themes identified in the last therapy session. Only generate new themes if new clinically meaningful patterns appear. Otherwise, retain the previous ones*"

This change would ensure greater continuity across sessions, reducing fragmentation and prompting a more coherent picture of patient development over time. In contrast, the second step, classifying psychological processes under the provided themes, would remain mostly unchanged. Since this step already maps processes to a given set of themes, we would need to ensure that the input includes both previously established and newly generated themes. To maintain temporal traceability, we propose tagging each psychological process with its originating session, enabling clinicians to distinguish the evolution of themes over time. This pipeline-level modification would enable the incremental development of patient networks across multiple sessions, providing a more continuous and interpretable view of the patient's therapeutic progress.

5.3.3. Creating Therapist Networks and Leveraging Them for Enhanced Therapeutic Insight

Another promising direction for future work involves the generation of a therapist network. The thesis focused exclusively on constructing patient networks primarily based on the clinical psychologists' preference to prioritize patient insight. The provided dataset includes annotations for therapist utterances that were identified as psychological processes.

As shown in the Appendix, we have access to several therapist-related data points. These include:

- Word Count per session B.1
- Utterance Count per session B.3,
- Speaking duration per session B.2,
- Psychological process count per session per patient from the therapist B.4, and
- Dimensions assigned to therapist-initiated psychological processes B.5.

By shifting the focus from the patient's psychological functioning to the therapist's intervention strategies, we can construct a separate therapist-specific network. This network aims to map the therapeutic techniques, patterns, or strategies employed during sessions, providing a structured view of the therapist's clinical behavior. In doing so, we would gain a dual network framework: one network representing the patient's internal psychological processes and another capturing the therapist's therapeutic approach.

To build such a therapist network, we would need to adapt the existing prompts, which were initially designed for patient networks. Specifically, the prompts must direct the model to identify therapeutic strategies rather than psychological symptoms or behaviors. This also means providing different thematic examples relevant to clinical intervention, which necessitates collecting domain-specific examples of high and low-quality therapist themes similar to the evaluation procedure established for patient networks. Iterative feedback from clinical psychologists plays a crucial role in refining the prompts and improving output consistency and clinical value.

Finally, a compelling future opportunity would be to generate explicit relationships between the patient and therapist networks. This would involve linking the patient's psychological functioning to specific therapeutic strategies, with the model providing explanations for why specific interventions were used in response to particular psychological states. Cross-network connections could significantly enhance the understanding of therapy sessions, providing a comprehensive view that integrates both perspectives. This integrative approach would not only deepen clinical insight but also pave the way for more informed, dynamic therapy planning.

5.3.4. Synthesizing Therapy Sessions from Patient and Therapist Networks

Another promising direction for future work involves synthesizing entire therapy sessions based on structured representations of both the patient's and therapist's networks. Recent work by Lee et al. [18] has demonstrated that LLMs can be effectively guided to simulate realistic interactions between clients and counselors.

Building on this foundation, our framework offers the capability to generate detailed network representations of a patient's psychological functioning. When paired with therapist networks that model strategic interventions, these could be used to generate fully synthetic therapy dialogues that evolve naturally over time. This would enable the creation of longitudinal, multi-session therapy datasets resources that are currently scarce but critically valuable for both research and clinical training.

An alternative or complementary approach could involve extracting only the therapeutic strategies employed in a given session and combining them with the corresponding patient network to generate synthetic therapeutic interactions. This may offer a more streamlined and targeted approach for simulating therapy while preserving the key dynamics of the therapeutic process.

6. Acknowledgment of AI Assistance

In the preparation of this thesis, generative AI and other language tools were used to support the writing process in the following ways:

- Rephrasing of my own sentences for improved clarity
- Grammar and spell checking of original content
- Assistance with LaTeX syntax and table formatting

The tools I used include ChatGPT, Grammarly, and DeepL Translator.

These tools were employed solely to enhance the readability and presentation of the thesis. All ideas and content are my own or have been appropriately cited.

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Appendices

A. Appendix A

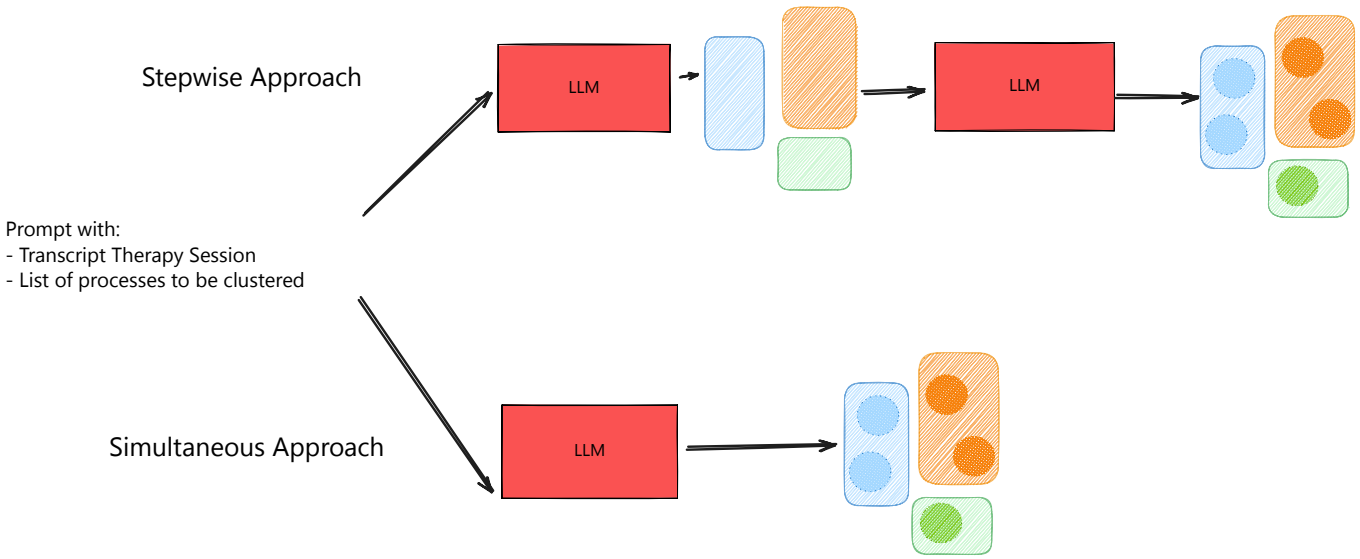


Figure A.1.: **Two-Stage Approach for Generating Clinically Meaningful Themes:** This figure compares two prompting strategies. The upper section represents the Stepwise approach used in our pipeline, while the lower section illustrates the Simultaneous approach used for comparison. In the Stepwise method, text generation and text classification are separated into two distinct steps, whereas the Simultaneous approach combines both tasks in a single prompt.

A.1. Ensemble Voting Prompts for Filler Detection

Utterance	Classification
you were saying like when you came up to your room it was easier to just sort of like here the thought is there and then I'm gonna do my thing.	Filler
Probably not as relevant now because I don't have a job, but also I think relevant because like, then I won't get out of the house.	Non-filler
And I'd like to do more of that, and I have observed myself thinking, well, it's going to be a whole lot easier for me to restart this whole thing than to go back to where I was.	Non-filler
So, homework were you able to do any, like, did you essentially the worksheet?	Filler
all it would be a hold.	Filler

Table A.1.: **Examples used as Input for Prompts detecting Filler-Utterances:** In this Table, we see the examples used in the prompts from Model 2 and Model 3 to detect Filler Sentences in the ensemble Voting approach.

Prompt for Binary Classification Model 1

A filler sentence in a counseling session is a statement used to fill a pause or create space in the dialogue, often without significant content. **Task:** Classify a given sentence as a filler or a not_filler. In addition,

you will be provided with the context before and after the given sentence. The output format should be one word only: "Filler" or "Not_filler". Do not add explanations, notes, punctuation, or extra text.

It is part of a bigger dialogue. For better context, here is the surrounding dialogue: Dialogue:

`{context_before}`

`{process_text}` (our main sentence)

`{context_after}`

Classify the sentence that is marked as (our main sentence) as a filler or a not_filler. `{process_text}`=>

Figure A.2.: **Prompt Template for Model 1:** The prompt includes a task description specifying the objective, along with the psychological process to be classified. It also incorporates contextual information from six utterances before and after the target instance. The model outputs either *not_filler* or *filler* as the final classification.

Prompt for Binary Classification Model 2

A filler sentence in a counseling session is a statement used to fill a pause or create space in the dialogue,

often without significant content. **Task:** Classify a given sentence as a filler or a not_filler. In addition,

you will be provided with the context before and after the given sentence. The output format should be

one word only: "Filler" or "Not_filler". Do not add explanations, notes, punctuation, or extra text.
A few examples:

- "you were saying like when you came up to your room it was easier to just sort of like here the thought is there and then I'm gonna do my thing." => Filler
- "Probably not as relevant now because I don't have a job, but also I think relevant because like, then I won't get out of the house." => Not_filler
- "And I'd like to do more of that, and I have observed myself thinking, well, it's going to be a whole lot easier for me to restart this whole thing than to go back to where I was." => Not_filler
- "all it would be a hold." => Filler

It is part of a bigger dialogue. For better context, here is the surrounding dialogue:

Dialogue:

`{context_before}`

`{process_text}` (our main sentence)

`{context_after}`

Classify the sentence that is marked as (our main sentence) as a filler or a not_filler. `{process_text}` =>

Figure A.3.: **Prompt Template for Model 2:** This prompt includes a task description, the psychological process to be classified, contextual information from six preceding and following utterances, and illustrative examples of both *filler* and *not_filler* sentences. The model returns either *filler* or *not_filler* as the final output.

Prompt for Binary Classification Model 3

A filler sentence in a counselling session is a statement used to fill a pause or create space in the dialogue, often without significant content.

Task: Classify a given sentence as "Filler" or "Not_filler". The output format should be one word only: "Filler" or "Not_filler". Do not add explanations, notes, punctuation, or extra text.

- "you were saying like when you came up to your room it was easier to just sort of like here the thought is there and then I'm gonna do my thing." => Filler
- "Probably not as relevant now because I don't have a job, but also I think relevant because like, then I won't get out of the house." => Not_filler
- "And I'd like to do more of that, and I have observed myself thinking, well, it's going to be a whole lot easier for me to restart this whole thing than to go back to where I was." => Not_filler
- "all it would be a hold." => Filler

Classify the sentence as a filler or a not_filler.

`{process_text}` =>

Figure A.4.: **Prompt Template for Model 3:** The prompt includes a task description specifying the objective, along with the psychological process to be classified. It also incorporates examples of filler and not_filler sentences but not contextual information. The model outputs either *not_filler* or *filler* as the final classification.

A.2. Generating Clinically Meaningful Themes and Classification

A.2.1. Stepwise Prompt First Round

First Round Prompt for Generating Clinically Meaningful Themes

You are a clinical psychologist analyzing a therapy session transcript.
Definitions:

- **Process:** A dynamic psychological mechanism that influences emotions, thoughts, and behavior, affecting mental health and therapy outcomes.

Input:

- Transcript of a therapy session dialogue.
- List of psychological processes.

Your Goal:

1. Review the provided dialogue for overall context. Do not extract utterances from the dialogue.
2. Generate clinically meaningful themes based on the listed psychological processes.
3. Consider the **freq** (X) for each process as an indicator of how often it appeared in a similar form within the dialogue.
    - A higher frequency suggests the process or theme is more important.
4. Do not return labels such as "new label 1" or "unknown topic"
5. Provide only the labels, no explanations.

Guidelines:

- Themes must be **specific and descriptive**, capturing clinically relevant themes.
- Themes should be distinct and cover unique aspects of the listed processes.
- Avoid generic or overly broad themes.
- Do not extract or list direct quotes from the dialogue. The transcript is only for contextual understanding.

Return in JSOn format like this:

```
{lables: ["lable name", "lable name"] }
```

Transcript: `{prompt_str}`
Listed Process: `{formatted_data_process_data}`
Do not include any examples or inspiration themes in the output. Now, generate the clinically meaningful themes based on the transcript and the listed processes without explanation! """

Figure A.5.: **First Round Prompt for Generating Clinically Meaningful Themes:** This Figure illustrates the prompt used in the first step of the stepwise approach. The model is provided with a definition of a psychological process and is prompted to interpret clinically meaningful themes. Additionally, goals and guidelines are included to help steer the model toward accurate and relevant outputs.

First Round Prompt for Clinical Process Classification

Task: You are a clinical psychologist categorizing psychological processes based on labels
Definitions:

- **Process:** A dynamic psychological mechanism influencing emotions, thoughts, and behavior.
- **Label:** A clinically meaningful theme representing psychological processes.

Your Goal:

1. Use the therapy session transcript as background context inly to understand the session's themes.
2. Classify each listed psychological process under one or more of the provided labels based on their thematic relevance
3. Do not assign a process to multiple labels unless it is highly relevant for that label.
4. Ensure classification accurately reflects clinical relevance

Guidelines:

- Each process must be assigned to at least one label.
- Do not introduce new labels or processes
- Return the results strictly in the JSON format below

Return JSON Fomrat

```
{{
"Theme 1": {{
    "Title": "[Theme Title]",
    "Processes": [
    {{"Process": "Process 1", "Frequency": X}},
    {{"Process": "Process 2", "Frequency": X}}
    ]
}},
"Theme 2": {{
    "Title": "[Theme Title]",
    "Processes": [
    {{"Process": "Process 1", "Frequency": Z}},
    {{"Process": "Process 2", "Frequency": N}}
    ]
}}
}}
```

Generated Labels: `{generated_labels}`
Therapy Session Transcript (Context Only!): `{prompt_str}`
List of Processes with Frequency (To Be Classified): `{formatted_data_process_data}`

Figure A.6.: **First Round Prompt for Classification of Processes Within Clinically Meaningful Themes:** This figure illustrates the prompt used in the second step of the stepwise approach. The model is provided with definitions—such as what constitutes a psychological process and the meaning of the provided labels. It also receives classification guidelines, outlining key considerations for making accurate and consistent labeling decisions.

A.2.2. Stepwise Prompt Second Round

Second Round Prompt for Generating Clinically Meaningful Themes

You are a clinical psychologist analyzing a therapy session transcript.
Task: Generate clinically meaningful themes
Definitions:

- **Process:** A dynamic psychological mechanism that influences emotions, thoughts, and behavior, affecting mental health and therapy outcomes.
- **Clinically meaningful theme:** A theme represents a meaningful pattern across psychological processes that reflects underlying functions or dynamics within the client's experience.

Input:

- Transcript of a therapy session dialogue.
- List of psychological processes.

Your Goal:

1. Review the provided dialogue for overall context. Do not extract utterances from the dialogue.
2. Generate clinically meaningful themes based on the listed psychological processes.
3. Consider the **freq** (X) for each process as an indicator of how often it appeared in a similar form within the dialogue.
    - A higher frequency suggests the process or theme is more important.

Guidelines:

- Themes must be **specific and descriptive**, capturing clinically relevant themes.
- Themes should be distinct and cover unique aspects of the listed processes.
- Avoid generic or overly broad themes.
- Do not return labels such as "new_label_1 or unknown_topic or theme1"
- Return output only as a list in this format: ‘["theme1", "theme2", ...]‘

The following are **inspiration-only** themes to help you understand the type of output we aim for. **Do not reuse, rephrase, or incorporate these into your answer.**
Poor/Invalid Themes (Too broad, external, or unfocused) with explanation of why:

- "Professional Networking and Collaboration" $->$ Focuses on external activities, not internal processes or functions.
- "Breathing and Behavioral Patterns" $->$ Groups physical and behavioral elements without a unifying psychological mechanism.
- "Therapy and Reflection" $->$ Describes the setting or activity it is not a meaningful psychological theme.

- "Decision Making and Consequences" $->$ Overly broad and generic it lacks a shared internal function.
- "Reflection and Understanding" $->$ No specific mechanism or insight about how the processes function together.

Valid Inspiration-Only Themes (Do not reuse):

- "Self confidence tied to competence in work role" $->$ "Impact of aspects of identity (professional role, age) on expectations for self and perceived expectations of others"
- "Evaluating thoughts to identify any cognitive distortions" $->$ "Assessing impact of reframed/alternate thought after identifying cognitive distortion" $->$ "Recognizing the role of thoughts and behaviors in emotional experiences"

Now, begin the task.
Transcript: `{prompt_str}`
Listed Process: `{formatted_data_process_data}`
Do not include any examples or inspiration themes in the output. Now, generate the clinically meaningful themes based on the transcript and the listed processes without explanation! """

Figure A.7.: **Second Round Prompt for Generating Clinically Meaningful Themes:** This Figure illustrates the prompt used in the first step of the stepwise approach in the second round. The model is provided with definitions of both psychological processes and also now clinically meaningful themes. Additionally, goals and guidelines are included to guide the model toward accurate and relevant outputs. This round also introduces qualitative examples illustrating both appropriate and inappropriate theme formulations.

Second Round Prompt for Clinical Process Classification

Task: You are a clinical psychologist classifying psychological processes based on provided themes.
Definitions:

- **Process:** A dynamic psychological mechanism influencing emotions, thoughts, and behavior.
- **Clinically meaningful theme:** It represents a meaningful pattern across psychological processes that reflects underlying functions or dynamics within the client's experience.

Input:

- Includes transcript of a therapy session dialogue
- A list of psychological processes
- Themes were processes should be classified

Your Goal:

1. Review the therapy session transcript for context.
2. Classify each listed psychological process under one or more of the provided themes based on their thematic relevance.
3. Do not assign a process to multiple themes unless it is highly relevant for that theme**.
4. Consider the freq (X) for each process as an indicator of how often it appeared in a similar form within the dialogue.
    - A higher frequency suggests the process or theme is more important.
5. Ensure classification **accurately reflects clinical relevance**.

Guidelines:

- Each process must be assigned to at least one theme.
- Each provided theme must have at least two relevant processes assigned to it.
- Do NOT introduce new processes.

Output Format:

```
{{
"Theme 1": {{
    "Title": "[Theme Title]",
    "Processes": [
    {{"Process": "Process 1", "Frequency": X}},
    {{"Process": "Process 2", "Frequency": X}}
    ]
}},
"Theme 2": {{
    "Title": "[Theme Title]",
    "Processes": [
```

```
        {{"Process": "Process 1", "Frequency": Z}},
        {{"Process": "Process 2", "Frequency": N}}
        ]
    }}
    }}
```

Therapy Session Transcript (Context Only!): `{prompt_str}`
List of Processes with Frequency (To Be Classified): `{formatted_data_process_data}`
Themes: `{generated_labels}`
Classify all of the given processes into one or more of the provided themes. Do not provide any explanation or modification to the provided data.

Figure A.8.: **Second Round Prompt for Classification of Processes Within Clinically Meaningful Themes:** This figure illustrates the prompt used in the second step of the stepwise approach during the second round. The model is provided with refined definitions, including what constitutes a psychological process and a clear definition of clinically meaningful themes. Additionally, the classification procedure is enhanced through revised goals and restructured guidelines to improve consistency and accuracy in labeling.

A.2.3. Stepwise Prompt Third Round

Third Round Prompt for Generating Clinically Meaningful Themes

You are a clinical psychologist analyzing a therapy session transcript.
Task: Generate clinically meaningful themes in a clear, one-sentence description.
Definitions:

- **Process**: A dynamic psychological mechanism that influences emotions, thoughts, and behavior, affecting mental health and therapy outcomes.
- **Clinically meaningful theme**: A theme represents a meaningful pattern across psychological processes that reflects underlying functions or dynamics within the client's experience.

Input:

- A transcript excerpt from a therapy session (for context only).
- List of psychological processes.

Your Goal:

1. Read the transcript excerpt to understand the emotional and interpersonal context. Do not quote or extract specific utterances.
2. Generate clinically meaningful themes based on the listed psychological processes.

Guidelines for Writing Themes:

- Each theme must be a complete, **short**, concise statement.
  - You will find under "Example:" how good and bad themes look like.
  - Do not start with "The client is.. " or "Exploring ..."
- Avoid generalities, labels that are not clinically informative.
- The **short** sentence must convey a specific psychological function, conflict, or transformation linking the processes in the cluster.
  - A strong theme goes beyond naming a topic.
  - It provides clinical insight into how or why the processes are showing up and what they reflect in the client's inner world.
- Do not simply list or rephrase the processes in the sentence.
- Together, the themes should cover the major psychological content and dynamics discussed in the transcript therapy session.
- Return output **only** as a list in this format: '["theme1", "theme2", ...]'
- **Do not** return themes such as $"new_label_1" or "unknown_topic" or "theme1"$
- **Do not** reuse the provided example themes in our output.

Examples

- "Self confidence tied to competence in work role" → good
- "Professional Networking and Collaboration" → bad
- "Impact of aspects of identity (professional role, age) on expectations for self and perceived expectations of others" → good
- "Emotional regulation and self-compassion" → bad
- "Evaluating thoughts to identify any cognitive distortions" → good
- "Breathing and Behavioral Patterns" → bad
- "Assessing impact of reframed/alternate thought after identifying cognitive distortion" → good
- "Therapy and Reflection" → bad
- "Decision Making and Consequences" → bad
- "Recognizing the role of thoughts and behaviors in emotional experiences" → good
- "Reflection and Understanding" → bad
- "Emotional Intimacy and Relationships" → bad
- "Difficulty with focus and concentration due to trauma and ADHD-like symptoms" → good
- "Tension between value of achievement and family obligations" → good
- "Therapy and Reflection" → bad
- "Decision Making and Consequences" → bad
- "Exploring the Impact of Surprises on Relationships" → bad

Transcript: `{prompt_str}`
Listed Processes: `{formatted_data_process_data}`
Now, generate clinically meaningful themes without additional explanations and no modifications to the provided data. Stick to the Guidelines.

Figure A.9.: **Third Round Prompt for Generating Clinically Meaningful Themes:** This figure illustrates the final prompt used in the first step of the stepwise approach for generating clinically meaningful themes. Key updates include a restructuring of the provided examples, explicit statements on how themes should not begin, guidance against reusing examples, and specific instructions on the proper structure of a theme.

Thrid Round Prompt for Clinical Process Classification

Task: You are a clinical psychologist classifying psychological processes based on provided themes.
Definitions:

- **Process:** A dynamic psychological mechanism influencing emotions, thoughts, and behavior.
- **Clinically meaningful theme**: It represents a meaningful pattern across psychological processes that reflects underlying functions or dynamics within the client's experience.

Input:

- Includes transcript of a therapy session dialogue
- A list of psychological processes
- Themes were processes should be classified

Your Goal:

1. Review the therapy session transcript** for context.
2. **Classify each listed psychological process under one or more of the provided themes** based on their thematic relevance.
3. Do not assign a process to multiple themes unless it is highly relevant for that theme.
4. Ensure classification accurately reflects clinical relevance.

Guidelines for Clustering:

- Use only the processes listed under "Listed Process to be Clustered".
- Every process **must appear at least in one cluster**.
- A process may belong to **multiple clusters** only if it is strongly relevant to each theme.
- Each cluster must:
  - Contain two or more processes.
  - Be grounded in what was clinically significant in the session (i.e., reflect key emotional or psychological dynamics the client is working through).
- Each provided theme must have at least two relevant processes assigned to it.
- Do not put all processes in one cluster
- If you were provided too many Themes like over 15, then use only the first 15 themes.

Output Structure:

```
    json
{{"Theme 1": {{
"Theme": "[Provided Theme A]",
"Processes": [
{{ "Process": "Process 1", "Frequency": X }},
```

```
{{ "Process": "Process 2", "Frequency": X }} ]}},
"Theme 2": {{
"Theme": "[Provided Theme B]",
"Processes": [{{ "Process": "Process 1", "Frequency": Z }},
{{ "Process": "Process 2", "Frequency": N }}]}}
}}
```

Therapy Session Transcript (Context Only!):`{prompt_str}`
List of Processes with Frequency (To Be Classified): `{formatted_data_process_data}`
Provided Themes: `{generated_labels}`
Now, classify all of the given processes into the provided themes. Do not generate any explanation or modification to the provided data. Stick to the Guidelines:

Figure A.10.: **Third Round Prompt for Classification of Processes Within Clinically Meaningful Themes:** This figure illustrates the prompt used in the second step of the stepwise approach during the third round. The model is provided with refined definitions, including what constitutes a psychological process and a clear explanation of clinically meaningful themes. The classification procedure has been further enhanced through updated goals and restructured guidelines to support greater consistency and accuracy. Additionally, the prompt now instructs the model to limit classifications to a maximum of 15 themes when an excessive number are detected.

A.2.4. Simultaneous Prompt

Simultaneous Prompt

Task: You are a clinical psychologist analyzing a therapy session transcript.
Definitions:

- **Process:** A dynamic psychological mechanism that influences emotions, thoughts, and behavior, affecting mental health and therapy outcomes.

Input: Includes transcript of a therapy session dialogue and a list of psychological processes.
Your Goal:

1. Review the provided dialogue for overall context. Do **not** extract utterances from the dialogue.
2. Cluster all listed processes into **distinct, clinically meaningful groups** based on thematic relevance.
3. Consider the **freq** (X) for each process as an indicator of how often it appeared in a similar form within the dialogue.
   - A higher frequency suggests the process or theme is more important.
4. Provide **clear, concise titles** for each cluster, reflecting clinically meaningful themes.

Guidelines:

- Use only the processes listed under "Processes to be Clustered".
  - Do not include any new processes from the dialogue.
  - Do not omit any listed processes. Every process *must appear in at least one cluster*.
  - Considering also their frequency
  - Use the full dialogue **only** for context. You do **not** need to label or cluster the dialogue.
  - A process may belong to multiple clusters only if it is strongly relevant to each theme.
    * Avoid excessive duplication—if a process appears in multiple clusters, it must have a distinct, meaningful connection to each.
  - Titles must be **specific and descriptive**, capturing each cluster's essence.
  - Ensure each cluster is meaningfully distinct rather than repeating the same processes under different labels.
  - **Do not alter the original text** of the Processes in any way.

Output Structure:

```
   ‘‘‘json
{{
    "Theme 1": {
    "Title": "[Theme Title]",
        "Processes": [
            {"Process": "Process 1", "Frequency": X },
```

```
            {"Process": "Process 2", "Frequency": X }
        ]
    },
    "Theme 2": {
    "Title": "[Theme Title]",
        "Processes": [
            {"Process": "Process 1", "Frequency": Z },
            {"Process": "Process 2", "Frequency": N }
        ]
    }
}}
```

Figure A.11.: **Simultaneous Prompt:** This figure illustrates the prompt used in the first round of evaluation, where the generation of clinically meaningful themes and the classification of psychological processes are combined within a single prompt.

A.2.5. Baseline Prompt

Baseline Prompt

You are a clinical psychologist and your task is to perform a full, end-to-end clinical reasoning analysis based on a therapy dialogue and a list of psychological processes.

Definitions:

- Psychological Process: A dynamic psychological mechanism that influences emotions, toughts, and behavior.
- Clinically Meaningful Theme: A meaningful pattern across psychological processes that reflects underlying psychological functions or dynamics within the client's experience.
- Process Relationship:
  - Connection: Indicates whether a meaningful relationship exists between two Clinically Meaningful Theme (1 = yes, 0 = no).
  - - Relationship Type:
    - Excitatory as in one process amplifies or reinforces another process. (e.g. Process A leads to Process B, Process A causes Process B)
    - Inhibitory as in one process suppresses or counteracts another process. (e.g. Process A stops Process B, Process A diminishes Process B)
  - Strength of Relationship:
    - "Strong": The two themes are closely and consistently related across contexts. One strongly influences or is inherently linked to the other in psychological functioning.
    - "Moderate": The two themes are meaningfully related, but the connection is less consistent or conditional. They are associated in many situations, but not tightly bound.
    - "Weak": The relationship between the two themes is minimal, indirect, or highly context-dependent. They may co-occur at times, but the link is loose or peripheral.

Input:

- Transcript: A therapy session dialogue (context only, do not extract utterances)
- Psychological Processes: A list of psychological processes

Guidelines for writing Themes:

- Each theme must be a **short, complete, concise sentence** that conveys clinical insight.
- The **short** sentence must convey a specific psychological function, conflict, or transformation linking the processes in the cluster.
- A strong theme goes beyond naming a topic.
- It provides clinical insight into how or why the processes are showing up and what they reflect in the client's inner world.

- **Do not** start the themes with "The client's ...", "Exploring ..." or "I ...""
- **Create new, original themes** based on the input only.

Guidelines for Process Classification:

- Classify all listed psychological process under one or more of the provided themes** based on their thematic relevance.
- Use **only** the processes listed under "Listed Process to be Clustered"
- Every process **must appear at least in one cluster**.
- Each Theme must contain at least two or more processes

Inter-Theme Relationships:

- Given Theme A and Theme B, determine whether or not a relationship exists between two them.
- Use only the Theme to generate these Relationships
- Provide a **brief explanation (a few words)** of the connection between Theme A and Theme B. Avoid restating the themes themselves.
- Analyze the relationship from Theme A to Theme B and separately from Theme B to Theme A. Do not assume the relationship is mutual, treat (A to B) and (B to A) as distinct.+
- Each theme needs at least one strong connection towards it or from it.

Your Goals

1. Theme Generation
2. Process Classification
3. Inter-Theme Relationships

Examples of Themes: (Note: These examples are for understanding style only, you must NOT reuse or adapt them!)

- "Self confidence tied to competence in work role" -> good
- "Professional Networking and Collaboration" -> bad
- "Impact of aspects of identity (professional role, age) on expectations for self and perceived expectations of others" -> good
- "Emotional regulation and self-compassion" -> bad
- "Evaluating thoughts to identify any cognitive distortions" -> good
- "Breathing and Behavioral Patterns" -> bad

- "Assessing impact of reframed/alternate thought after identifying cognitive distortion" -> good
- "Therapy and Reflection" -> bad
- "The client's struggle to articulate their feelings is a manifestation of the unexpressed emotions from their past." -> bad
- "Decision Making and Consequences" -> bad
- "Recognizing the role of thoughts and behaviors in emotional experiences" -> good
- "Reflection and Understanding" -> bad
- "Emotional Intimacy and Relationships" -> bad
- "Difficulty with focus and concentration due to trauma and ADHD-like symptoms" -> good
- "Tension between value of achievement and family obligations" -> good
- "Exploring the Impact of Surprises on Relationships" -> bad
- "The client's self-blame is a coping mechanism for feelings of overhelm and anxiety" -> bad

Output Format:

```
   {
  "classified_processes": {
    "Theme 1": { "Title": "Theme A", "Processes":

    [{"Process": "Process 1"}, {"Process": "Process 3"}] },
    "Theme 2": { "Title": "Theme B", "Processes":

    [{"Process": "Process 2"}] }
  },
  "theme_relationships": [
    { "input_themes": ["Theme A", "Theme B"], "connection": [1], "relationship_type":

    ["excitatory"], "strength_of_relationship": ["strong"], "explanation": "..." },
    { "input_themes": ["Theme B", "Theme A"], "connection": [1], "relationship_type":

    ["inhibitory"], "strength_of_relationship": ["moderate"], "explanation": "..." }
    {input_themes": ["Theme A", "Theme C"], "connection": [0] }
  ]
}
```

Transcript: `{prompt_str}`

Listed Process to be Clustered: `{{formatted_data_process_data}`

Do not reuse the provided examples, generate new Clinically Meaningful Themes. Only return your response as single complete JSON file using the structure above. Do not include commentary or multiple JSON blocks. Now begin your task:

Figure A.12.: **Prompt Baseline:** This Figure presents the prompt used as a baseline for comparison against our end-to-end pipeline. The prompt guides the generation of clinically meaningful themes, the classification of psychological processes under these themes, and the articulation of expressive relationships between them. It is provided with the same information as the pipeline, including the transcribed therapy session and the associated psychological processes.

A.3. Generating Clinically Meaningful Connections

Prompt for Generating Clinically Meaningful Connections

You are a psychologist and your task is to identify relationships between psychological processes.
Task: Your task is to determine whether a relationship exists between **Process A** and **Process B**, and to generate information on the nature of that relationship.
Definitions:

- **Psychological Process:** A psychological process refers to the mental activities and mechanisms through which individuals perceive, think, feel, and respond to their environment, influencing (overt) behavior, cognition, affect, attention, sense of self, motivation.
- **Connection:**
  - 1: there is a relationship
  - 0: there is no relationship
- **Type of relationships between two psychological processes:**
  - Excitatory as in one process amplifies or reinforces another process. (e.g. Process A leads to Process B, Process A causes Process B)
  - Inhibitory as in one process suppresses or counteracts another process. (e.g. Process A stops Process B, Process A diminishes Process B)
- **Strength of Relationship:**
  - 1) "Strong": The two processes are closely and consistently related across contexts. One strongly influences or is inherently linked to the other in psychological functioning.
  - 2) "Moderate": The two processes are meaningfully related, but the connection is less consistent or conditional. They are associated in many situations, but not tightly bound.
  - 3) "Weak": The relationship between the two processes is minimal, indirect, or highly context-dependent. They may co-occur at times, but the link is loose or peripheral.

Input: Two psychological processes
Guidelines:

- Given process A and process B, determine whether or not a relationship exists between two processes.
- Provide a **brief explanation (a few words)** of the connection between Process A and Process B. Avoid restating the processes themselves

Output Structure: If relationships exist from process A to process B, return a structured output containing details for each identified relationship:

```
json
{{"relationship":
[{{ "input_processes": ["Process A", "Process B"],
"connection": [1],
"relationship_type": ["excitatory", "inhibitory"],
"strength_of_relationship": ["strong", "moderate", "weak"],
"explanation":
"A concise natural language explanation of why this relationship exists."}}]
}}
```

If no relationships exist from Process A to Process B, return:

```
 json
{{"relationship":
[{{ "input_processes": ["Process A", "Process B"],
"connection": [0]}}]
}}
```

Process A: {process_1_str}
Process B: {process_2_str}
Now, generate the clinically meaningful relationship between the provided Processes in the specified format. Only output the relationship, and nothing else.

Figure A.13.: **Full Version Prompt for Generating Clinically Meaningful Connections:** This figure presents the prompt used to generate clinically meaningful connections. The model is provided with definitions for each attribute of the relationship, a description of what constitutes a psychological process, and clear instructions on how the explanation of the relationship should be structured.

B. Appendix B

B.1. Detailed Word Count from Patient and Therapist

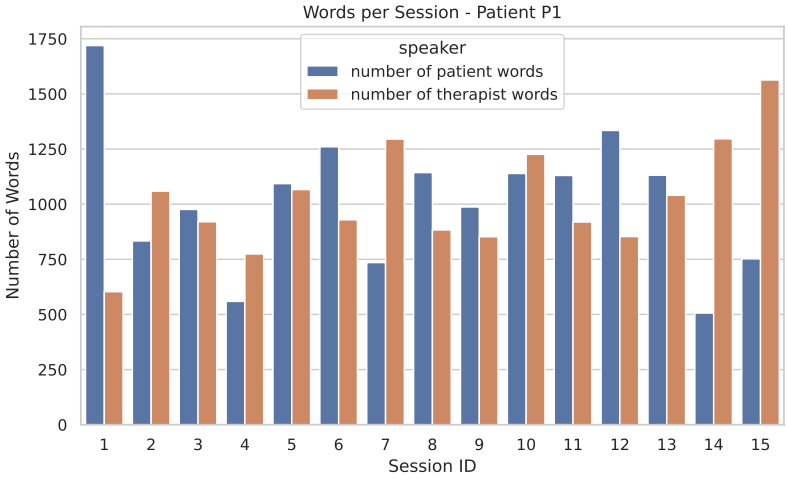


Figure B.1.: **Word Count Across Session - Patient P1 and Therapist:** This figure shows the Word Count for Patient P1 and the therapist across all sessions, displayed separately. The patient is represented in blue and the therapist in orange.

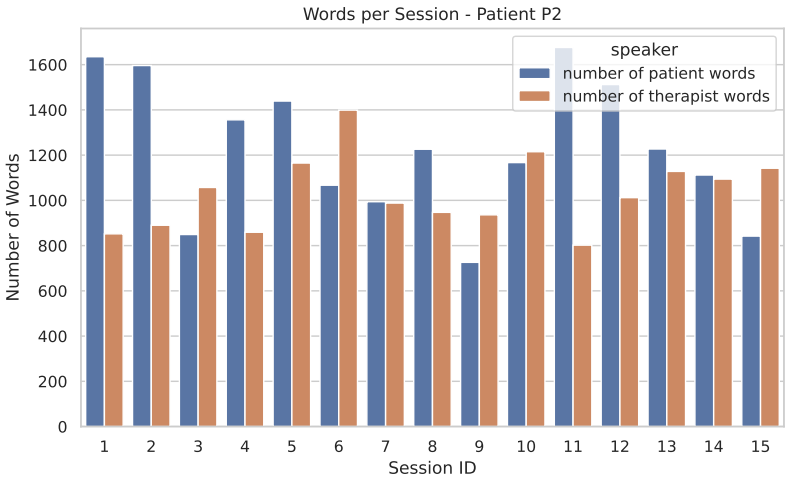


Figure B.2.: **Word Count Across Session - Patient P2 and Therapist:** This figure shows the Word Count for Patient P2 and the therapist across all sessions, displayed separately. The patient is represented in blue and the therapist in orange.

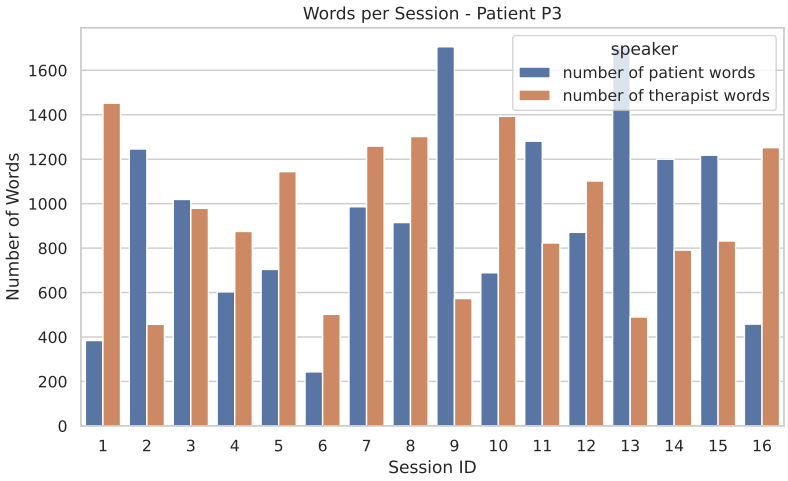


Figure B.3.: **Word Count Across Session - Patient P3 and Therapist:** This figure shows the Word Count for Patient P3 and the therapist across all sessions, displayed separately. The patient is represented in blue and the therapist in orange.

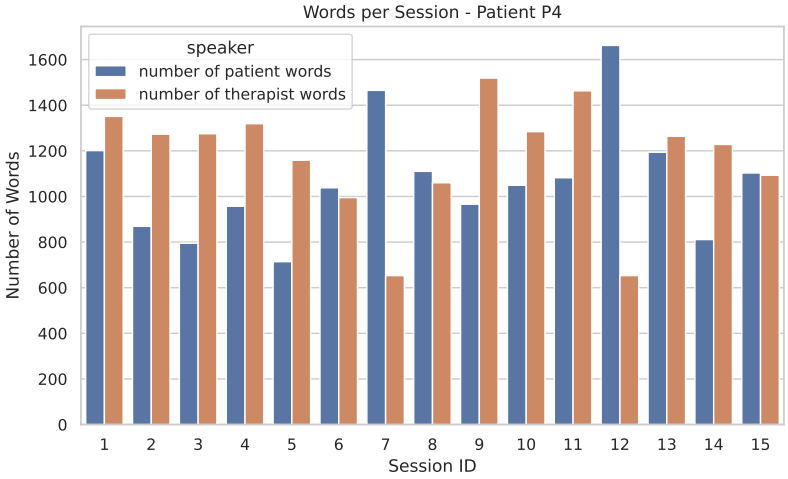


Figure B.4.: **Word Count Across Session - Patient P4 and Therapist:** This figure shows the Word Count for Patient P4 and the therapist across all sessions, displayed separately. The patient is represented in blue and the therapist in orange.

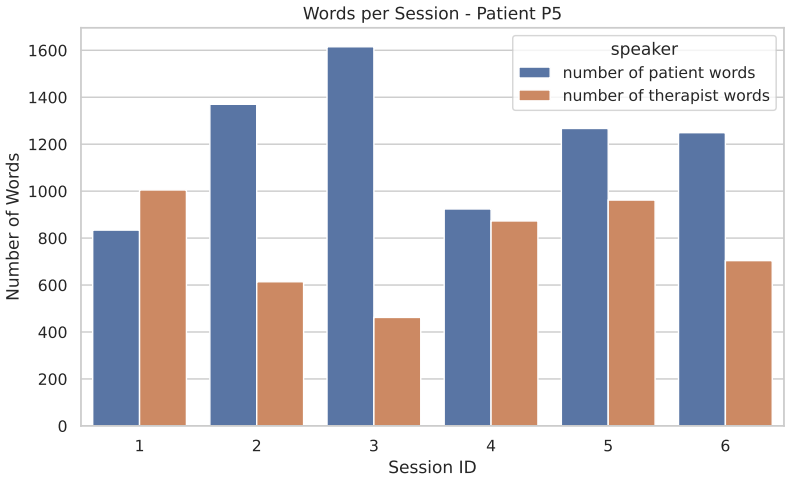


Figure B.5.: **Word Count Across Session - Patient P5 and Therapist:** This figure shows the Word Count for Patient P5 and the therapist across all sessions, displayed separately. The patient is represented in blue and the therapist in orange.

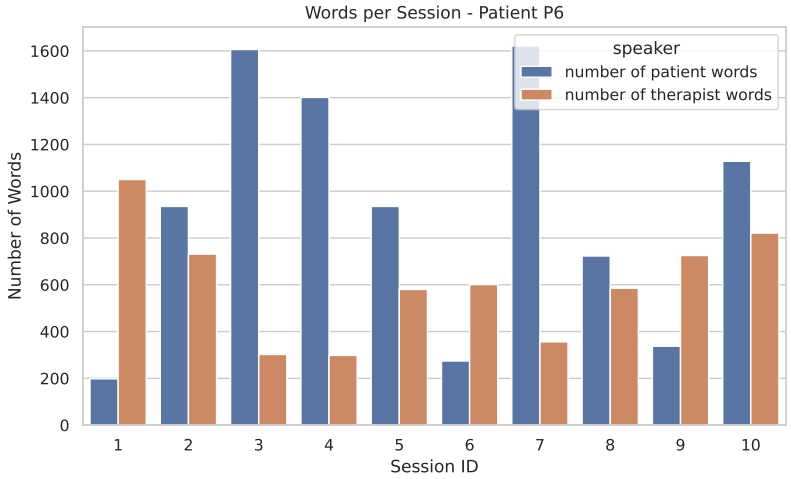


Figure B.6.: **Word Count Across Session - Patient P6 and Therapist:** This figure shows the Word Count for Patient P6 and the therapist across all sessions, displayed separately. The patient is represented in blue and the therapist in orange.

B.2. Detailed Speaking Duration from Patient and Therapist

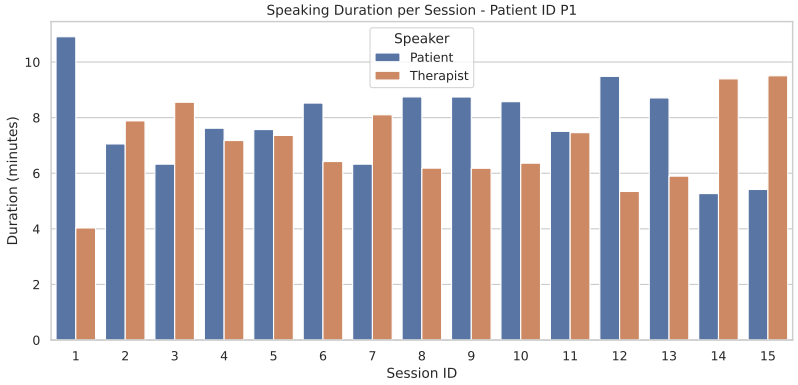


Figure B.7.: **Speaking Duration Across Sessions - Patient P1 and Therapist:** This figure shows the speaking duration in minutes for Patient P1 and the therapist across all sessions, displayed separately. The patient is represented in blue and the therapist in orange.

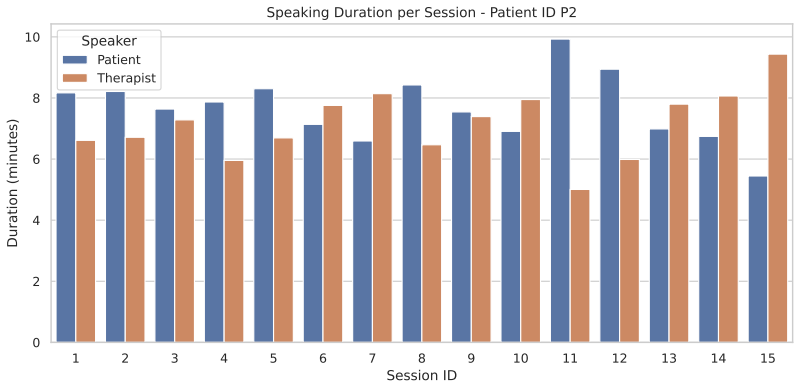


Figure B.8.: **Speaking Duration Across Sessions - Patient P2 and Therapist:** This figure shows the speaking duration in minutes for Patient P2 and the therapist across all sessions, displayed separately. The patient is represented in blue and the therapist in orange.

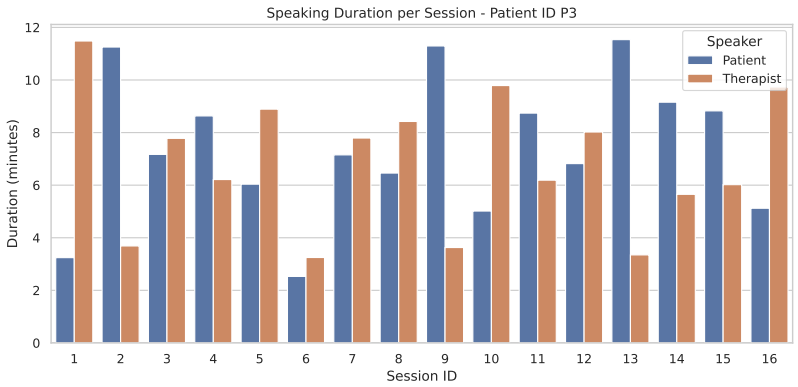


Figure B.9.: **Speaking Duration Across Sessions - Patient P3 and Therapist:** This figure shows the speaking duration in minutes for Patient P3 and the therapist across all sessions, displayed separately. The patient is represented in blue and the therapist in orange.

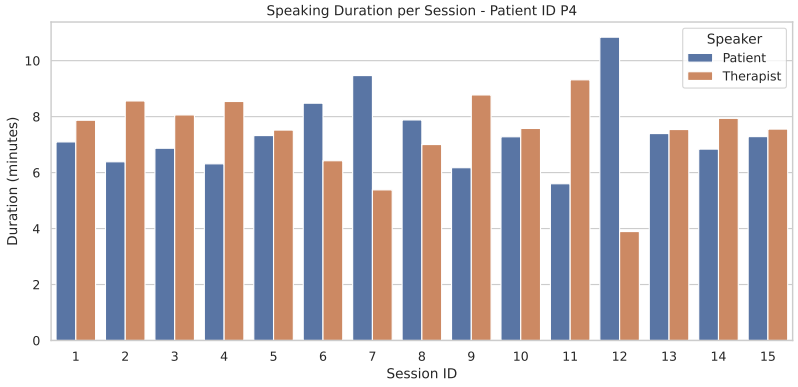


Figure B.10.: **Speaking Duration Across Sessions - Patient P4 and Therapist:** This figure shows the speaking duration in minutes for Patient P4 and the therapist across all sessions, displayed separately. The patient is represented in blue and the therapist in orange.

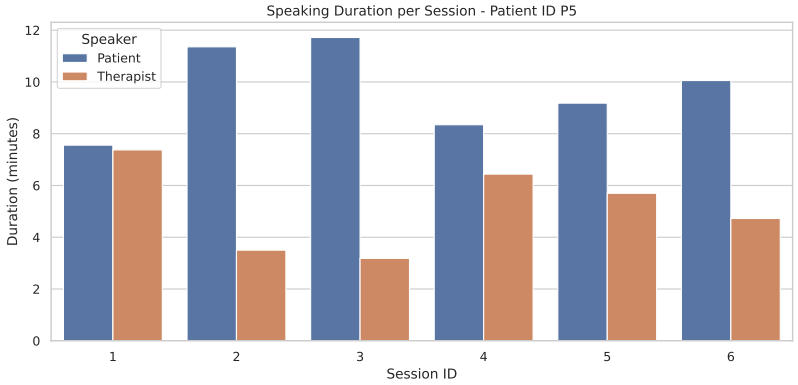


Figure B.11.: **Speaking Duration Across Sessions - Patient P5 and Therapist:** This figure shows the speaking duration in minutes for Patient P5 and the therapist across all sessions, displayed separately. The patient is represented in blue and the therapist in orange.

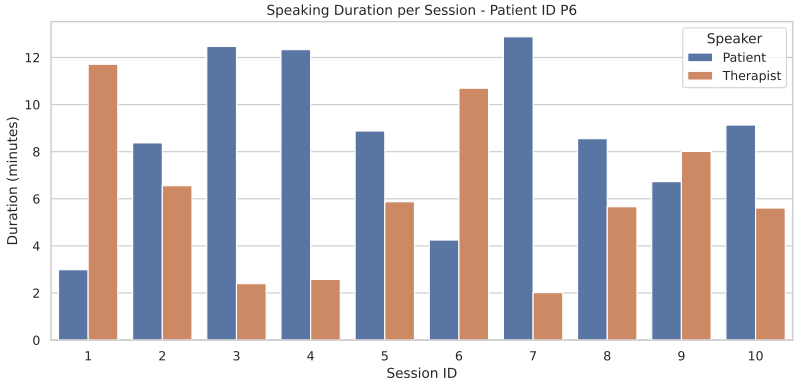


Figure B.12.: **Speaking Duration Across Sessions - Patient P6 and Therapist:** This figure shows the speaking duration in minutes for Patient P6 and the therapist across all sessions, displayed separately. The patient is represented in blue and the therapist in orange.

B.3. Detailed Utterances Count from Patient and Therapist

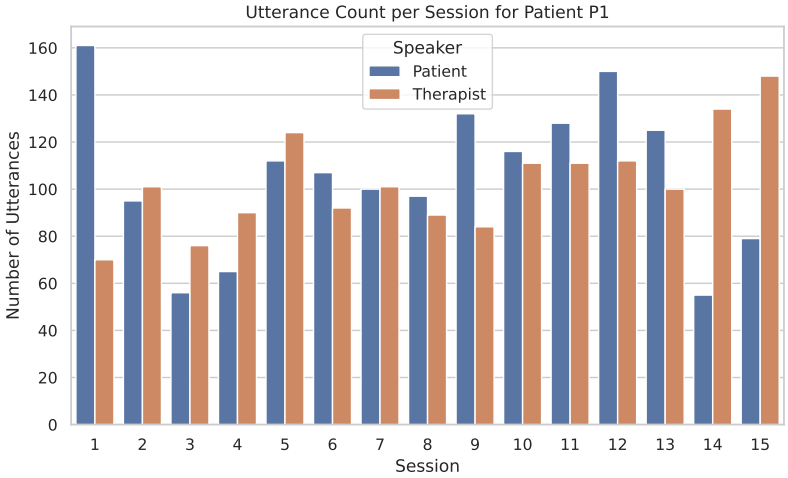


Figure B.13.: **Utterance Count Across Sessions - Patient P1 and Therapist:** This figure shows the number of utterances per session for Patient P1 and the therapist. The utterance counts are displayed separately for each session, with the patient shown in blue and the therapist in orange.

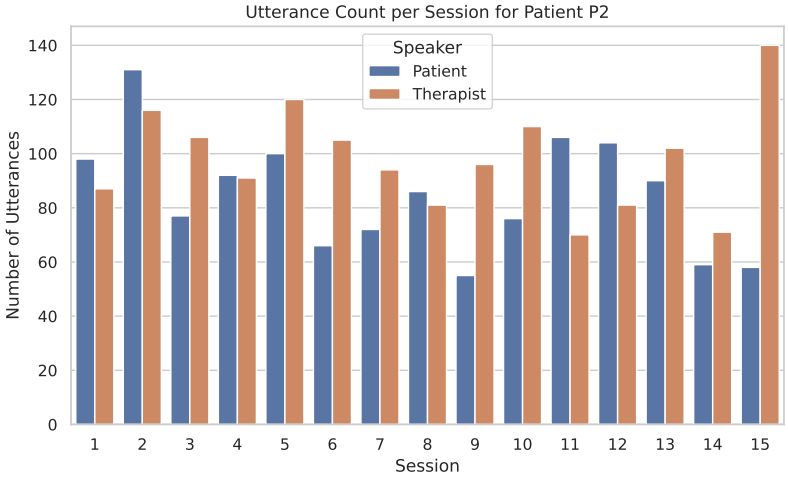


Figure B.14.: **Utterance Count Across Sessions - Patient P2 and Therapist:** This figure shows the number of utterances per session for Patient P2 and the therapist. The utterance counts are displayed separately for each session, with the patient shown in blue and the therapist in orange.

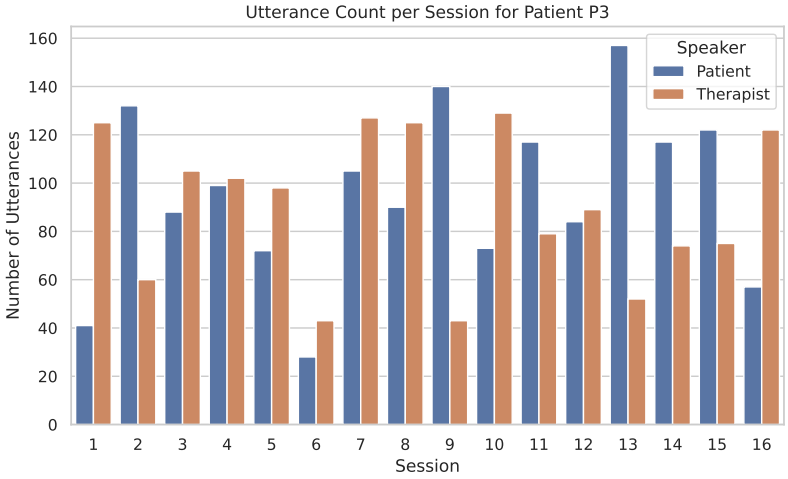


Figure B.15.: **Utterance Count Across Sessions - Patient P3 and Therapist:** This figure shows the number of utterances per session for Patient P3 and the therapist. The utterance counts are displayed separately for each session, with the patient shown in blue and the therapist in orange.

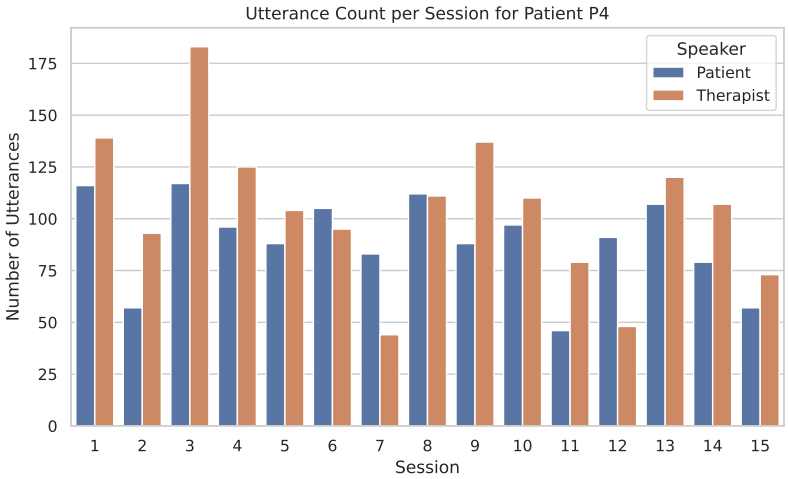


Figure B.16.: **Utterance Count Across Sessions - Patient P4 and Therapist:** This figure shows the number of utterances per session for Patient P4 and the therapist. The utterance counts are displayed separately for each session, with the patient shown in blue and the therapist in orange.

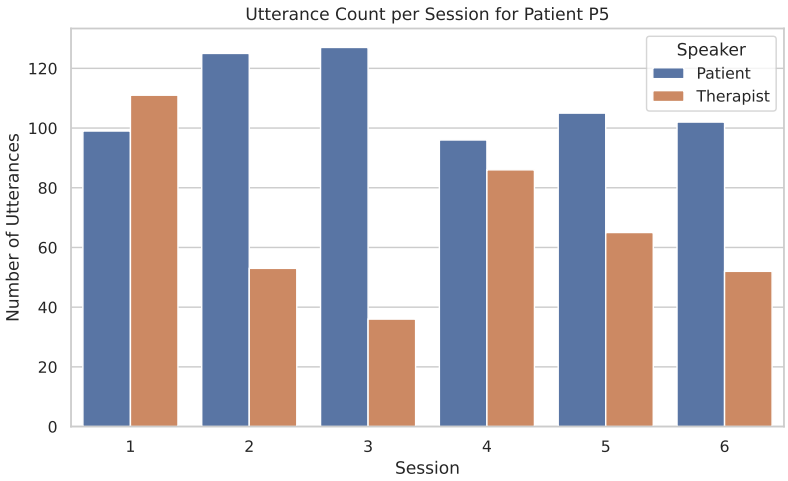


Figure B.17.: **Utterance Count Across Sessions - Patient P5 and Therapist:** This figure shows the number of utterances per session for Patient P5 and the therapist. The utterance counts are displayed separately for each session, with the patient shown in blue and the therapist in orange.

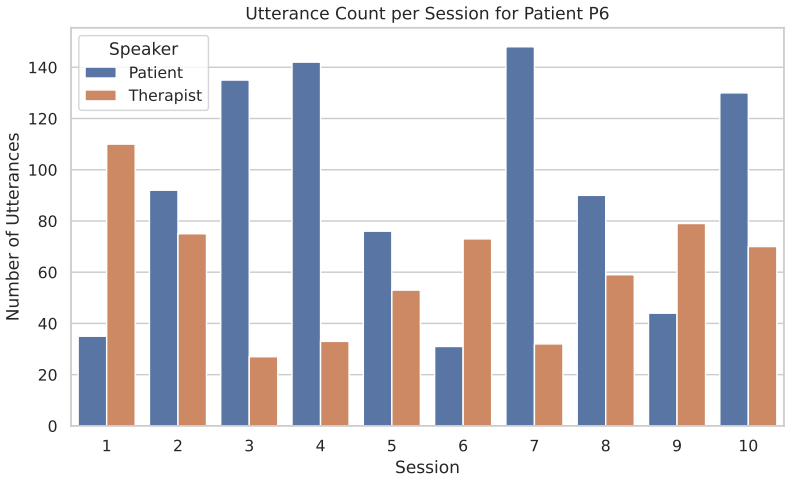


Figure B.18.: **Utterance Count Across Sessions - Patient P6 and Therapist:** This figure shows the number of utterances per session for Patient P6 and the therapist. The utterance counts are displayed separately for each session, with the patient shown in blue and the therapist in orange.

B.4. Detailed Process Counts from Patient and Therapist

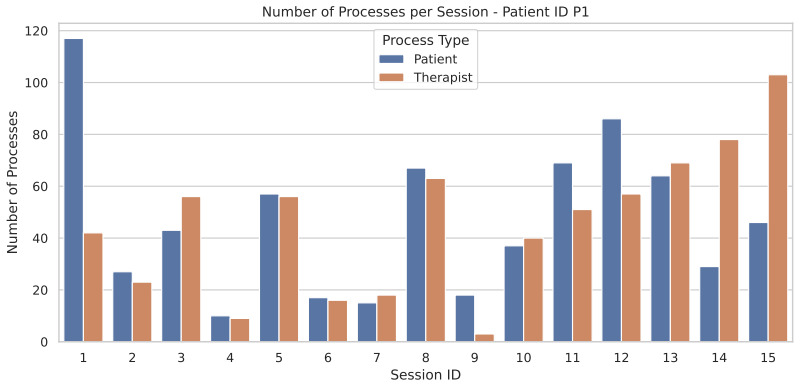


Figure B.19.: **Psychological Processes Across Sessions - Patient P1 and Therapist:** This figure shows the number of psychological processes per session for Patient P1 and the therapist. The process counts are displayed separately for each session, with the patient shown in blue and the therapist in orange.

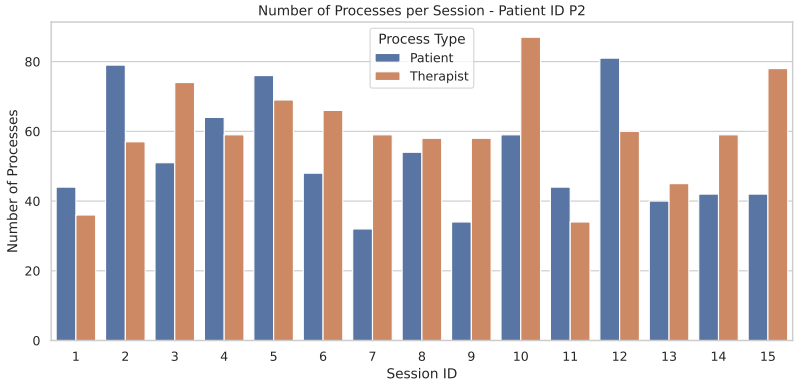


Figure B.20.: **Psychological Processes Across Sessions - Patient P2 and Therapist:** This figure shows the number of psychological processes per session for Patient P2 and the therapist. The process counts are displayed separately for each session, with the patient shown in blue and the therapist in orange.

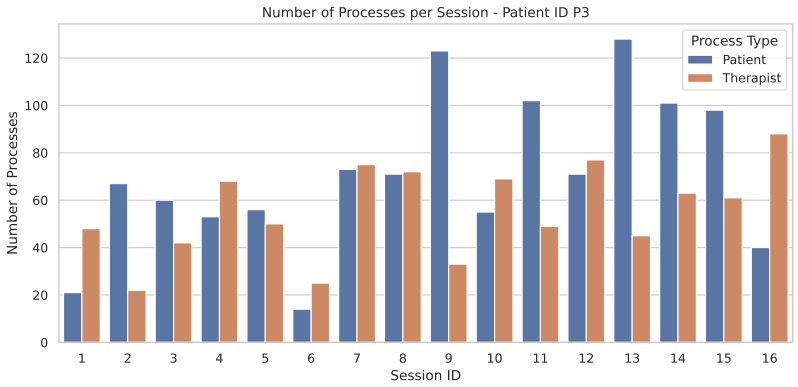


Figure B.21.: **Psychological Processes Across Sessions - Patient P3 and Therapist:** This figure shows the number of psychological processes per session for Patient P3 and the therapist. The process counts are displayed separately for each session, with the patient shown in blue and the therapist in orange.

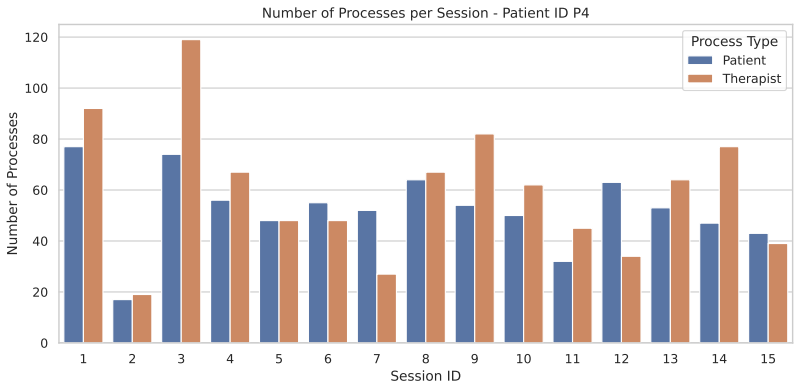


Figure B.22.: **Psychological Processes Across Sessions - Patient P4 and Therapist:** This figure shows the number of psychological processes per session for Patient P4 and the therapist. The process counts are displayed separately for each session, with the patient shown in blue and the therapist in orange.

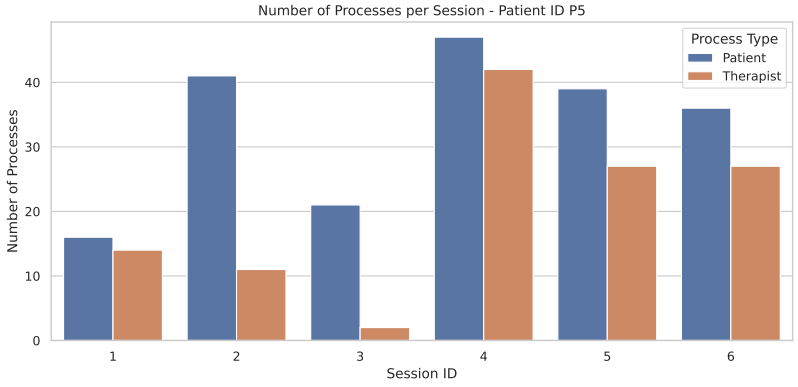


Figure B.23.: **Psychological Processes Across Sessions - Patient P5 and Therapist:** This figure shows the number of psychological processes per session for Patient P5 and the therapist. The process counts are displayed separately for each session, with the patient shown in blue and the therapist in orange.

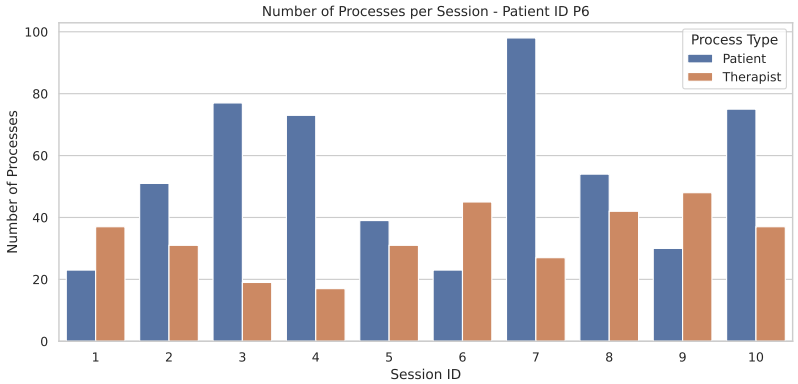


Figure B.24.: **Psychological Processes Across Sessions - Patient P6 and Therapist:** This figure shows the number of psychological processes per session for Patient P6 and the therapist. The process counts are displayed separately for each session, with the patient shown in blue and the therapist in orange.

B.5. Detailed Dimension Distribution from Patient and Therapist

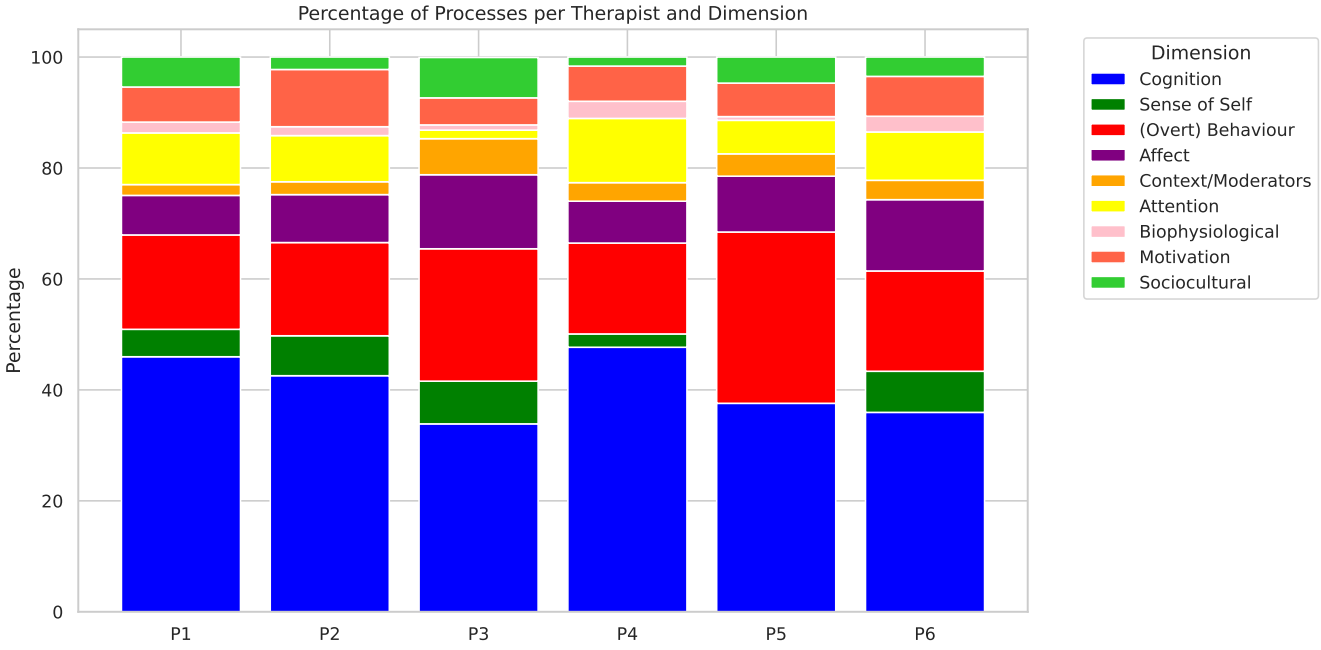


Figure B.25.: **Total Therapist Dimension Percentages:** This figure shows the percentage distribution of psychological processes across all therapy sessions for each therapist, based on the predefined psychological dimensions. Each bar sums to 100%, representing how a therapist's processes are distributed across the different dimensions. Note that a single psychological process may be associated with multiple dimensions.

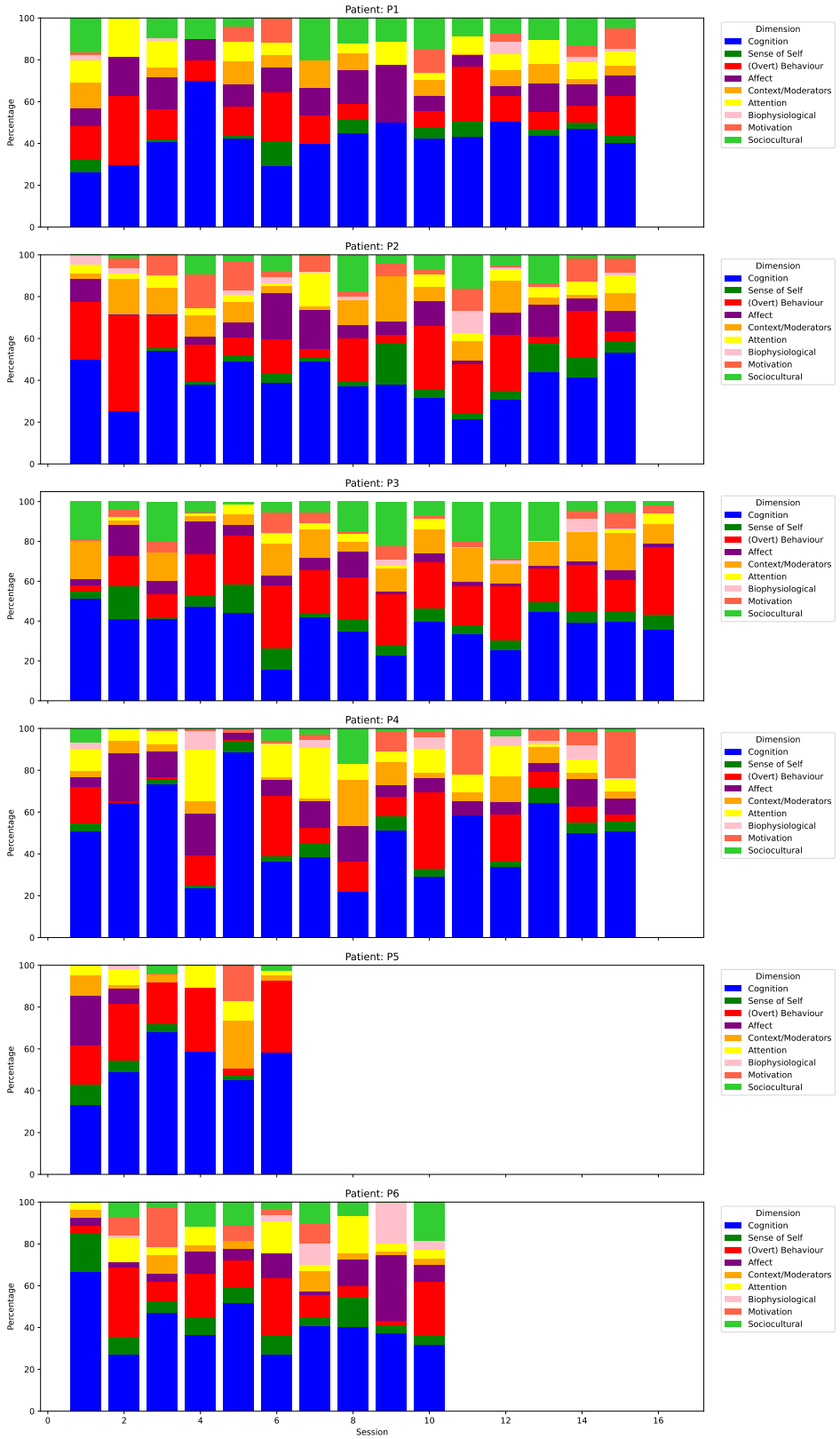


Figure B.26.: **Detailed Distribution of Psychological Process Dimensions Across Sessions (Patient):** This figure shows the percentage distribution of all psychological processes by their assigned dimensions in each session for the patient. Each bar represents one session, illustrating which labels were applied to the processes within that session.

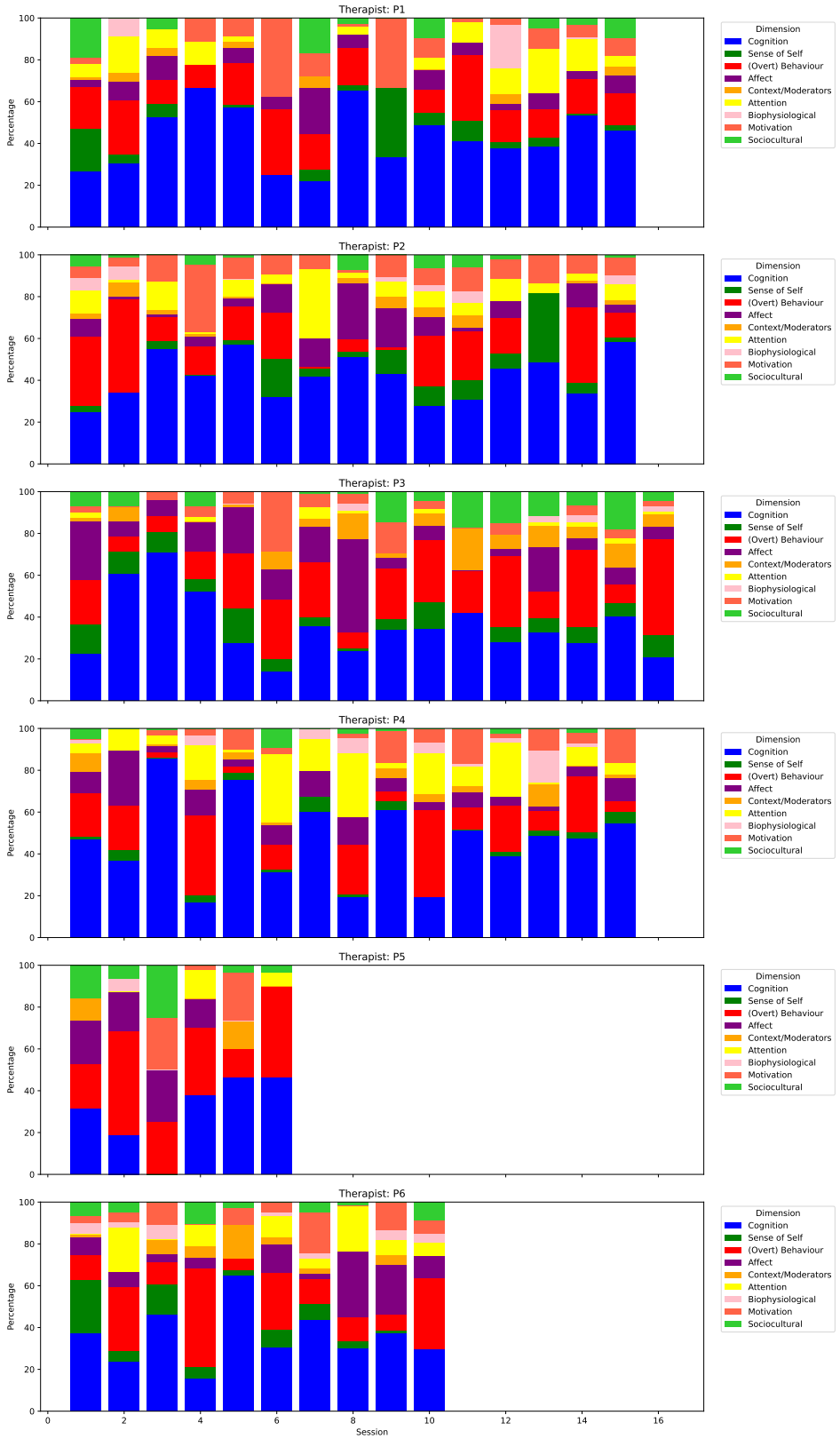


Figure B.27.: **Detailed Distribution of Psychological Process Dimensions Across Sessions (Therapist):** This figure shows the percentage distribution of all psychological processes by their assigned dimensions in each session for the therapist. Each bar represents one session, illustrating which labels were applied to the processes within that session.

C. Appendix C

C.1. Formula Scores

The formula used to calculate **Total Score**, **Insightfulness** and **Trustworthiness** are:

$$\begin{aligned}\textbf{Total Score} = {} & (0.25 \times \text{Clinical Relevance}) + (0.20 \times \text{Novelty}) + (0.15 \times \text{Usefulness}) + \\ & (0.10 \times \text{Specificity}) + (0.10 \times \text{Coverage}) + (0.08 \times \text{Completeness}) + \\ & (0.07 \times \text{Intuder}) + (0.05 \times \text{Redundancy})\end{aligned} \tag{C.1}$$

$$\begin{aligned}\text{Insightfulness} = {} & 0.45 \times \text{Clinically Relevant} + \\ & 0.35 \times \text{Novelty} + 0.20 \times \text{Usefulness}\end{aligned} \tag{C.2}$$

$$\begin{aligned}\textbf{Trustworthiness} = {} & 0.275 \times \text{Specificity} + 0.275 \times \text{Coverage} + \\ & 0.20 \times \text{Completeness} + 0.15 \times \text{Intruder} + 0.10 \times \text{Redundancy}\end{aligned} \tag{C.3}$$

C.2. Example of Self-Made Agreements Calculation

Two evaluators rate three items using the scale **1–3**:

- **Evaluator A (Ground Truth):** For Session x: [3, 3, 3] $\bar{x_{a}}_A = 3.0$ and for Session Y: [3, 1, 2] $\bar{y}_A = 2.00$
- **Evaluator B (Prediction):** For Session x: [2, 3, 3] $\bar{x}_B = 2.67$ and for Session Y: [1, 1, 2] $\bar{y}_B = 1.33$

For **label-based agreement**, we compare the rounded means for each session:

- Session X: round(3.0) = round$(2.67) = 3$
- Session Y: round$(2.0) \neq$ round(1.33)
- So the label-based agreement is 0.5

For **hard agreement**, we check for exact equality: $3.0 \neq 2.67$, and $2.0 \neq 1.33 \rightarrow$ agreement is 0

Selection agreement is computed item-by-item: (3 vs 2) is a mismatch, (3 vs 3) is a match, (3 vs 3) is a match, (3 vs 1) is a mismatch, (1 vs 1) is a match and (2 v 2) is a match. This gives $4/6 = 0.67$.

For the F1 agreement, we treat the scores assigned by Evaluator A as the true labels and those from Evaluator B as the predicted labels. In this case, Evaluator A assigned the scores [3,3,3,3,1,2] [3,3,3,3,1,2], while Evaluator B assigned [2,3,3,1,1,2] [2,3,3,1,1,2]. We interpret each score (1, 2, or 3) as a distinct class label and compute the F1 score for each class individually, averaging the results to obtain a macro F1 score. For class 1, there is one true instance and two predicted instances, with one correct match, resulting in a precision of 0.5, recall of 1.0, and an F1 score of approximately 0.67. For class 2, there is again one true instance and two predicted instances, with one correct match, leading to the same F1 score of approximately 0.67. For class 3, there are four true instances and two predicted instances, with two correct matches, yielding a precision of 1.0, recall of 0.5, and an F1 score of approximately 0.67. Taking the average of these three class-specific F1 scores gives a final macro F1 agreement score of approximately 0.67.

C.3. Detailed Evaluation Results

C.3.1. Evaluation of Generated Clinically Meaningful Themes and Classification per Evaluator

Metric	Simultaneous R1	Stepwise R1	Stepwise R2	Stepwise R3
Clinically Relevant	1.94	1.92	1.98	1.97
Novelty	1.00	2.00	1.71	1.94
Usefulness	1.25	1.75	1.64	2.00
Specificity	1.50	2.33	2.07	2.38
Coverage	2.00	2.00	2.21	2.00
Completeness	2.25	2.00	2.07	2.42
Intruder	2.67	2.25	2.51	2.01
Redundancy	2.25	2.50	2.14	1.93
Insightfulness	1.41	1.87	1.79	1.99
Trustworthiness	2.04	2.18	2.18	2.18
Total Score	1.70	2.02	1.96	2.06

Table C.1.: **Domain Expert KA Ratings for Theme Generation Methods and Classification Across Rounds:** This table presents expert ratings for the Simultaneous and Stepwise theme generation and assignment methods across multiple evaluation rounds. Scores range from 1 (best) to 3 (worst). The **Total Score** reflects the overall quality ranking provided by domain experts. **Insightfulness** is calculated based on the first three metrics, while **Trustworthiness** is derived from the remaining metrics.

Metric	Simultaneous R1	Stepwise R1	Stepwise R2	Stepwise R3
Clinically Relevant	1.74	1.85	1.77	2.21
Novelty	1.83	1.75	1.75	2.43
Usefulness	1.50	1.75	1.58	2.43
Specificity	1.67	2.00	2.00	2.71
Coverage	1.83	2.00	1.75	2.29
Completeness	2.33	2.25	2.17	2.43
Intruder	1.80	1.58	1.67	1.70
Redundancy	2.83	1.00	1.83	1.71
Insightfulness	1.69	1.79	1.72	2.34
Trustworthiness	1.98	1.89	1.90	2.29
Total Score	1.83	1.81	1.79	2.30

Table C.2.: **Domain Expert CA Ratings for Theme Generation Methods and Classification Across Rounds:** This table presents expert ratings for the Simultaneous and Stepwise theme generation and assignment methods across multiple evaluation rounds. Scores range from 1 (best) to 3 (worst). The **Total Score** reflects the overall quality ranking provided by domain experts. **Insightfulness** is calculated based on the first three metrics, while **Trustworthiness** is derived from the remaining metrics.

Metric	Simultaneous R1	Stepwise R1	Stepwise R2	Stepwise R3
Clinically Relevant	2.17	2.30	1.93	2.56
Novelty	2.20	1.80	2.00	2.57
Usefulness	2.20	2.20	1.62	2.43
Specificity	2.20	2.20	1.85	2.86
Coverage	2.00	3.00	1.92	2.57
Completeness	1.00	1.00	2.00	2.06
Intruder	2.29	2.22	2.10	1.74
Redundancy	2.00	1.20	1.69	1.71
Insightfulness	2.19	2.09	1.86	2.46
Trustworthiness	1.90	2.08	1.92	2.34
Total Score	2.07	2.08	1.89	2.43

Table C.3.: **Domain Expert ES Ratings for Theme Generation Methods and Classification Across Rounds:** This table presents expert ratings for the Simultaneous and Stepwise theme generation and assignment methods across multiple evaluation rounds. Scores range from 1 (best) to 3 (worst). The **Total Score** reflects the overall quality ranking provided by domain experts. **Insightfulness** is calculated based on the first three metrics, while **Trustworthiness** is derived from the remaining metrics.

C.3.2. Evaluation of Generated Clinically Meaningful Themes and Classification Results per Patient

Metric	Simultaneous R1	Stepwise R1	Stepwise R2	Stepwise R3
Clinically Relevant	1.88	1.77	1.79	2.29
Novelty	1.75	1.67	1.60	2.75
Usefulness	1.75	1.50	1.40	2.50
Specificity	1.75	1.67	2.00	2.88
Coverage	2.25	2.00	2.00	2.50
Completeness	1.75	2.33	2.40	2.50
Intruder	2.35	1.53	2.58	1.85
Redundancy	2.25	1.33	2.00	1.50
Insightfulness	1.79	1.63	1.63	2.58
Trustworthiness	2.03	1.84	2.17	2.41
Total Score	1.90	1.72	1.85	2.44

Table C.4.: **Domain Expert Ratings for Patient P1 Theme Generation Methods and Classification Across Rounds:** This table presents expert ratings for the Simultaneous and Stepwise theme generation and classification methods across multiple evaluation rounds for the Patient P1. Scores range from 1 (best) to 3 (worst). The **Total Score** reflects the overall quality ranking provided by domain experts. **Insightfulness** is calculated based on the first three metrics, while **Trustworthiness** is derived from the remaining metrics.

Metric	Simultaneous R1	Stepwise R1	Stepwise R2	Stepwise R3
Clinically Relevant	2.28	1.80	2.09	2.14
Novelty	2.00	2.00	2.08	1.67
Usefulness	2.33	2.00	1.92	2.00
Specificity	2.33	2.00	2.15	2.67
Coverage	2.00	2.00	2.15	2.33
Completeness	1.67	2.00	2.23	3.00
Intruder	2.06	2.17	2.00	1.71
Redundancy	2.67	3.00	1.92	1.67
Insightfulness	2.20	1.90	2.03	1.95
Trustworthiness	2.10	2.13	2.12	2.40
Total Score	2.16	2.01	2.07	2.11

Table C.5.: **Domain Expert Ratings for Patient P2 - Theme Generation Methods and Classification Across Rounds:** This table presents expert ratings for the Simultaneous and Stepwise theme generation and classification methods across multiple evaluation rounds for the Patient P2. Scores range from 1 (best) to 3 (worst). The **Total Score** reflects the overall quality ranking provided by domain experts. **Insightfulness** is calculated based on the first three metrics, while **Trustworthiness** is derived from the remaining metrics.

Metric	Simultaneous R1	Stepwise R1	Stepwise R2	Stepwise R3
Clinically Relevant	2.00	2.12	1.74	2.17
Novelty	2.00	2.00	1.67	2.25
Usefulness	2.00	2.00	1.33	2.00
Specificity	2.00	2.50	1.67	2.67
Coverage	2.00	2.00	1.78	2.33
Completeness	2.00	1.50	1.78	2.31
Intruder	1.33	2.00	2.10	1.79
Redundancy	3.00	2.50	2.11	2.00
Insightfulness	2.00	2.08	1.57	2.14
Trustworthiness	2.00	2.09	1.83	2.31
Total Score	2.00	2.07	1.70	2.20

Table C.6.: **Domain Expert Ratings for Patient P3 - Theme Generation Methods and Classification Across Rounds:** This table presents expert ratings for the Simultaneous and Stepwise theme generation and classification methods across multiple evaluation rounds for the Patient P3. Scores range from 1 (best) to 3 (worst). The **Total Score** reflects the overall quality ranking provided by domain experts. **Insightfulness** is calculated based on the first three metrics, while **Trustworthiness** is derived from the remaining metrics.

Metric	Simultaneous R1	Stepwise R1	Stepwise R2	Stepwise R3
Clinically Relevant	2.08	2.38	2.00	2.07
Novelty	1.33	2.00	1.60	2.00
Usefulness	1.33	2.50	1.60	2.33
Specificity	1.33	3.00	2.00	2.33
Coverage	2.00	3.00	2.20	1.89
Completeness	1.67	1.50	1.80	1.83
Intruder	2.27	2.62	1.91	2.04
Redundancy	2.67	1.50	1.80	1.89
Insightfulness	1.63	2.31	1.78	2.11
Trustworthiness	1.86	2.49	1.98	2.02
Total Score	1.74	2.35	1.84	2.07

Table C.7.: **Domain Expert Ratings for Patient P4 - Theme Generation Methods and Classification Across Rounds:** This table presents expert ratings for the Simultaneous and Stepwise theme generation and classification methods across multiple evaluation rounds for the Patient P4. Scores range from 1 (best) to 3 (worst). The **Total Score** reflects the overall quality ranking provided by domain experts. **Insightfulness** is calculated based on the first three metrics, while **Trustworthiness** is derived from the remaining metrics.

Metric	Simultaneous R1	Stepwise R1	Stepwise R2	Stepwise R3
Clinically Relevant	1.60	1.77	1.50	1.93
Novelty	2.00	2.00	1.50	2.33
Usefulness	1.33	1.75	1.50	2.33
Specificity	2.00	2.25	2.00	3.00
Coverage	1.67	2.50	2.00	2.33
Completeness	2.33	1.50	2.00	3.00
Intruder	2.37	1.90	2.00	2.37
Redundancy	1.67	1.25	1.50	2.00
Insightfulness	1.67	1.84	1.50	2.20
Trustworthiness	2.00	2.02	1.95	2.62
Total Score	1.80	1.90	1.68	2.34

Table C.8.: **Domain Expert Ratings for Patient P5 - Theme Generation Methods and Classification Across Rounds:** This table presents expert ratings for the Simultaneous and Stepwise theme generation and classification methods across multiple evaluation rounds for the Patient P5. Scores range from 1 (best) to 3 (worst). The **Total Score** reflects the overall quality ranking provided by domain experts. **Insightfulness** is calculated based on the first three metrics, while **Trustworthiness** is derived from the remaining metrics.

Metric	Simultaneous R1	Stepwise R1	Stepwise R2	Stepwise R3
Clinically Relevant	1.33	2.61	1.81	2.20
Novelty	1.00	1.50	2.00	2.20
Usefulness	1.00	2.00	1.60	2.20
Specificity	1.00	2.00	2.00	2.20
Coverage	1.00	2.50	1.60	1.80
Completeness	2.00	1.50	2.20	2.00
Intruder	2.33	1.89	2.19	1.75
Redundancy	3.00	1.00	1.60	1.80
Insightfulness	1.17	2.07	1.84	2.20
Trustworthiness	1.60	1.92	1.92	1.94
Total Score	1.37	2.01	1.86	2.09

Table C.9.: **Domain Expert Ratings for Patient P6 - Theme Generation Methods and Classification Across Rounds:** This table presents expert ratings for the Simultaneous and Stepwise theme generation and classification methods across multiple evaluation rounds for the Patient P6. Scores range from 1 (best) to 3 (worst). The **Total Score** reflects the overall quality ranking provided by domain experts. **Insightfulness** is calculated based on the first three metrics, while **Trustworthiness** is derived from the remaining metrics.

C.3.3. Evaluation Results Pairwise Comparison per Evaluator

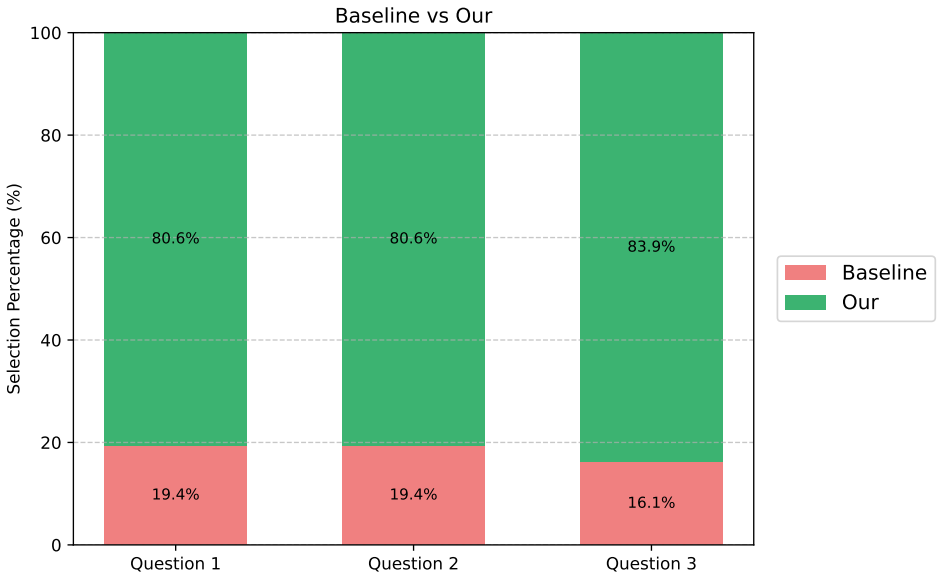


Figure C.1.: **Evaluation Results from CA for Pairwise Comparison:** This figure presents the results of the pairwise comparison between patient networks generated by our pipeline and the baseline, as evaluated by expert CA. The evaluator selected between the two networks based on three predefined questions. The figure shows the distribution of how often each type of patient network was chosen. This evaluation included 32 patient network pairs.

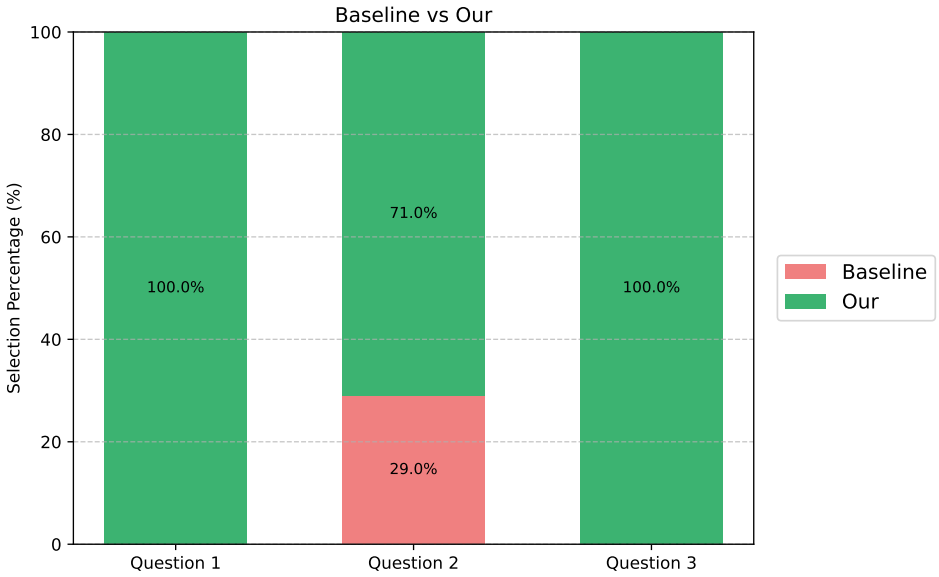


Figure C.2.: **Evaluation Results from KA for Pairwise Comparison:** This figure presents the results of the pairwise comparison between patient networks generated by our pipeline and the baseline, as evaluated by expert KA. The evaluator selected between the two networks based on three predefined questions. The figure shows the distribution of how often each type of patient network was chosen. This evaluation included 32 patient network pairs.

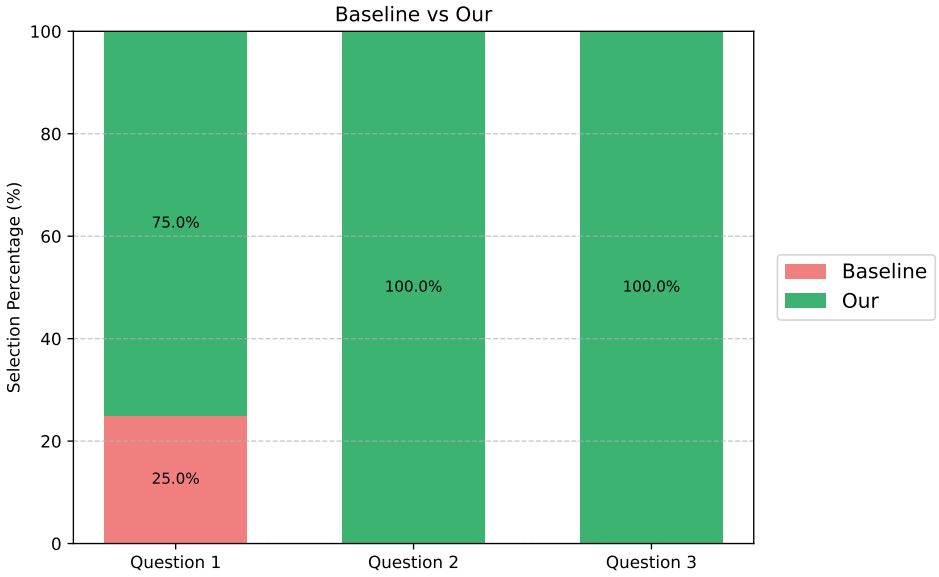


Figure C.3.: **Evaluation Results from ES for Pairwise Comparison:** This figure presents the results of the pairwise comparison between patient networks generated by our pipeline and the baseline, as evaluated by expert ES. The evaluator selected between the two networks based on three predefined questions. The figure shows the distribution of how often each type of patient network was chosen. This evaluation included 5 patient network pairs.

D. Appendix D

D.1. Evaluation Generation of Meaningful Themes and Classification

Evaluation Generated Clinically Meaningful Clusters
(Evaluation guidelines)

Eugen Owtscharow
Computer Science Department
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Outline

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1. Introduction
   Why is the Evaluation of the Cluster important?

2. Overview of Evaluation
   Cluster
   Prompt
   Annotation tool
   Key definitions
   Annotation Steps
   Evaluation Criteria Network Level: Definition and Example
   Evaluation Criteria Cluster Level: Definition and Example
   Evaluation Criteria General Assessment: Definition and Example

3. Evaluation steps in details

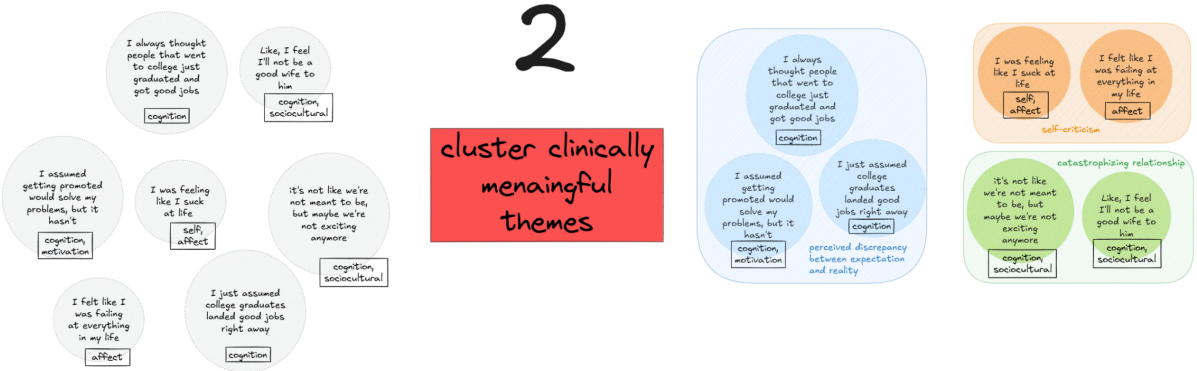
1. Introduction

Why is the Evaluation of the Cluster important?

Mental health issues represent a major global public health challenge, and while numerous therapeutic strategies are available, current clinical psychology often relies on standardized protocols that may fail to fully address the unique needs of individual patients. Processed-Based Therapy (PBT) offers a novel approach by integrating multiple effective treatment strategies tailored to individual needs, guided by the Extended Evolutionary Meta-Model (EEMM). The EEMM serves as an assessment tool (graph network) that transcends specific theoretical or intervention frameworks (e.g., CBT, ACT), and instead outlines areas of coverage and principles of change that clinical theories must address when considering humans as evolving systems.

However, constructing EEMM networks requires therapists to manually annotate therapy sessions to identify key psychological processes and their interrelationships, posing significant scalability challenges. This project seeks to overcome these limitations by automating the entire process of building EEMM (aka client networks) from annotated therapy sessions. In particular, this thesis is set up to identify and consolidate redundant processes (nodes in the client network), cluster processes that share highly similar clinical themes, and predict explainable relationships between these themed clusters, thereby enhancing the efficiency and scalability of PBT implementation.

We are currently in work package two, where we aim to generate clinically meaningful themes through the use of large language models (no connection between themes, this will be the next Work Package). By evaluating the clusters generated from the mini-session, you will help to determine how effectively the model groups related processes and generates clinically meaningful themes for each cluster. This evaluation will provide valuable insight into the performance and overall usefulness of the model, which will be used to produce the EEMM.



2. Overview of the evaluation

Cluster

The cluster you are going to evaluate was generated by a large language model using input from a single mini-session from a patient. This mini-session includes pre-labeled processes and a prompt that guides the model in producing the cluster.

Prompt:

Here is the prompt (instruction) currently used by the large language model to generate the output you are going to evaluate.

Task:
You are a clinical psychologist analyzing a therapy session transcript.

Definitions:
*- **Process**: A dynamic psychological mechanism that influences emotions, thoughts, and behavior, affecting mental health and therapy outcomes.*

Your Goal:
*1. Review the provided dialogue for overall context. Do **not** extract utterances from the dialogue.*
*2. Cluster all listed processes into **distinct, clinically meaningful groups** based on thematic relevance.*
*3. Consider the **freq** (X) for each process as an indicator of how often it appeared in a similar form within the dialogue.*
- A higher frequency suggests the process or theme is more important.
*4. Provide **clear, concise titles** for each cluster, reflecting clinically meaningful themes.*

Guidelines:
- Use only the processes listed under "Processes to be Clustered".
- Do not include any new processes from the dialogue.
*- Do not omit any listed processes. Every process *must appear in at least one cluster*.*
- Considering also their frequency
*- Use the full dialogue **only** for context. You do **not** need to label or cluster the dialogue.*
- A process may belong to multiple clusters only if it is strongly relevant to each theme.

- Avoid excessive duplication—if a process appears in multiple clusters, it must have a distinct, meaningful connection to each.
*- Titles must be **specific and descriptive**, capturing each cluster's essence.*
- Ensure each cluster is meaningfully distinct rather than repeating the same processes under different labels.
*- **Do not alter the original text** of the Processes in any way.*

The Output Format:\n
Cluster 1: [Title of the Cluster]\n
- Process 1 [Freq = X] \n
- Process 2 [Freq = X] \n
...

Cluster 2: [Title of the Cluster]\n
- Process 1 [Freq = Z] \n
- Process 2 [Freq = N] \n
...

Full Dialogue (Context Only!):
{prompt_str}

Listed Process to be Clustered:
{formatted_data_process_data}

Generate the output strictly in the specified format without additional explanations or modifications to the provided data.
"""
Note: The parameter *"primpt_str"* is the dialogue from the minisession and "*formatted_data_process_data*" are the processes listed with frequency

Provided Material

To evaluate the generated Cluster from a mini-session, you will be provided with an Excel spreadsheet along with the dialogue from the mini-session. This will give you an overall context of the mini-session's content. The dialogues are also labeled with the corresponding processes.

Key Definitions

- **Large Language Model (LLM)**: A LLM is an advanced artificial intelligence system trained on vast amounts of text data to understand, generate and respond to human language in a meaningful way.

- **Process:** A process is a series of connected actions or changes that happen over time and are influenced by the situation. In psychology, processes explain how and why problems develop and change, helping guide personalized treatment.

- **Dimension:** A broad category representing different aspects of psychological function (e.g., cognitive, emotional, behavioral), within which processes operate.

- **Cluster:** A cluster is a group of related nodes (processes with labels) that the large language model has grouped together based on its interpretation of clinical relevant themes. Each cluster reflects a specific clinical theme derived from these nodes.

- **Frequency**: Each process has a frequency, indicating how often a semantically similar process occurred in the dialogue. A higher frequency suggests that the process itself holds greater significance.

- **Dialogue**: The provided dialogue is a transcript from a mini-session, specifically the one you are going to evaluate. It includes processes labeled with dimension and frequency.

- **Generated Cluster Name (Clinically Theme)**: This is the name assigned to a Cluster. The name reflects the overall clinical theme or concept that unifies these nodes based on their content, making the cluster's subject/topic/idea clear.

- **Network:** The network represents the complete set of clusters produced by the model. It shows how all the identified processes and themes are interconnected.

- **Node (Process, Frequency, and Dimension):** A Node represents a specific process identified during a patient’s session (mini session), paired with a label that categorizes this process and frequency.

- **Evaluation Criteria Network Level**: This level evaluates the overall structure and quality of the network.

- **Evaluation Criteria Cluster Level**: This level evaluates the quality and relevance of individual clusters, including the Generated Cluster Name.

- **Evaluation Criteria General Assessment**: This level assesses the combination of Cluster and Network Level

Evaluation Criteria Network Level: Definition and Example

Criteria	Definition
Specificity	Specificity measures whether the network identifies only relevant processes/nodes, avoiding overly broad or unrelated information.
Novelty	Novelty evaluates whether the network provides new, unexpected, or useful insights beyond existing knowledge of the client/therapy session.
Completeness	Completeness measures whether all processes have been incorporated into the network. (Automatic, not evaluated by you)
Coverage	Coverage measures whether the network covers all relevant clinical topics in the current mini-session, ensuring that nothing important is missing.
Propose Clinically Theme (Optional)	The evaluator can suggest an alternative clinical theme/title for a cluster if the generated theme does not fully capture the meaning/theme of the nodes. This helps us to refine the model's output by incorporating clinical expertise and improving the relevance and accuracy of the themes.
Redundancy	Redundancy assesses whether the network contains duplicate or overlapping clinical themes, reducing efficiency and clarity.
Clinical Relevance	Assesses whether the identified clinical theme is meaningful to client well-being/functioning/goals.
Comments	Here you can add some comments about the Network itself

Criteria Specificity:

Criteria	Definition
Specificity	Specificity measures whether the network identifies only relevant processes/nodes, avoiding overly broad or unrelated information.

Value	Example
1 Too Broad	The network clusters sentences like *"I feel like a failure," "I'm always stressed,"* and *"I don't sleep well"* under a broad theme like "Mental Health Issues." This is too general to provide useful clinical insights.
2 Moderately Specific	The network clusters sentences like *"I'm overwhelmed by work"* and *"I can't relax because of my job"* under a theme called "Work Stress." This is somewhat specific but could be refined further.
3 Highly Specific	The network clusters sentences like *"I worry constantly about meeting deadlines," "I feel paralyzed by fear before presentations,"* and *"I avoid talking to my boss"* under the theme "Work-Related Anxiety." This is highly specific and clinically useful.

Criteria Novelty:

Criteria	Definition
Novelty	Novelty evaluates whether the network provides new, unexpected, or useful insights beyond existing knowledge of the client/therapy session.

Value	Example
1 None	The clusters simply reflect common categories like "Stress" and "Anxiety" without highlighting new insights.
2 Moderately Insightful	The network groups *"I can't stop worrying about my performance"* with *"I'm afraid of being judged,"* suggesting a common theme of "Performance Anxiety," which is useful but expected.
3 Highly Novel	The network groups *"I feel like I'm failing as a parent"*

	with *“I feel guilty when I relax,”* revealing a novel theme of “Parental Perfectionism,” providing a previously unknown clinical insight.

Criteria Coverage:

Criteria	Definition
Coverage	Coverage measures whether the network covers all relevant clinical topics in the current mini-session, ensuring that nothing important is missing.

Value	Example
1 Incomplete	The network captures themes like “Work Stress” but misses themes like “Family Conflict” and “Self-Worth Issues,” even though the patient mentioned statements like *“I feel like a bad mother”* or *“My partner and I argue constantly.”*
2 Partially Complete	The network includes themes like “Work Stress” and “Anxiety” but misses “Low Self-Esteem,” despite the patient saying things like *“I never feel good enough”* or *“I think I’m a failure.”*
3 Fully Complete	The network captures all relevant themes, such as “Work Stress,” “Family Conflict,” “Self-Worth Issues,” and “Anxiety,” covering statements like *“I feel like a failure,” “I can’t balance work and family,”* and *“I’m always anxious about my performance.”*

Criteria Redundancy:

Criteria	Definition
Redundancy	Redundancy assesses whether the network contains duplicate or overlapping clinical themes, reducing efficiency and clarity.

Value	Example
1 Highly Redundant	The network has separate clusters for *"I feel like a failure"* and *"I never feel good enough,"* both under themes like "Low Self-Esteem" and "Insecurity," which overlap significantly
2 Moderately Redundant	The network clusters *"I can't handle my job"* under "Work Stress" and *"I'm overwhelmed by my workload"* under "Burnout," which share some overlap but have subtle differences.
3 Minimal Redundancy	Each cluster is distinct: *"I can't relax because of work"* under "Work Stress" and *"I don't feel competent in my role"* under "Low Self-Esteem." No themes are duplicated, but the same node is allowed to be included in multiple clusters.

Criteria Clinical Relevance:

Criteria	Definition
Clinical Relevance	Assesses whether the identified clinical theme is meaningful to client well-being/functioning/goals.

Value	Example
1 Clinically Insignificant	The network produces clusters with vague or irrelevant themes, such as grouping "mood swings" and "sleep patterns" without any clear clinical connection or practical use.

	The network clusters *“I feel tired”* and *“I don’t enjoy things”* under a vague theme like “General Discomfort,” which isn’t useful for clinical diagnosis.
2 Partially Meaningful	The network contains some useful clinical themes, like “anxiety disorders,” but are too broad or lack focus, making it difficult to apply them directly in clinical decision-making. The network clusters *“I can’t sleep”* and *“I’m constantly worried”* under the theme “Stress,” which is somewhat helpful but lacks specificity.
3 Clinically Significant	The network identifies clear, actionable clinical themes, such as “excessive worry about health,” and “hypervigilance for threat” providing meaningful insights that can support diagnosis or treatment planning. The network clusters *“I worry excessively about my kids,” “I feel like a bad mother,”* and *“I’m afraid of making mistakes with parenting”* under the theme U“Parental Anxiety,” which is clinically actionable.

Evaluation Criteria Cluster Level: Definition and Example

Criteria	Explanation
Consistency	Consistency checks whether the generated clinically theme aligns with the nodes. **Focus:** The overall alignment between the theme and the nodes
Intruder	Intruder identifies whether (if any) node in the cluster does not fit or is irrelevant to the cluster overall theme **Focus:** Identifying specific nodes that do not fit within a cluster
Comments	Here you can add some comments about each Cluster

Note: Consistency measures overall thematic alignment, which can still be high even if a few intruders are present, provided they do not dominate the cluster. Intruder detection, however, pinpoints specific out-of-place nodes to evaluate how well a cluster is accurate. In the comments section, you can indicate where you would like to see the node.

Criteria Consistency:

Criteria	Definition
Consistency	Consistency checks whether the generated clinically theme aligns with the nodes. **Focus:** The overall alignment between the theme and the nodes

Value	Example
1 Poorly Coherent	The cluster is labeled **"Self-Esteem Issues"** but contains statements like *"I worry about meeting deadlines"* and *"I feel stressed at work,"* which are more about work anxiety.
2 Moderately Coherent	The cluster is labeled **"Work Stress,"** and most statements fit, such as *"I can't relax because of my job,"*. However, it also contains "I feel like I don't deserve success.". While this node isn't irrelevant, it reflects self-esteem issues, suggesting the label Work Stress does not fully capture the theme.
3 Highly Coherent	The cluster is labeled **"Low Self-Esteem,"** and all

	nodes fit the theme, such as *“I never feel good enough,” “I think I’m a failure,”* and *“I feel unworthy of success.”*

Criteria Intruder:

Criteria	Definition
Intruder	Intruder identifies whether (if any) node in the cluster does not fit or is irrelevant to the cluster overall theme **Focus:** Identifying specific nodes that do not fit within a cluster

Value	Example
1 Multiple Intruders	The cluster is labeled **“Work Stress”** but contains nodes like *“I feel like a bad parent”* and *“I don’t enjoy social events,”* which do not fit.
2 Few Intruders	The cluster is labeled **“Anxiety Issues”** and contains mostly relevant nodes, like *“I worry about everything,”* but includes one out-of-place node, *“I feel worthless.”*
3 All Relevant	The cluster is labeled **“Work Burnout”** and contains only fitting nodes, such as *“I feel exhausted by my job,” “I can’t keep up with my workload,”* and *“I dread going to work.”*

Evaluation Criteria General Assessment: Definition and Example

Criteria	Definition
Usefulness	Usefulness evaluates whether the network offers valuable insights or supports clinical decision making effectively
Comments	Here you can add some comments about the overall Performance of Network and Clusters

Criteria Usefulness:

Value	Example
1 Limited Insight	The network and cluster provide very general or surface-level insights, making it difficult to derive meaningful clinical conclusions. Example: Clusters statements like *"I feel tired," "I can't concentrate,"* and *"I feel sad"* under a vague theme like "General Mental Health Issues." The themes are too broad and do not help clinicians identify specific problems or guide treatment decisions.
2 Some Insight	The network and clusters offer some useful information, but the insights are incomplete or lack depth, limiting their clinical usefulness Example: The Network identifies clusters such as "Work Stress" and "Family Issues". While these themes are helpful, they miss deeper insights or connections. For instance, a cluster on "Low Self-Esteem" could have been more detailed.
3 Valuable Insight	The network and clusters provide detailed, specific, and actionable insights that effectively support clinical decision-making. Example: The network identifies distinct and clinically useful themes like "Parental Anxiety," "Work-Related Burnout," and "Low Self-Worth." Each cluster contains statements like *"I feel guilty when I relax"* (Parental Anxiety), *"I dread going to work"* (Work Burnout), and *"I never feel good enough"* (Low Self-Worth). These

	insights help clinicians identify targeted interventions and understand the patient's unique struggles.

Annotation steps:

1. Read the provided dialogue from the current mini-session.
2. Read all the processes of the mini-session and the generated clinically theme
3. Evaluate the criteria at the Network Level **in order**
4. Evaluate the criteria at Cluster Level at Cluster Level **in order**
5. Evaluate the criteria at General Assessment

Evaluation steps in details

Before starting the evaluation, you will open the evaluation sheet that has been assigned to you.

The first two rows of the evaluation sheet are reserved for the heading, **do not change them!**

	A	B	C	D	E	F	G
1		Node					
2	**Generated Cluster Name (Clinnically Theme)**	**Process**	**Frequency**	**Dimension**	**Specificity:** Is the network too broad or sufficently specific to be clinically usefull?	**Novelty:** Does the network reveal any new, unexpected, or useful insights?	**Coverag** network clinicly t

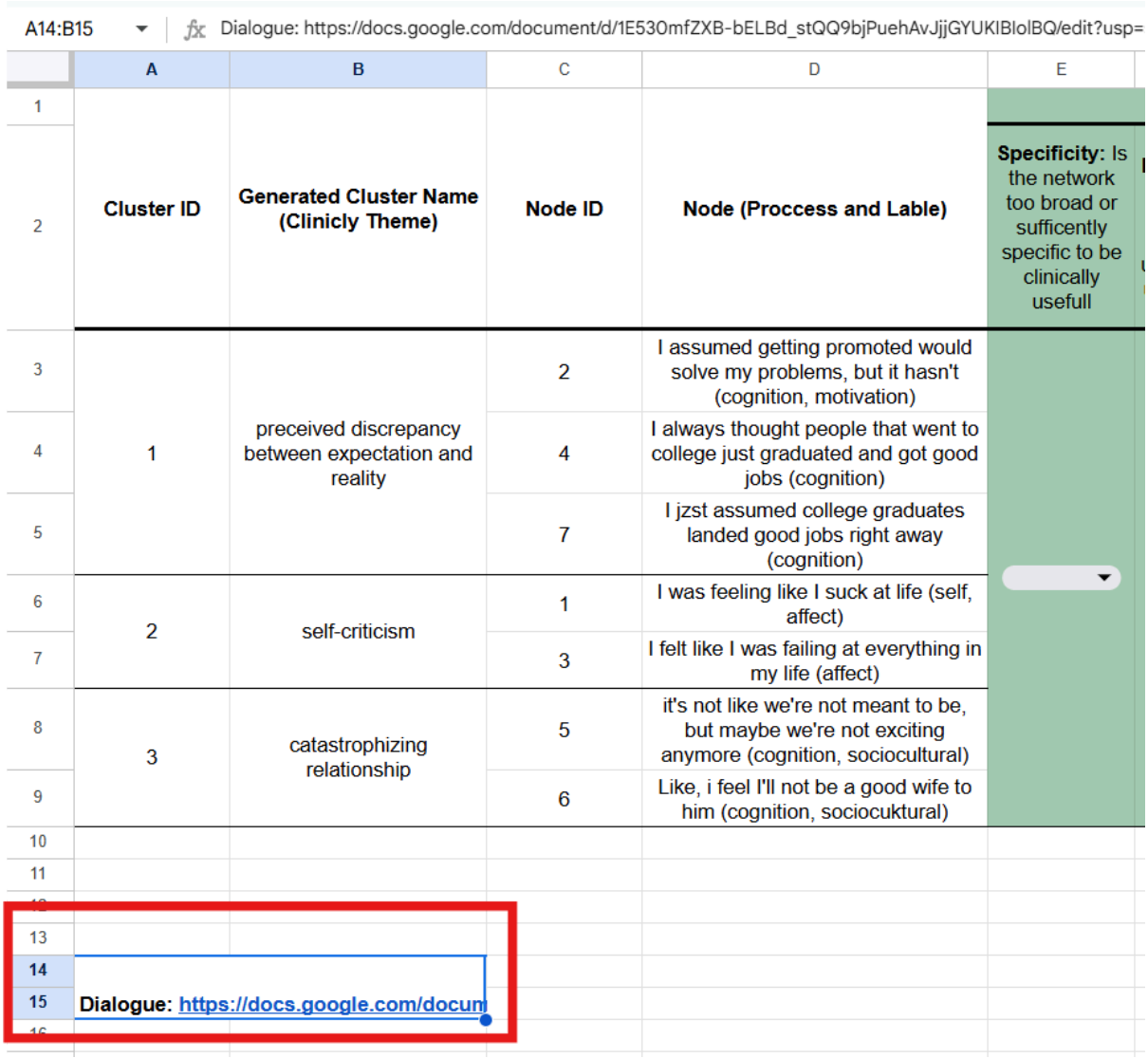
J	K	L	M	N	O	P
		Evaluation Criteria Cluster Level			**Evalaution Criteria General Assessment**	
Clinical Relevance: Is the identified clinicly theme meaningful?	Comments	**Consistency:** Is the theme consistent with the content?	**Indruder:** Is there an intruder node in the Cluster	Comments	**Usefulness:** Can this cluster provide insight or support clinical decision-making?	Comments

The headings are colour coded. Colourless means that this column is the given input and should **not be changed!**
Green represents the network level assessment, **yellow** represents the cluster level assessment, and **purple** represents the overall assessment. The Evaluation criteria in each of those sections is described above (Chapter 2).

Steps:

1. Read the provided dialogue from the current mini-session

Each dialogue can be found at the bottom of the evaluation sheet. Click on the link to go to the text document



You will find all the listed dialogs assigned to you in this text document.
Each dialogue starts with the patient number and session, followed by dashes. You will find these dashes at the end of each dialogue

Session: P11_S09

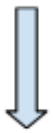
Client: [process dimension=['Cognition'], freq=1]So I don't know how much this will relate to like the next time we go out and do something big.[/process]
Therapist: I think it'll be a good test because it's like, oh, I was also going to say, I think, like, you kind of keeping at it for the past couple of weeks.

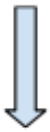
Therapist: Analogy for how to.
Therapist: Respond to anxiety.
Therapist: Okay.

Find the dialogues you are currently evaluating and read through them. This will give you a general idea of what this mini-session is about. To find out which mini-session you are currently evaluating, look at the sheet name at the bottom.

2. Read the processes of the mini-session and the generated clinically theme

After reading the dialogue, read the colourless section where the nodes (process, frequency, dimension) and the generated cluster name are written to be able to evaluate the network and the clusters.





	A	B	C	D	E
1		**Node**			
2	**Generated Cluster Name (Clinnically Theme)**	**Process**	**Frequency**	**Dimension**	**Specificity:** Is the network too broad or sufficently specific to be clinically usefull?

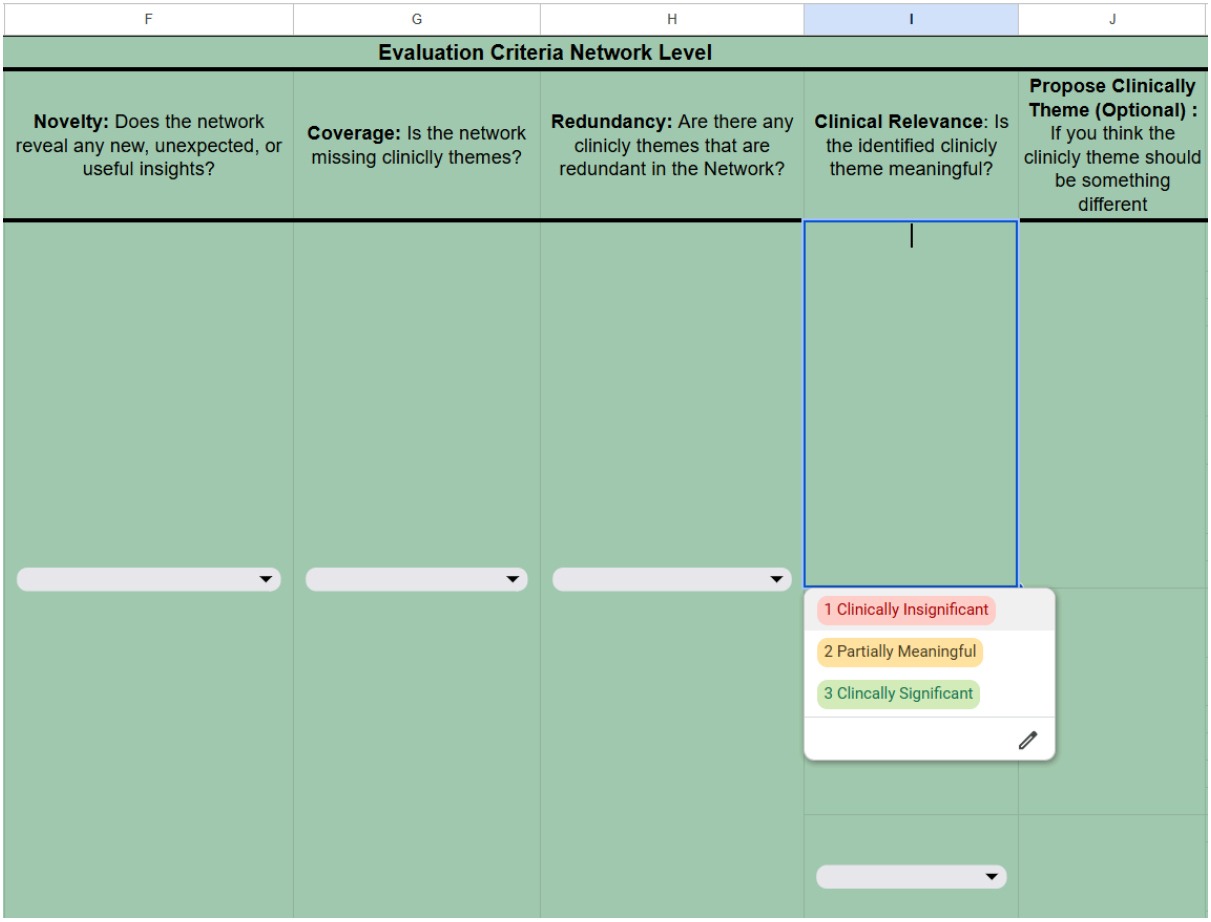
Those are in the columns **A** (Generated Cluster Name/Clinically Theme) and **B-D** (Node with Processes, Frequency and Dimension).

3. Evaluate the criteria at the Network Level in order

Once you have gained an overview of the network, the next step is to evaluate the network level using the criteria provided **in the given order**. These criteria are presented in columns **E** through **J**. In addition, a question is posed for each criterion in order to obtain a more comprehensive understanding.

E	F	G	H	I	J	K
Evaluation Criteria Network Level						
Specificity: Is the network too broad or sufficently specific to be clinically usefull?	**Novelty:** Does the network reveal any new, unexpected, or useful insights?	**Coverage:** Is the network missing clinicly themes?	**Redundancy:** Are there any clinicly themes that are redundant in the Network?	**Clinical Relevance**: Is the identified clinicly theme meaningful?	**Propose Clinically Theme (Optional) :** If you think the clinicly theme should be something different	Comments

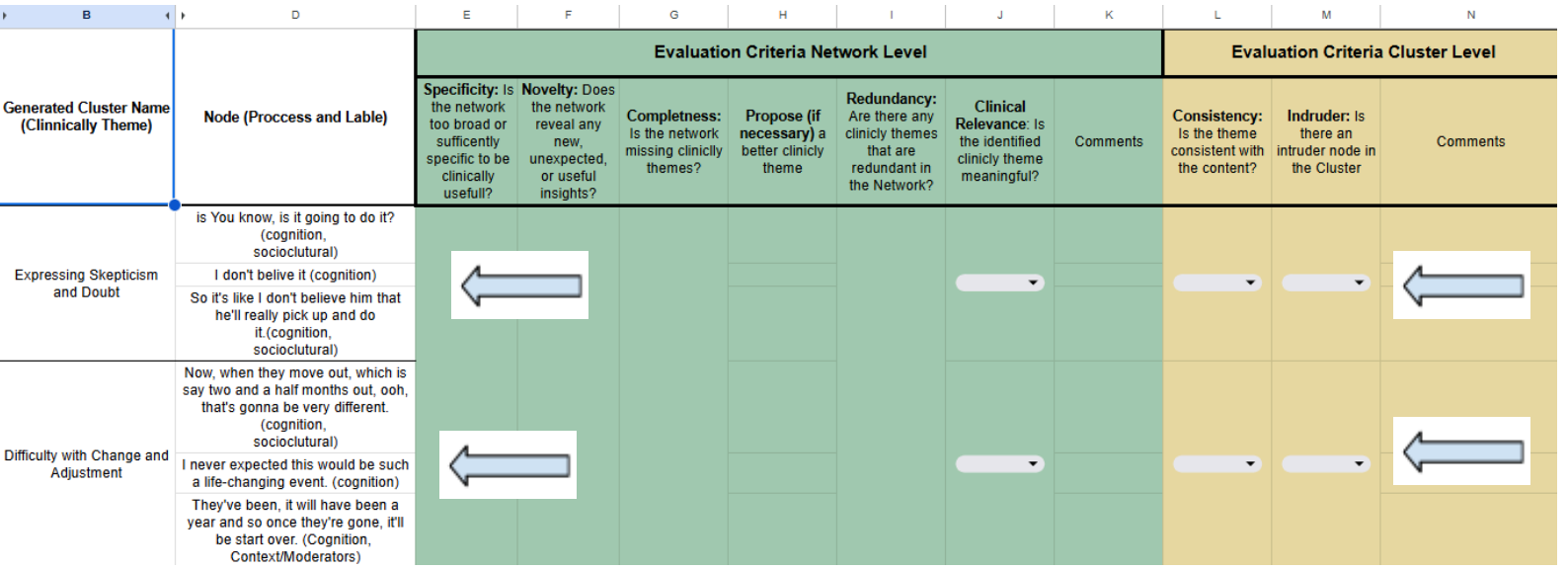
Each of the criteria is accompanied by a drop down, each of which is assigned a value between 1 and 3 (plus a few words definition of the number). The second chapter provides a detailed explanation of the criteria, as well as illustrative examples.



While criteria such as specificity (Column E), novelty (Column F), coverage (Column G) and redundancy (Column I) are common to all clusters, clinical relevance (Column J) is specified for each cluster.
In addition, you also have the possibility to make some comments in column K for the evaluation of the Network Level.

4. Evaluate the criteria at Cluster Level at Cluster Level in order

Once the network has been evaluated, you will evaluate each cluster in the network, which will be shown in yellow (Evaluation Criteria Cluster Level). Columns from L trough M

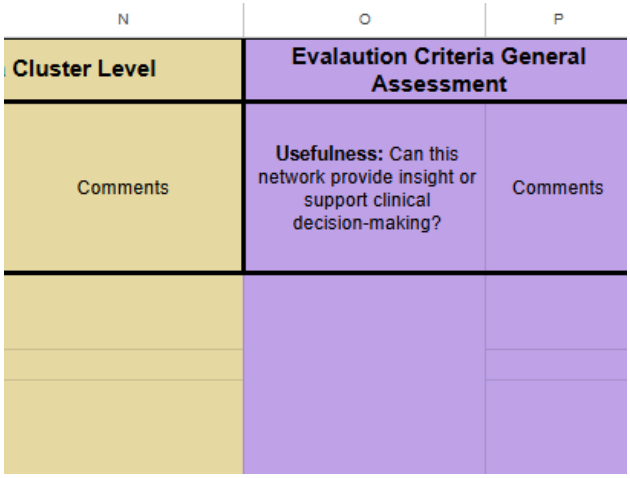


B	D	E	F	G	H	I	J	K	L	M	N
		Evaluation Criteria Network Level							Evaluation Criteria Cluster Level		
Generated Cluster Name (Clinnically Theme)	**Node (Proccess and Lable)**	**Specificity:** Is the network too broad or sufficently specific to be clinically usefull?	**Novelty:** Does the network reveal any new, unexpected, or useful insights?	**Completness:** Is the network missing clinicly themes?	**Propose (if necessary)** a better clinicly theme	**Redundancy:** Are there any clinicly themes that are redundant in the Network?	**Clinical Relevance**: Is the identified clinicly theme meaningful?	Comments	**Consistency:** Is the theme consistent with the content?	**Indruder:** Is there an intruder node in the Cluster	Comments
	is You know, is it going to do it? (cognition, socioclutural)										
Expressing Skepticism and Doubt	I don't belive it (cognition)										
	So it's like I don't believe him that he'll really pick up and do it.(cognition, socioclutural)										
	Now, when they move out, which is say two and a half months out, ooh, that's gonna be very different. (cognition, socioclutural)										
Difficulty with Change and Adjustment	I never expected this would be such a life-changing event. (cognition)										
	They've been, it will have been a year and so once they're gone, it'll be start over. (Cognition, Context/Moderators)										

This means that you must select a number between 1-3 for **each cluster** in the network in the drop down menu. Additionally like before you have also the possibility to add some Comments into this Category in Column N.

5. Evaluate the criteria at General Assessment

This step is the final part of the evaluation. In this category, which is coded as purple, you will evaluate the network and the cluster together.



N	O	P
Cluster Level	**Evalaution Criteria General Assessment**	
Comments	**Usefulness:** Can this network provide insight or support clinical decision-making?	Comments

Again there is a drop down menu in column O where you can select a score from 1 to 3. There is also a comment section in column P.

D.2. Evaluation Pairwise Comparison

Pairwise Comparison of Patient Networks
(Evaluation guidelines)

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1. Overview of Evaluation

Dear Evaluator,
Thank you for participating in this evaluation. You will be analyzing two clinical networks per session and deciding which one performs better according to specific questions. Below are instructions on how to conduct the evaluation using the materials provided.

What You will Receive:

1) A **folder** specific for you
    - Within this folder, you'll find a set of session-specific subfolders (e.g. P15_S05, P11_S08, etc)
    - Each of the Session folder contains two html files named with Network_A.html and Network_B.html

2) An **Google Sheets file**
    - Link to the file will be provided via email
    - Each **row** corresponds to a session
    - The sheet contains 3 evaluation questions with dropdowns and optional comment fields

3) Google Doc **with annotated Transcripts**
    - This document contains transcripts from all sessions, each annotated for clarity and reference

Here's a refined and well-structured version of your instructions:

How to Evaluate
For each session:

1. **Select the session folder** you want to evaluate.
2. **Open both HTML files** (`Network_A.html` and `Network_B.html`) in your web browser.
3. **Access the evaluation sheet** in the Google Sheets file:
    - Locate the row that corresponds to the session ID.
4. **Open the Google Doc** with annotated transcripts:
    - Find and read the transcript for the specific therapy session.
5. **Review the three evaluation questions** on the Google Sheet.
    - Carefully analyze both Network A and Network B before responding.

For each question:

- Use the dropdown menu to select **either Network A or Network B** as your answer.

(Optional) Comments Section:

- Add any observations in the comment columns for Network A and Network B.
- You may note:
    - Strengths and weaknesses
    - Unusual or unclear connections

- Clinically irrelevant themes or anomalies

Note:

- Each question must be answered using the dropdown menu
- Comments are encouraged but optional

Questions:

Question
Which Network has better clinically meaningful themes?
Which Network has better connections?
Which network helps you understand this patient better and could assist you more in conducting therapy sessions?

Key Definitions:

Relationship Attributes:

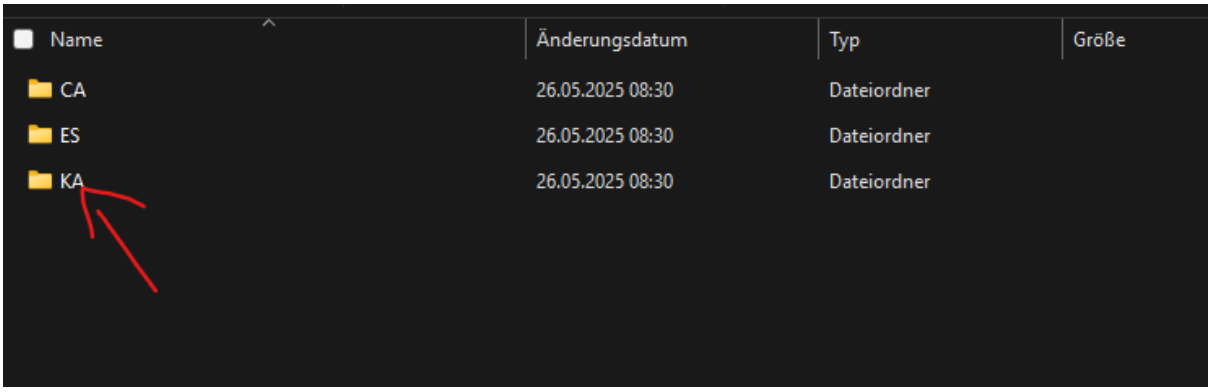
- **Type of relationships between Themes**
  1) Excitatory as in one process amplifies or reinforces another process. (e.g. Process A leads to Process B, Process A causes Process B)
  2) Inhibitory as in one process suppresses or counteracts another process. (e.g. Process A stops Process B,

Process A diminishes Process B)

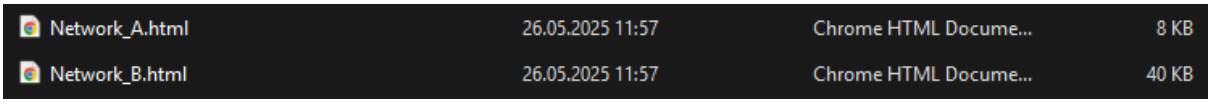
- **Strength of Relationship**

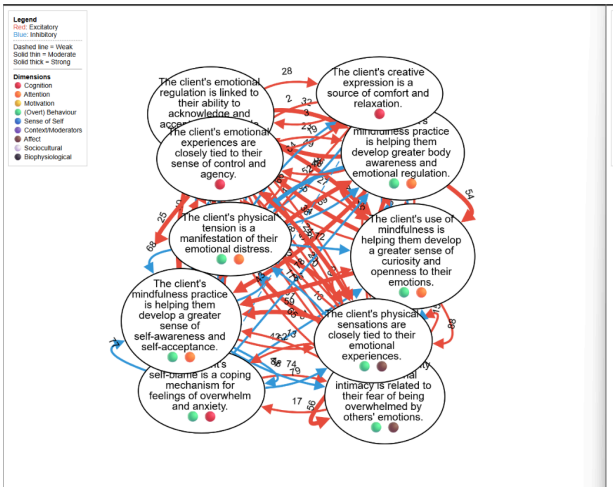
1) "Strong": The two processes are closely and consistently related across contexts. One strongly influences or is inherently linked to the other in psychological functioning.
2) "Moderate": The two processes are meaningfully related, but the connection is less consistent or conditional. They are associated in many situations, but not tightly bound.
3) "Weak": The relationship between the two processes is minimal, indirect, or highly context-dependent. They may co-occur at times, but the link is loose or peripheral.

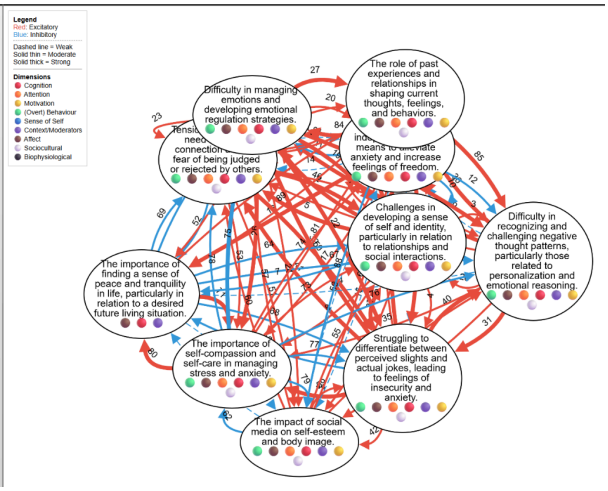
Example Pictures:



Open both html files







Network A

Network B

Google Sheet

	A	B	C	D	E	F
		Questions			Comments (Optional)	
	Session ID	Which Network has better clinically meaningful themes?	Which Network has better connection?	Which network helps you understand this patient better and could assist you more in conducting therapy sessions?	Network A	Network B
	P16_S10					
	P16_S08					

Click for example at B3 a drop down Menu will com where you can select Network A or Network B

A	B	C	D	E	F
Session ID	**Questions**			**Comments (Optional)**	
	Which Network has better clinically meaningful themes?	Which Network has better connection?	Which network helps you understand this patient better and could assist you more in conducting therapy sessions?	Network A	Network B
P16_S10					
P16_S08					
P12_S05					
P11_S01					

Network A

Network B

Generated Cluster Name (Clinically Theme)	Node					Evaluation Criteria Network Level							Evaluation Criteria Cluster Level		Evaluation Criteria General Assessment	
	Process ID	Process	Frequency	Dimension	Context	Specificity: Is the network too broad or sufficiently specific to be clinically useful?	Novelty: Does the network reveal any new, unexpected, or useful insights?	Coverage: Is the network missing clinically themes?	Redundancy: Are there any clinically themes that are redundant in the Network?	Clinical Relevance: Is the identified clinically theme meaningful?	Comments for Network Level	Intruder: Is there an intruder node in the Cluster	Comments for Cluster Level	Usefulness: Can this network provide insight or support clinical decision-making?	General Comments
Theme A	PXX_SYY															
	PXX_SYX															
Theme B																
Theme C																

Figure D.1.: **Example of an Evaluation Sheet for Evaluating the generated Themes and Classification:**

Session ID	Questions			Comments (Optional)	
	Which Network has better clinically meaningful themes?	Which Network has better connection?	Which network helps you understand this patient better and could assist you more in conducting therapy sessions?	Network A	Network B
P11_S01					
P11_S03					
P11_S05					
P11_S08					
P11_S10					
P11_S11					
P11_S13					
P12_S01					

Figure D.2.: **Example of an Evaluation Sheet for Pairwise Comparison**